



**Medtronic**

# **Stealth AXiS™ Cranial Optical and EM Clinical Application User Manual**

Instructions for cranial procedures using optical and electromagnetic (EM) localization



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Stealth AXIS™ clinical application version 1.0

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## Intended use / Indications for use

The Stealth AXiS™ Surgical System, with the Stealth AXiS™ Cranial clinical application, is intended for precise positioning of surgical instruments and as an aid for locating anatomical structures in open, minimally invasive, and percutaneous neurosurgical procedures. Their use is indicated for medical conditions in which the use of stereotactic surgery may be appropriate, and where reference to a rigid anatomical structure, such as the skull, can be identified relative to images of the anatomy.

This can include, but is not limited to, the following cranial procedures (including stereotactic frame-based and stereotactic frame alternatives-based procedures):

- Tumor resections
- General ventricular catheter placement
- Pediatric ventricular catheter placement
- Depth electrode, lead, and probe placement
- Cranial biopsies

## Contraindications

None

## Device description

The Stealth AXiS™ Surgical System is a computer-assisted surgery system that is composed of a platform, clinical application, surgical instruments, and a referencing system (which includes patient and instrument trackers). The system tracks the position of instruments in relation to the surgical anatomy, known as localization, and then identifies this position on preoperative or intraoperative images of a patient. The Stealth AXiS™ Cranial clinical application helps guide surgeons during cranial procedures such as biopsies, tumor resections, and shunt placements.

Patient images can be displayed by the Cranial clinical application from a variety of perspectives (axial, sagittal, coronal, oblique) and 3-dimensional (3D) renderings of anatomical structures can also be displayed. During navigation, the system identifies the tip location and trajectory of the tracked instrument on images and models the user has selected to display. The surgeon may also create and store one or more surgical plan trajectories before surgery and simulate progression along these trajectories. During surgery, the clinical application displays how the actual instrument tip position and trajectory relate to the pre-surgical plan, helping to guide the surgeon along the planned trajectory. While the surgeon's judgment remains the ultimate authority, real-time positional information obtained through the Stealth AXiS™ Surgical System can serve to guide this judgment.

Stealth AXiS™ Core is composed of the workstation cart and the camera cart, which contain the key navigation components including the localizers and the computer with the operating system. The camera cart is used for optical procedures, and EM localizers are used for EM procedures.

## Compatibility and installation

The clinical application may be used on a Stealth AXiS™ planning station in an office environment, a Stealth AXiS™ computer-assisted surgery system in an operating room or office environment, or optionally from a web browser or a connected computer if the remote planning feature has been enabled on a Stealth AXiS™ planning station.

To use the remote planning feature, the connected computer must meet the minimum system performance requirements, operating system and browser compatibility requirements, and be able to access the planning server on the network. Configuring a planning station as a planning server requires the installation of a software license by qualified Medtronic personnel and further configuration by hospital staff. See the *Stealth AXiS™ IT Guide* for server and web browser configuration instructions.

The clinical application is compatible with Stealth AXiS™ systems and Medtronic navigational instruments.

The clinical application is installed by Medtronic service personnel only. If you have any questions about your clinical application installation, contact Medtronic Technical Services.

**Note:** When the Stealth AXiS™ clinical application is used with other medical devices, see the user instructions for those devices for additional labeling limitations.

## Navigation accuracy

Under representative worst-case configuration, the Stealth AXiS™ Surgical System with the Stealth AXiS™ Cranial clinical application has demonstrated performance in 3D positional accuracy with a mean error  $\leq 2.0$  mm and in trajectory angle accuracy with a mean error  $\leq 2.0$  degrees. The positional error is defined as the Euclidean distance from target to instrument tip at final point of insertion. The trajectory angle error is the angle between the planned surgical trajectory and trajectory of the neurosurgical instrument at final point of insertion. Depending on the surgical procedure and the particular neurosurgical instruments used, the resulting navigational accuracy may be negatively impacted.

This performance was determined using an anatomically representative phantom and utilizing a subset of system components and features that represent the worst-case combination of all potential system components in a clinically relevant workflow. The test configuration included CT images with slice spacing and thickness of 1.25 mm and T2-weighted MR images with slice spacing and thickness of 1.0 mm.

## User profile

The intended user profile for the system with the Cranial clinical application is neurosurgeons, their support staff, hospital biomedical engineering support staff, medical imaging staff, and hospital IT.

## Intended patient population

The intended patient population for the Cranial clinical application is patients for whom stereotactic image-guided surgery is appropriate.

## Side effects

The system and its associated applications are auxiliary surgical tools and do not independently produce side effects. When used in conjunction with navigated instruments for surgical procedures, the associated side effects are the same as those for the procedure itself.

## Potential complications

The potential complications for Stealth AXiS™ Surgical System Cranial Procedures identified by Medtronic include, but are not limited to:

General surgical complications, pain, edema, hydrocephalus, insufficient treatment, infection, tissue or bone damage, burn, hearing loss, vision impairment, electric shock, cerebrospinal fluid leakage or fistula, extracranial or intracranial bleeding, brain or nervous system injury, permanent or transient neurological deficit, paralysis, seizure, death.

## Clinical benefits

Navigated procedures performed using the Stealth AXiS™ Surgical System with Stealth AXiS™ Cranial clinical application potentially confer the following clinical benefits:

- Provides consistent accuracy that contributes to the achievement of surgical goals and reduces the risk of procedure-related complications for patients.
- When used with the Stealth Autoguide™ System, the system enables targeted, accurate acquisition of intracranial biopsy specimens, thereby reducing the risk of repeat procedures.
- Electromagnetic-based tracking system enables pinless cranial procedures, eliminating the need for rigid head fixation, thereby reducing patient discomfort and postoperative pain.
- Provides detailed anatomical information, including 3D models and white matter tracts visualization, supporting surgeons in developing patient-specific surgical approaches for deep-seated and/or eloquent anatomic areas, which may lessen neurological deficits.
- The use of less invasive cranial procedures may result in shorter length of stay (LOS) and reduced postoperative pain, thereby facilitating faster postoperative recovery.

## Optical localization and electromagnetic localization

The system can track instruments either optically or electromagnetically. Some systems may have one or both of the tracking methods.

- With optical tracking, a camera (also called a localizer) detects infrared light from optical markers (either reflective spheres or LEDs) on the specialized instrument. The infrared light is either emitted by the camera and reflected off spheres on the instrument or emitted directly by LEDs on the instrument. The camera transmits the instrument's location to the navigation software.
- With electromagnetic (EM) tracking, an emitter and an instrument interface work together as a localizer. The emitter emits a low-energy magnetic field with unique characteristics at every point. The electromagnetic instruments contain sensors which allow the navigation software to identify the instrument's location within the electromagnetic field.

This document employs the following conventions to distinguish information that is specific to either optical or EM localization:

- When a section of this document applies to optical localization only, the heading for that section contains **(optical only)**.
- When a section of this document applies to EM localization only, the heading for that section contains **(EM only)**.

## Warnings

Refer to the applicable package inserts and system manual for additional instructions, warnings, and cautions.

**Warning:** The system hardware and clinical application should be used only by qualified medical professionals who are trained in performing surgery and familiar with computer-assisted surgery systems.

**Warning:** Abort use of the system if navigation seems inaccurate and steps to restore accuracy are unsuccessful.

**Warning:** Inspect all visible system components before use including the camera. If visibly damaged, do not use the system. Abandon use of any component damaged during the procedure. Contact Medtronic technical services.

**Warning:** Inspect sterile device packaging and expiration date before use. Do not use any pre-sterilized device whose sterile packaging appears to be compromised or whose expiration date has passed. Contact Medtronic technical services.

**Warning:** Prior to use, examine devices for damage and deterioration. Do not use any bent, damaged, or compromised devices. Abandon use of any device bent or damaged during the procedure. Contact Medtronic technical services.

**Warning:** Do not reprocess, resterilize, or reuse any single-use device. Attempts to reprocess or resterilize a single-use device may be ineffective and may compromise its structural integrity. Any reuse creates a risk of contamination which could result in patient injury, illness, or death.

**Warning:** The system and its associated applications contain no user-repairable parts. For repair or replacement of any part of the system or application, contact Medtronic technical services.

**Warning:** Disposable devices are provided in sterile condition for single use only. After use, dispose of single-use devices according to local and national regulations.

**Warning:** Device components may be supplied in a non-sterile condition and must be cleaned and sterilized prior to use. Verify that all necessary instrumentation has been properly cleaned and sterilized before surgery. For sterilizable components, refer to the cleaning and sterilization instructions that accompany the component. For non-sterilizable components, refer to the *Stealth AXiS™ Surgical System Cleaning and Disinfection Instructions*.

**Warning:** Do not open sterile-barrier packages or containers until surgical use.

**Warning:** Do not modify the Stealth AXiS™ Surgical System, components or accessories, without prior written consent from Medtronic Navigation. Prohibited modifications include but are not limited to: altering, repairing, or replacing system components; altering software provided by Medtronic; updating the operating system or drivers; and installing any software that is not provided by Medtronic. Secure the system to prevent unauthorized modifications. Unauthorized modifications could render the system unsafe or ineffective for its intended use.

**Warning:** Use care during handling, storage, and cleaning of instruments to prevent physical injury and deformation of the instrument tip.

**Warning:** Do not allow fluid to enter the electrical components of tracked instruments.

**Warning:** Do not reuse or reprocess a device where transmissible spongiform encephalopathy agents or Creutzfeldt-Jakob disease contamination is suspected or confirmed. Incinerate the potentially contaminated device according to national guidelines.

**Warning:** To prevent cross-contamination, always handle, transport, and reprocess devices that contact the central nervous system separately from other devices.

**Warning:** To avoid potential exposure to blood-borne pathogens and chemicals, use appropriate Personal Protective Equipment when handling or disposing of Medtronic devices.

## Precautions

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**Warning:** Before transporting the system carts, shut down the system and stow all components, dock the monitors, dock the camera on the camera cart, close all storage compartments, and remove any loose items from the top of the carts.

### Warnings (EM only)

**Warning:** RF emitters typically found in an operating room can affect medical electrical equipment performance, such as the Stealth AXiS™ electromagnetic localization system.

**Warning:** The side emitter and the flat emitter are not intended for sterile patient contact or interaction. If you place these items in the sterile field, follow proper aseptic technique guidelines to bag or drape them prior to use.

**Warning:** Do not use the side emitter or the flat emitter in ambient (room) temperatures greater than 30°C (86°F). Exceeding this temperature limit could result in navigational inaccuracy or patient or user harm from excessive temperatures.

## Precautions

**Caution:** Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.

**Caution:** The system mouse is not designed for sterilization and may be damaged if sterilization is attempted.

**Caution:** Before use, clean the camera lenses according to the *Stealth AXiS™ Surgical System Cleaning and Disinfection Instructions*. Dirty camera lenses could impair optimal performance of the camera.

**Caution:** The system is not suitable for use in the presence of a flammable, anesthetic mixture with air, oxygen, or nitrous oxide.

### Precautions (EM only)

**Caution:** The system mouse, side emitter, flat emitter, and emitter holder are not designed for sterilization and may be damaged if sterilization is attempted.

**Caution:** For procedures performed with the flat emitter, a surgical table with a radiolucent table top is recommended. If the surgical table has a metal frame or metal side rails, position the flat emitter at least 5 cm (2 in) away from the metal. Positioning the emitter closer may result in reduced navigational accuracy and volume.

**Caution:** If the surgical table to be used with the side emitter has a metal frame or metal side rails, position the side emitter at least 25 cm (10 in) away from the metal. Positioning the emitter closer may result in reduced navigational accuracy and volume.

**Caution:** Procedures performed with the side emitter require a surgical bed featuring side rails because the arm assembly that supports the emitter connects to the side rails.

**Caution:** Electrical noise in or near the electromagnetic navigation field can degrade navigational accuracy and the navigation volume. If electrical noise introduces excessive error, the system may automatically disable navigation. To restore navigation, move devices that produce electrical noise (such as electrocautery equipment and electric drills) away from the navigation field.

**Caution:** Metallic or conductive objects in or near the electromagnetic navigation field can degrade navigational accuracy and the navigation volume. If metallic distortion causes excessive error, navigation may be disabled. To restore navigation, move metallic and conductive objects away from the navigation field.

## Contact information

If a serious incident occurs in relation to the use of the system, report it to Medtronic Navigation. If a serious incident occurs in the European Union, also report it to the competent authority in the Member State where the incident occurred.

For instructions about return shipping or about transporting system components or accessories, contact Medtronic technical support.

|                  |                              |                 |
|------------------|------------------------------|-----------------|
| <b>Telephone</b> | Technical support worldwide  | +1 720 890 3160 |
|                  | Technical support in the USA | 1 800 595 9709  |
|                  | General                      | +1 720 890 3200 |
|                  | Fax                          | +1 720 890 3500 |

|                        |                            |
|------------------------|----------------------------|
| <b>Mailing address</b> | Medtronic Navigation, Inc. |
|                        | 200 Medtronic Drive        |
|                        | Lafayette, Colorado 80026  |
|                        | USA                        |

**World Wide Web**      [www.medtronic.com](http://www.medtronic.com)  
[www.medtronic.com/manuals](http://www.medtronic.com/manuals)

**E-mail**                [rs.navtechsupport@medtronic.com](mailto:rs.navtechsupport@medtronic.com)

## Interacting with the clinical application: Touchscreen monitor and mouse

The touchscreen monitor is the primary interface for the system; therefore, these instructions include monitor interactions such as tap, double-tap, and drag. Instructions to tap an object on the screen mean to touch the monitor with a finger, a gloved finger, or a capacitive stylus and then lift the finger or stylus quickly.

You can also use a mouse to interact with the clinical application. When using a mouse, instructions to “tap” an object on the screen are equivalent to click on the object with the left mouse button.

For a list of touchscreen interactions and their equivalent mouse actions, see “Touch and mouse interactions.”

## Document conventions used in this manual

This document employs the following conventions:

- Warnings are preceded by the word **Warning**. Failure to observe a warning may result in serious injury, death, or other serious adverse reactions to the patient or operator. Pay special attention to these items.
- Precautions are preceded by the word **Caution**. Failure to observe a precaution could result in injury, damaged equipment, forfeited time or effort, or the need to abort use of the system.
- References to buttons that appear on the touchscreen monitor are enclosed in square brackets. For example: Tap **[Edit...]**.
- References to buttons that are icons without text are capitalized. For example: Tap the Expand button...
- References to menu options that appear on the touchscreen monitor are printed in bold letters. For example: Select **Clear** from the list.
- References to titles of specific instructions for use documents are in italics.
- Electronic instructions for use can be accessed in two different ways: at [www.medtronic.com/manuals](http://www.medtronic.com/manuals) and within the clinical application. To view the instructions for use in the clinical application, tap **[Help]**, and tap **[Go to list of instructions for use]**.

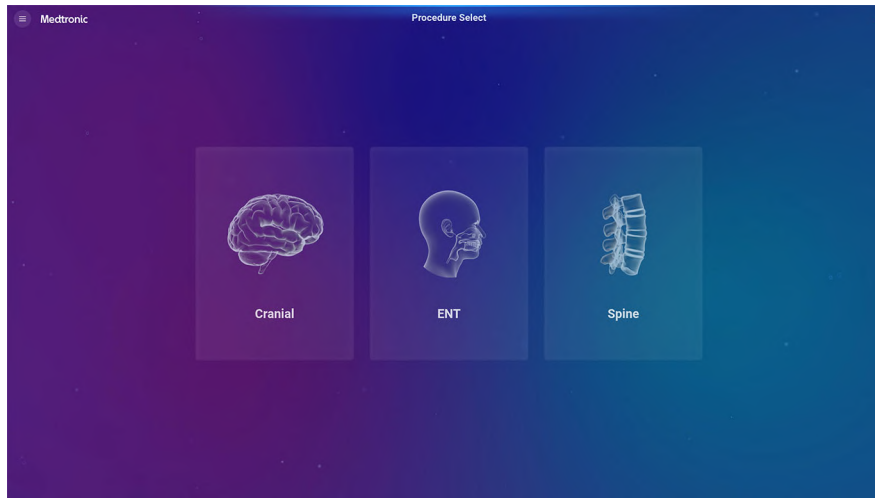
## Getting started with optical and EM based procedures

### Planning options

You can create and modify surgical plans in the **Planning** task by using the workstation cart, the Planning Station, or in a remote planning session. For more information about the Remote Planning feature, see “Remote planning.” For more information about the Planning Station, see the *Stealth AXIS™ Planning Station System Manual*.

### Logging in to the system from a cart

1. At the login screen, enter your username and password.
  - Tap **[Log In]**.
  - If you are unable to log in to the system, contact your hospital's Stealth Admin or IT department to confirm login information. Medtronic technical services can provide a temporary passcode for emergency system access. Tap **[EMERGENCY AND SERVICE LOGIN]** and call Medtronic technical services for assistance.
2. To launch the clinical application, tap the desired clinical application icon or tap **[View Recent Cases]**.



### Remote planning

The Remote Planning feature allows you to plan procedures both inside and outside the operating room, provided your computer has hospital network access and the facility has set up a Planning Station server. Using a supported web browser, you can view patient data and create surgical plans in the clinical application. No patient data is stored on your computer. Plans are saved to the Planning Station server and can later be imported to a Stealth AXiS™ workstation for use in the operating room.

While a remote planning session is active, the selected patient case is locked to prevent simultaneous edits by multiple users. If another user selects the same surgeon profile, the procedure is also locked. Users are encouraged to log in with their own credentials or create their own unique profile.

Browser sessions time out after a period of inactivity, with a warning message displayed beforehand. The timeout duration is set by hospital IT staff and can last between 10 and 60 minutes.

**Warning:** Review patient data created using remote planning before making surgical decisions and actions.

**Note:** Images may take a longer time to load when using the Remote Planning feature.

**Note:** In a remote planning session, you cannot use a CD/DVD or USB drive to import or export images or surgeon profiles.

### Minimum laptop or desktop requirements

To use remote planning, you only need to install the Google Chrome™ browser on your laptop or desktop—no additional client or software installation is required. Refer to a Windows or Apple user manual or technical support to confirm what your hardware and software specifications are.

**Note:** If you don't comply with the following recommendations, remote planning might not work on your laptop or desktop.

**Note:** If your network connection is poor, confirm that you have both a stable and fast internet connection. Consider moving your desktop or laptop to a new location.

Minimum computer screen and resolution recommendations:

- Minimum screen size: 13 inches
- The following screen resolutions have been tested:
  - 2560 x 1440
  - 1920 x 1080
  - 1440 x 810

Minimum Windows hardware and software recommendations:

- Windows 11 (64-bit) operating system
- Intel® Core™ i5\* 10th generation or later processor, or equivalent AMD processor
- Memory: 16 GB RAM
- Graphics:
  - Integrated Graphics Card: Intel® Arc™\* graphics card or equivalent AMD graphics card
  - Dedicated Graphics Card: NVIDIA RTX™ A1000\* graphics card or equivalent AMD graphics card

Minimum Apple hardware and software recommendations

- Operating System: macOS Sonoma 14, macOS Sequoia 15
- Processor: Apple M3 or Apple M1 Pro
- Memory: 8 GB RAM
- Graphics: Apple M3 or Apple M1 Pro

### Guidance for optimizing Windows laptops or desktops

The following guidelines are recommendations for Windows laptops or desktops running Windows 11 equipped with two GPUs. Visualization performance is optimized with remote planning by using the dedicated GPU instead of the integrated GPU.

1. Confirm that the Chrome™\* browser is installed on your laptop.
  - a. Press the Windows key on your keyboard, type **Graphics settings**, and then press “<Enter>.”
  - b. Click **[Add desktop app]**.
  - c. Click in the address bar, type **C:\Program Files\Google\Chrome\Application\** and press “<Enter>.”
  - d. Click **chrome.exe** and then **[Add]**. Depending on your system settings, you may not see the .exe suffix. It may only be labeled “chrome”.
2. Select **Google Chrome** and click **[Options]**.
3. Select the dedicated GPU that you want to utilize.
4. Close all Chrome™\* windows and restart the Chrome™\* browser for the changes to take effect.

## Selecting a surgeon and procedure

After selecting the desired clinical application icon, the **Procedure Select** screen displays.

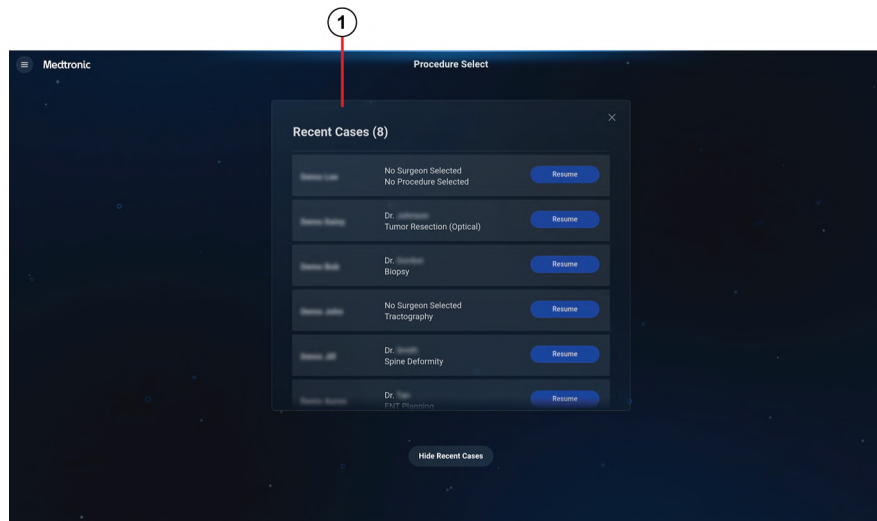
Follow these steps to select a surgeon and a procedure.

1. Tap **[Standard]** or select a surgeon name.  
 Alternatively, tap **[Add Profile]** to add a surgeon profile in the **Preferences** task. If a procedure is locked by another user, you can create a copy of the procedure.  
 Surgeon profiles allow you to save surgeon-specific preferences such as instruments, navigation options, and view settings. These preferences are organized by procedure. For each surgeon profile, you can also control the display of the on-screen keyboard and the colorblind aid.
2. After selecting a surgeon profile, select the surgical procedure to be performed.  
 Alternatively, tap **[Add Procedure]** to create or customize procedures in the **Preferences** task. See “Customizing procedures” for more details.

## Resuming recent cases

1. Tap **[View Recent Cases]** in the **Procedure Select** screen to view a list of patients imported or selected from the last 36 hours.
2. Tap **[Resume]** to select the desired patient case.

The **Images** task displays for the last surgeon and procedure used with this patient.



1 Recent Cases

## Images

### Importing images

You can import patient images from multiple sources (for example, CD/DVD or USB media, other Stealth AXiS™ systems, or a hospital PACS). Images must meet certain requirements. For information on scan requirements, see the *Stealth AXiS™ Imaging Protocols for Navigated Procedures*.

If you plan to create fiber tracts, both a diffusion scan and the Stealth AXiS™ Cranial Tractography license for the Stealth AXiS™ Cranial clinical application are required. Refer to the *Stealth AXiS™ Tractography Imaging Protocol* for guidance on developing imaging protocols for diffusion imaging and tractography processing in the Stealth AXiS™ clinical application. Also refer to the *Stealth Tractography User Manual* for information about importing, processing, and merging diffusion scans and creating fiber tracts.

You can import DICOM (Digital Imaging and Communications in Medicine) patient images in the **Images** task. Tap a tab at the top of the screen:

- Use the **Import** tab to import images from a PACS, another Stealth AXiS™ system, CD/DVD or USB media, or to import Demo Exams.
- Use the **Patients On This Stealth** tab to select images that are already on the system.

To import NIfTI (Neuroimaging Informatics Technology Initiative) images, see “Importing NIFTI images.”

**Note:** A Stealth AXiS™ archive or a StealthStation™ S8 application v2.1 or later archive can also be imported.

**Note:** To transfer an unregistered scan from the O-arm™ system to the Stealth AXiS™ system, refer to the user manual that accompanies the O-arm™ Imaging System.

### Importing images from a network source (DICOM)

**Note:** If an image has been manually pushed from a network source (DICOM) to the Stealth AXiS™ system, it will automatically appear under the **Patients On This Stealth** tab.

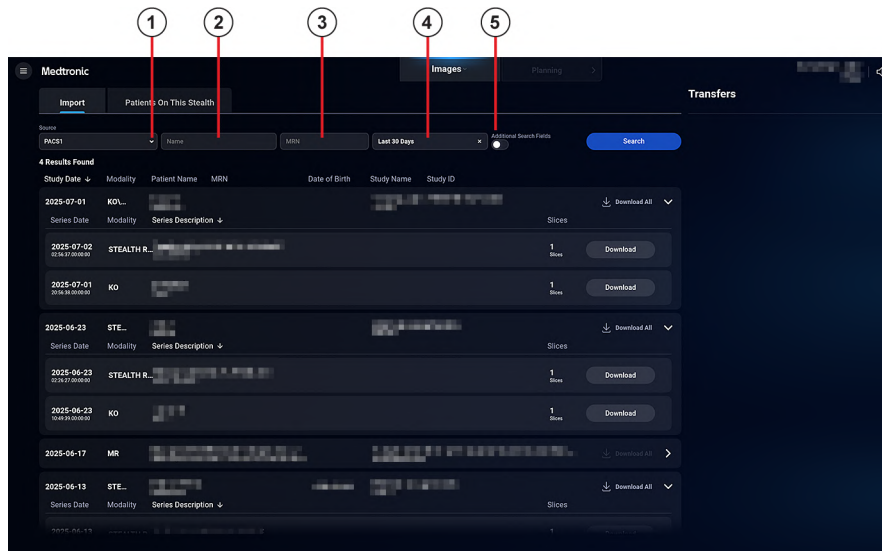
1. Tap the **Import** tab, if it is not already selected.
2. Select a network source from the **Source** drop-down menu.  
If you need to add a network source, see “DICOM Setup.”
3. Enter search criteria.
  - a. Enter information in any combination of the **Name** and **MRN** (Medical Record Number) fields. The patient name search is not case sensitive.
  - b. Tap the date filter field (**Last 30 Days** is the default filter) to display date search options (select **Last 24 Hours**, **Last 7 Days**, **Last 30 Days**, **All Time**, or a custom date range).
  - c. Turn on the **Additional Search Fields** toggle to show the **Referring Physician** field and the **All Modalities** toggle.

d. Turn off the **All Modalities** toggle to filter by MR, CT, or Stealth modalities.

**Note:** To see all data from the network source, leave the search fields blank.

**Note:** Refine search criteria for faster searches. Leaving all search fields blank may result in long search times.

4. Tap **[Search]**.



- 1 **Source** drop-down menu
- 2 **Name** filter field
- 3 **MRN** filter field
- 4 Date filter field (**Last 30 Days** is the default)
- 5 **Additional Search Fields** toggle (turns on and off additional search filters)

5. Import the images needed for the procedure. The first study in the list is expanded by default. Tap other studies in the list to display associated series.

- To import an individual series, tap **[Download]** beside the series.
- To import all series for the study, tap **[Download All]**.

A list of patients displays in the **Transfers** list after a download is complete. The **Active Transfers** provides a download status for each image series.

6. After downloading all necessary images, proceed to preview the images.

## Importing images from media

On a Stealth AXIS™ system, you can import patient images located on CD/DVD or USB media. This option is not available for remote planning.

1. In the **Images** task, tap the **Import** tab.
2. Insert the CD/DVD or USB media that contains DICOM images into the system.
3. If more than one media type is available, or media was already inserted before tapping the **Import** tab, select the desired media from the **Source** drop-down menu and tap **[Search]**.
4. In the **Select a directory to search** dialog, search the folders to find the desired images.

**Note:** The clinical application searches for DICOM images in subfolders, but manually selecting a lower level folder will result in faster search times.

5. Tap **[OK]**.

The clinical application searches for DICOM images in the selected folder.

6. When the search results appear, select a patient.

A list of the patient image series appears.

7. Import the images needed for the procedure.
  - To import an individual series, tap **[Download]** beside the series.
  - To import all series for the selected study, tap **[Download All]**.
8. After downloading all necessary images, proceed to preview the images.

### Importing NIfTI images

On a Stealth AXIS™ system, you can import NIfTI images located on CD/DVD or USB media. This feature is not available for remote planning.

1. In the **Images** task, tap the **Import** tab.
2. Insert the CD/DVD or USB media that contains NIfTI images into the system.
3. If more than one media type is available, or media was already inserted before tapping the **Import** tab, select the desired media from the NIfTI section of the **Source** drop-down menu and tap **[Search]**.
4. Browse the media file system and select a folder that contains NIfTI images.
5. Tap **[OK]**.

The clinical application searches for NIfTI files in the selected folder.
6. When the search results appear, select a file.

**Note:** You can tap **[Select All]** to select all the available files.
7. You can import NIfTI files as new patients or existing patients.
  - Tap **[Create New Patient]** and enter a **Name** and **MRN** to import the files to a new patient.
  - Tap **[Select Existing Patient]** and select a patient from the drop-down menu to import the file to a patient who is already in the system.
8. Tap **[Import]**.
9. After downloading all necessary images, proceed to preview the images.

### Viewing patient status and history

If your hospital facility is running a planning server, a lock icon next to a patient case indicates that another user has the patient case selected. If so, the system prevents you from selecting the same patient. From the **Patients On This Stealth** tab you can view the patient status on the patient card. The name of the user who has the patient currently selected displays and the patient is not selectable.

If a procedure is locked, a second user might be using the same surgeon profile to look at that procedure. For this reason, it is recommended that you create your own unique profile that is not used by others.

If you want to make a copy of a patient case while it is currently in use by someone else, you can clone the patient. See “Cloning patients” for more information.

Tap the History button to show or hide a list of entries for each previous user session. The history list includes username, time stamp, and certain actions such as plans created. This may be helpful to detect unexpected edits to the plan by someone else. The History button is disabled if the patient is currently selected by another user on the system.

### Previewing images

When your patient's name appears in the list of imported images:

1. Tap the patient's name. The clinical application displays all of the series associated with the patient in the **Patient Images** list.

If images are not under the same patient, see “Managing patients and images” for instructions on how to combine patients or move images.
2. Tap on the series that you plan to use for navigation. The clinical application displays preview images and image details.

Options to browse and view additional slices include:

  - Tap and drag the slice indicator on the vertical scroll bar or use the mouse wheel.
  - Tap above or below the slice indicator to move one slice in the indicated direction.
  - Tap the arrow at the top or bottom of the scroll bar to move one slice in the indicated direction.
3. In **Exam Details**, tap the Expand button to view more image details.
4. Preview the images:
  - a. Make sure that the anatomy of interest is captured in the images.

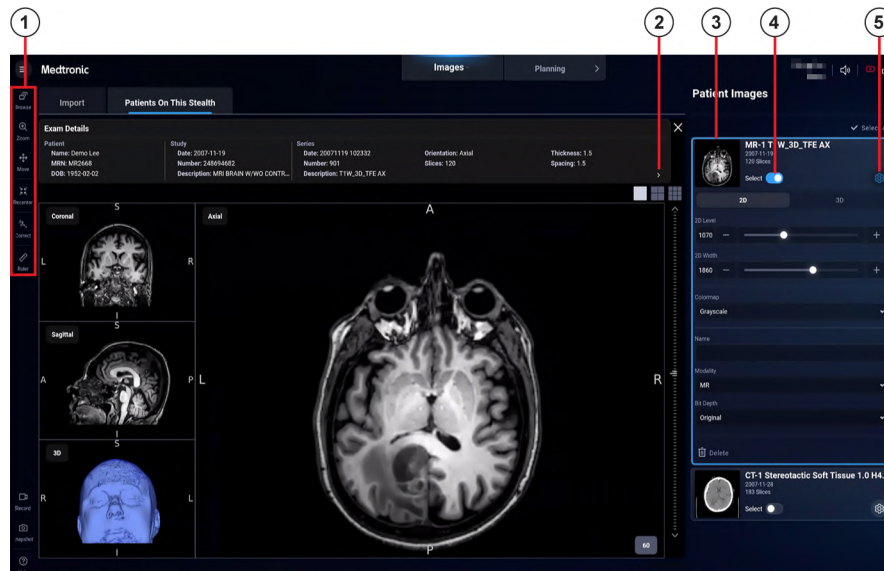
- b. Make sure that the images do not contain geometric distortion or intensity distortion (intense brightness in parts of the image compared to the rest of the image). If images contain distortion, contact radiology.
  - c. Check the slice spacing and slice thickness. Series with slice spacing equal to slice thickness are recommended.
  - d. To change the view orientation of a selected procedure, see “Customizing procedures.” If you have not selected a procedure, you can change the view and patient orientation while in this task.
5. If the image orientation is incorrect in the scan, you can correct it:
 

**Warning:** Only correct the image orientation if it is incorrect in the scan. Verify that all aspects of the patient images are correct, including patient data and image orientation. Make sure that the image orientation matches the patient's anatomy. Incorrect image orientation can result in inaccurate navigation and loss of data.

  - a. To rotate the image, tap **[Correct]** and tap the desired icon in the **Rotate** tab. Tap **[Confirm]**.
  - b. To reorient the image, tap **[Correct]** and tap the **Reorient** tab. Then tap the desired orientation button. Tap **[Confirm]**.
6. If you need to adjust the level or width, tap the Settings button on the series in the **Patient Images** list. Then tap the **2D** or **3D** tab and use the **Level** and **Width** sliders.
 

For additional information about the buttons on this screen, see “Buttons in Images, Planning, or Navigation.”

Merging multiple series from the same patient requires the Cranial StealthMerge™ license. See “Using image merge” for more details.
7. If you plan to merge multiple series from the same patient:
  - a. Tap each series in the **Patient Images** list to preview the images and check the image quality.
  - b. Turn on the **Select** toggle in an image series to add the images to the merge set.
8. After confirming that everything is correct in the patient's images, tap the next desired task at the top of the screen.



- 1 Tools for modifying images
- 2 The Expand button displays additional exam details
- 3 Patient images
- 4 The **Select** toggle is only available when the Cranial StealthMerge™ license is installed
- 5 The Settings button shows image display options such as level and width adjustments

## Equipment

### Setting up and managing equipment

The **Equipment** task displays system components and external devices used in the procedure. You can manage the equipment for the procedure and confirm that the equipment is communicating with the system.

To enter the **Equipment** task, tap the Menu button in the upper left corner of the screen and then tap **Equipment**.

The **Add Equipment** list contains the equipment available for the procedure.

- To add equipment to the procedure, tap the Add button.
- Some equipment requires additional information before you can add it to the procedure. Complete the necessary fields.

The equipment status bar and connection lines display communication status. Solid green lines show verified communication, and dashed orange lines show lack of communication.

- To remove equipment from the procedure, tap the Delete button.
- Tap the Settings button on the equipment card to:
  - display information for a part of the system that is not communicating.
  - display information about the equipment.
  - tap some equipment fields to edit them.

When all the necessary equipment is communicating, proceed to the next task in the clinical application.

## Instruments

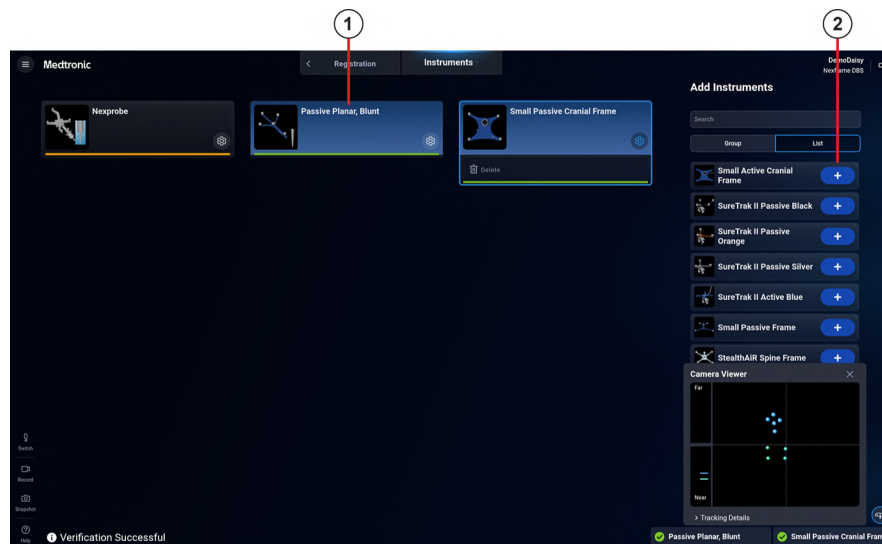
### Instrument management

In the **Instruments** task, you can choose the appropriate instruments for your procedure and verify instruments. All instruments used in a procedure must be verified prior to clinical use.

The **Instruments** task displays tool cards for the instruments added to the procedure.

The **Add Instruments** section contains a list of instruments that you can add to your procedure. You can drag with one finger up or down on the touch screen or scroll through the list using a mouse wheel.

**Note:** (EM only) Any instruments that are plugged into the EM instrument interface will appear automatically.



- 1 Instrument tool card
- 2 Add Instruments button

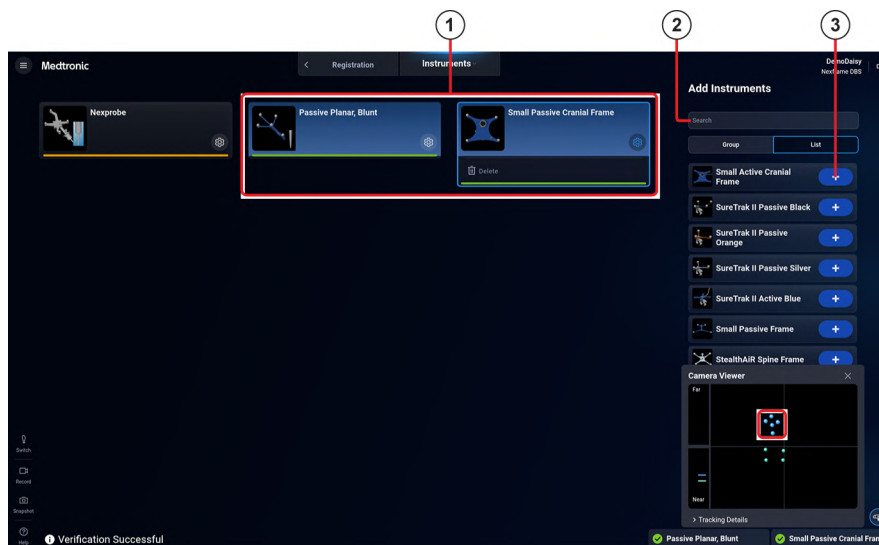
## Adding or removing instruments

You can add or remove instruments from your procedure in the **Instruments** task. To edit the instrument set for your procedure, follow these steps:

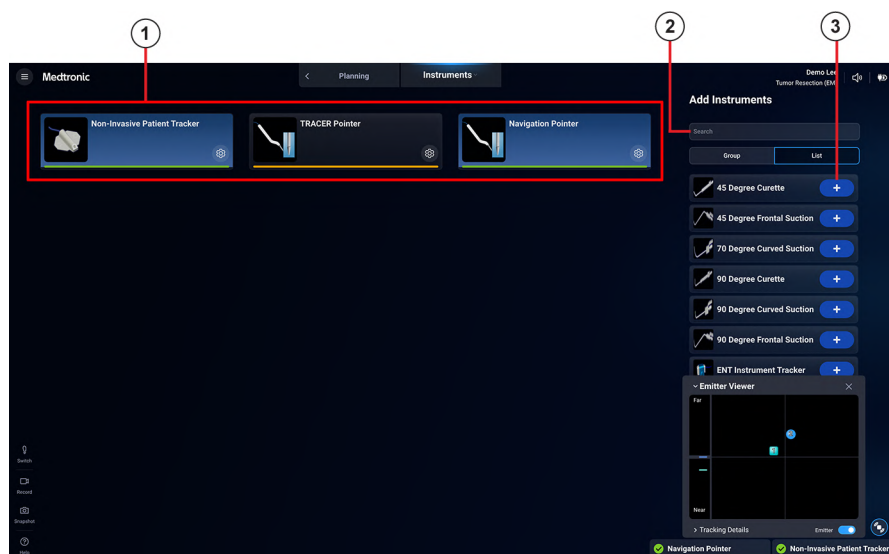
1. In the **Add Instruments** menu, scroll through the list of available instruments or enter the name of an instrument in the search bar.
2. Tap either **[Group]** or **[List]**.

**Note:** Results for searches entered in the search bar display as a list of instruments and are not grouped.

3. Tap the Add button to add the instrument to the procedure. An instrument card will display in the **Instruments** task.
4. To remove an instrument, swipe the associated tool card to the right and tap the Delete button. Alternatively, tap the tool card, tap the Settings button, and then tap the Delete button within the card.



- 1 Optical Instruments in Procedure
- 2 Search bar
- 3 Add button



- 1 EM Instruments in Procedure
- 2 Search bar
- 3 Add button

## Selecting instrument tips (optical only)

For Stealth-Midas™ MR8™ instruments, you can select tips (which are also called “tools”) in the **Instruments** task or in the **Navigation** task.

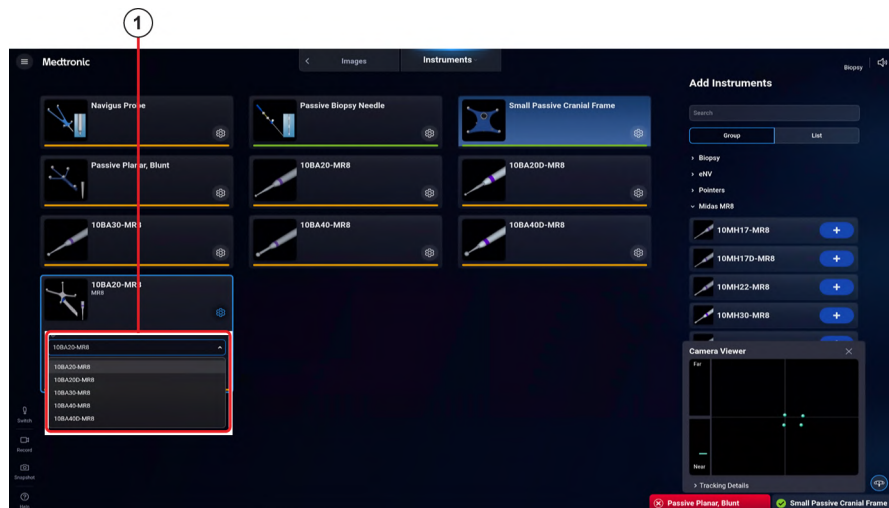
**Warning:** If a tool is changed during a procedure, you must also change the tool in the clinical application and reverify before navigating. Failure to update and reverify could result in inaccurate navigation.

To select a tip (or “tool”), follow these steps:

1. In the **Instruments** task, tap the Settings button on the instrument's tool card.
 

**Note:** You can change instrument tips in the **Navigation** task by opening the Instruments control panel and tapping **[TOOL]** on the instrument's tool card.
2. Select a tip from the tip list below the instrument.
 

**Note:** Size options are determined by the selected instrument and are not adjustable.



- 1 Tip list

3. Verify the tool.

## Aiming the camera (optical only)

Position the camera so that it is aimed at the instruments and trackers, ensuring it is at a distance that allows for the detection of each device's LEDs or reflective spheres. Optically navigated instruments and trackers must be in the camera's field of view with a clear line of sight for the system to track them.

1. Clear the line of sight between the camera lenses and the reflective spheres or LEDs on the patient reference frame.
2. Grasp the camera handle and position the camera approximately 1.0 m to 2.4 m (3.3 ft to 7.9 ft) from the patient reference frame. Aim the camera at the patient reference frame.
3. Press the laser trigger button on the camera handle and aim the camera head so that the laser beam passes through the center of the desired field of view.

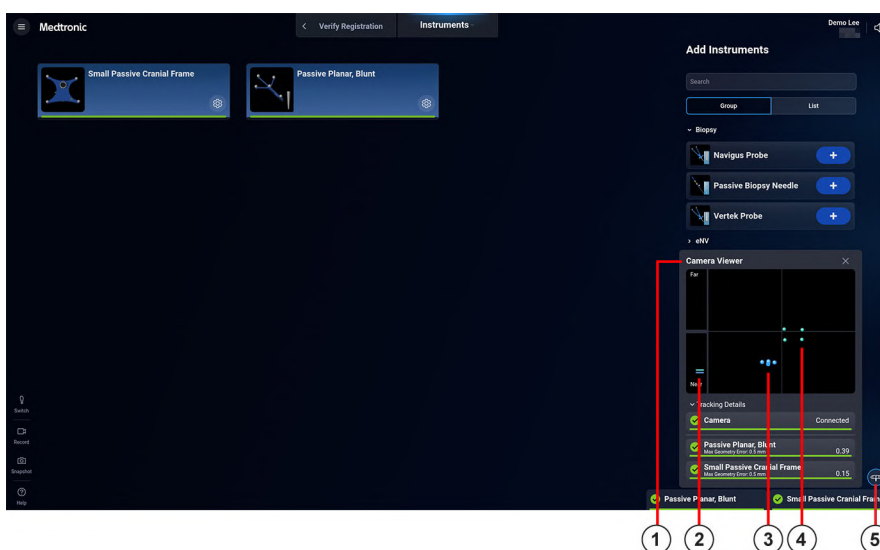
**Warning:** The laser aiming system is a Class 2 laser that transmits laser radiation. Use caution when operating the device, and never allow the laser beam to enter someone's eye. Laser radiation, even at low levels, can damage the eyes.

4. Tap the Tracking View button to open the tracking view if it is not already open.
5. Adjust the left/right and up/down aim of the camera head so that the reflective spheres or LEDs on the patient reference frame and navigated instruments will stay within the camera's view during navigation.
6. Move the camera closer to or further from the patient reference frame until the distance indicator is near the middle of the bar in the clinical application. Reposition the patient reference frame, if necessary, to position it in the camera's field of view.

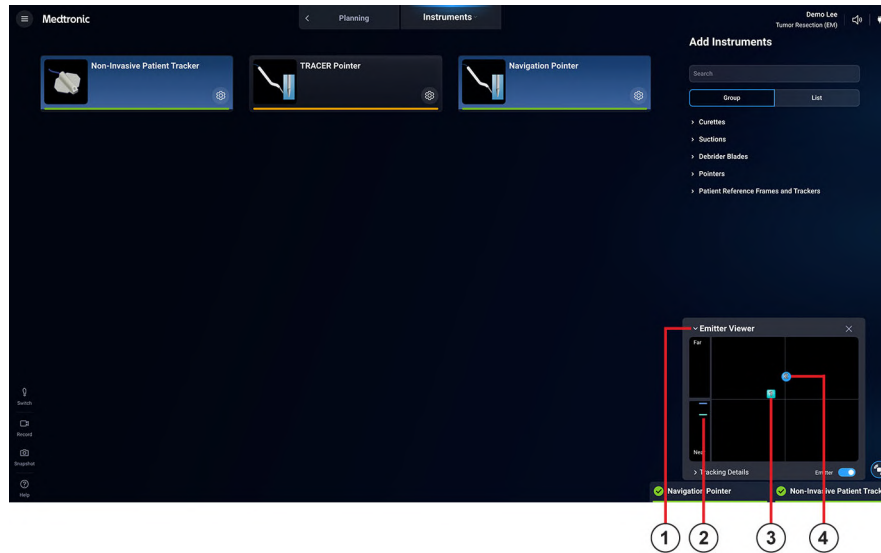
## Tracking view

The tracking view shows the location of tracked instruments and the patient reference frame in the navigation field. The distance indicator shows the instrument's and patient reference frame's distance from the localizer.

- To open the tracking view, tap the instrument status bar, the patient reference frame status bar, or the Tracking View button.
- For more detailed information about the navigated instrument and patient reference frame, tap **[Tracking Details]**.



- 1 Camera Viewer
- 2 Distance indicator
- 3 Instrument
- 4 Patient reference frame



- 1 Emitter Viewer
- 2 Distance indicator
- 3 Patient reference frame
- 4 Instrument

### Verifying instruments (optical only)

You can verify instruments in the **Instruments**, **Registration**, and **Navigation** tasks. There is a green bar beneath the instrument name in the tool card if it has been verified or does not require verification. Unverified instruments display an orange bar.

**Warning:** Physically inspect instruments for any defects. Never attempt to use a bent or damaged instrument. Use of a bent or damaged instrument can cause patient harm.

**Warning:** Always verify an instrument before use. Failure to verify instruments before use can result in patient harm.

**Warning:** Always reverify an instrument if you have changed tools or attachment orientation.

**Warning:** Do not apply force when verifying the Stealth-Midas™ MR8™ dissecting tool.

**Warning:** Only use Medtronic supported sterile spheres on patient trackers and navigated instruments to prevent inaccurate navigation information.

If you need to verify a SureTrak™ II instrument, see “Performing SureTrak™ II calibration (optical only).”

If you plan to use an ultrasound transducer, see “Using an ultrasound system (optical only).”

To verify instruments, follow these steps:

1. Snap sterile spheres onto the instrument and the patient reference frame. Push each sphere onto the stem until it “clicks” into place. Make sure the sphere is firmly seated on the stem.
2. Make sure that the screen shows the correct patient reference frame.
3. Insert the tip of the instrument into the bottom of the divot in the patient reference frame. Keep the instrument as perpendicular to the divot as possible.



**Warning:** If you are verifying a semi-rigid instrument (such as the PCI tip), make sure the tip does not bend during verification.

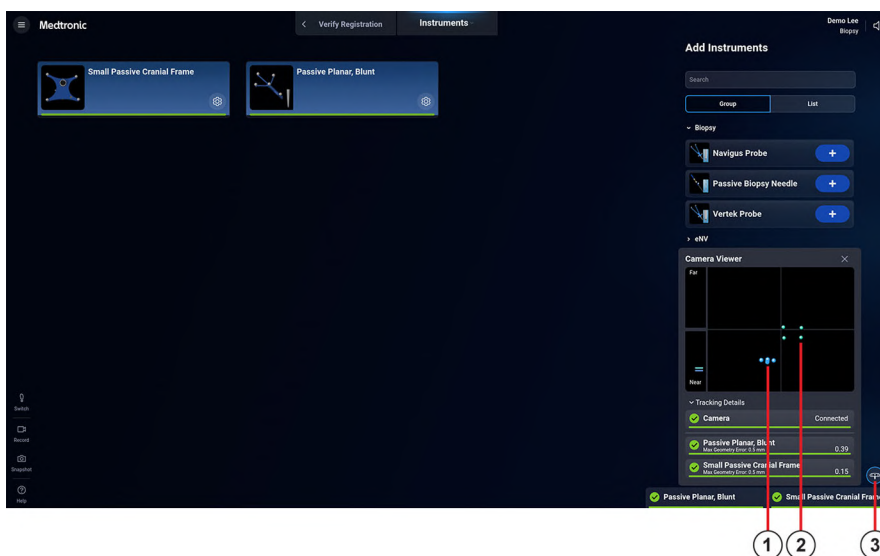
4. Tap the Tracking View button to open the tracking view and then confirm that both the instrument and the patient reference frame are in the camera's field of view.

5. Hold the instrument steady until you hear a chime, or press the footswitch.

When an instrument is verified, the system makes a positive sound (chime). When an instrument verification fails, the system makes a negative sound (bonk). See "Troubleshooting instruments (optical only)" for instrument verification troubleshooting tips.

6. Repeat this process for each instrument that you plan to use in the procedure.

To add or remove an instrument from the procedure, see "Adding or removing instruments."



- 1 Instrument
- 2 Patient reference frame
- 3 Tracking View button

## Using automatically verified electromagnetic (EM) instruments (EM only)

**Warning:** Verify that the navigation system displays the name of the tracked tool and patient tracker.

**Warning:** Physically inspect instruments for any defects. Never attempt to use a bent or damaged instrument.

Medtronic EM instruments with integrated sensors are automatically verified. When you want to use an automatically verified instrument, follow these steps:

1. Plug the instrument and patient reference frame into open ports on the EM instrument interface.
2. Confirm that the clinical application displays the name of the tracked tool and patient reference frame.
3. Place the instrument and patient reference frame in the navigation field.
4. Confirm that both the instrument and the patient reference frame are visible in the tracking view.
5. Repeat this process for each instrument that you plan to use in the procedure.

**Note:** It is unnecessary to put the tip of automatically verified EM instruments into the patient reference frame divot.

**Note:** Plugging automatically verified EM instruments into the EM interface adds them to the active procedure.

## Planning

In the **Planning** task and in the **Navigation** task, you can use a variety of tools to define surgical plans to target structures appropriate for your procedure.

After you define a surgical plan, you can change the plan in the following ways:

- Edit the target point
- Edit the entry point
- Edit the plan trajectory
- Change the plan name
- Change the plan color
- Lock the plan to prevent inadvertent changes
- Show or hide the plan
- Display a cylindrical margin around the plan

Refer to “Editing a surgical plan” for instructions on editing plans.

**Warning:** Verify that a plan's target point, entry point, and trajectory are clinically valid before using it for a procedure. Traverse the surgical plan to check the entry, target, and trajectory after any edit. If you change the merge reference image after the plan is created, review your surgical plan to avoid inaccurate navigation. Some existing plan data is remerged to the new reference image and may have shifted.

In the **Planning** and **Navigation** tasks, the 3D model automatically rotates to look down the selected plan. If you prefer to set the 3D model to a fixed orientation, change the **Auto-rotate 3D View to Plan** setting to OFF on the **Navigation** tab in your surgeon profile. See “Editing a procedure.”

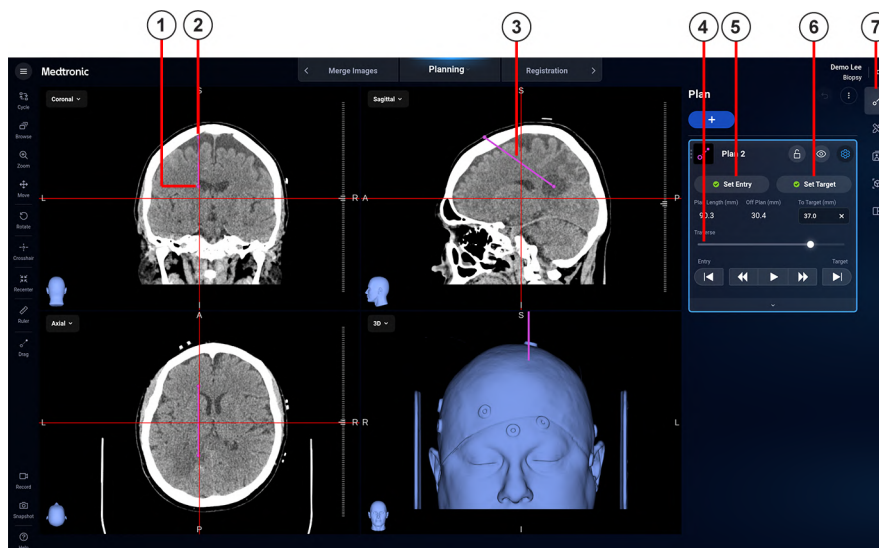
## Defining a surgical plan

To define a surgical plan:

1. Tap the Plans button to open the **Plan** drawer.
2. Scroll through the 2D image slices and find the slice that contains the surgical target.
3. Tap a target point in one of the 2D views. Refine its location by tapping in any of the other views.
4. Tap **[Set Target]**.
5. Tap an entry point on one of the 2D views or on the surface of the 3D model.
6. Tap **[Set Entry]**.
7. Simulate movement along the plan by doing one of the following:
  - Drag the **Traverse** slider.
  - Tap the Backward and Forward buttons to move along the plan by -1.0 mm or +1.0 mm, or tap the Skip Backward or Skip Forward buttons to move the plan to the entry or target location.
  - Use the left, right, up, or down arrow keys on the keyboard.
  - Tap the Play button to automatically traverse the plan.

If you move a point by mistake, tap the Undo button to undo one change.

8. If you want to create another plan, tap **[+]** in the **Plan** drawer and repeat the steps above.



- 1 Plan target
- 2 Plan entry
- 3 Plan trajectory
- 4 Traverse slider
- 5 **[Set Entry]**
- 6 **[Set Target]**
- 7 Plans button

## Cone of reachability (optical only)

(Biopsy procedures only)

The cone of reachability displays a graphical overlay of the area within reach from the top of the burr hole along a surgical plan. In a biopsy procedure, the cone of reachability is an optional feature that helps determine whether a surgical plan is mechanically feasible with a Navigus™ trajectory guide straight base or an angled base. You can display and adjust the cone of reachability in the 2D and 3D image views from the **Planning** and **Navigation** tasks.

### Displaying the cone of reachability (optical only)

1. In the **Planning** task, the **Navigation** task, or the **Tractography** task, tap the Plans button.
2. Select a plan from the **Plan** drawer.
3. Tap the Expand button to display advanced options for the plan.
4. To enable the Cone of Reachability toggle, set the plan entry and target.
5. Turn on the **Cone of Reachability**.

The clinical application displays the cone of reachability and angles in the Trajectory 1, Trajectory 2, Guidance, Probes Eye, and 3D views.

6. To adjust the cone of reachability, proceed to aligning the cone of reachability.

## Aligning the cone of reachability (optical only)

(Biopsy procedures only)

You can adjust the cone of reachability overlay to align with the patient anatomy and the surgical plan.

1. In each trajectory view (**Trajectory 1** and **Trajectory 2**), drag the point of the cone of reachability to the location where the plan intersects the patient's skull.
2. In each trajectory view, drag the base of the cone of reachability overlay to rotate the overlay and align the burr hole line with the outer surface of the patient's skull.

The burr hole lines represent a 14 mm burr hole.

The two trajectory views are at 90° from one another. As you rotate the cone overlay in one trajectory view, the width of the overlay in the other view automatically adjusts to represent the shape of the cone in three dimensions.

3. Refine the position of the cone of reachability so that it is aligned to the patient's skull in both views.

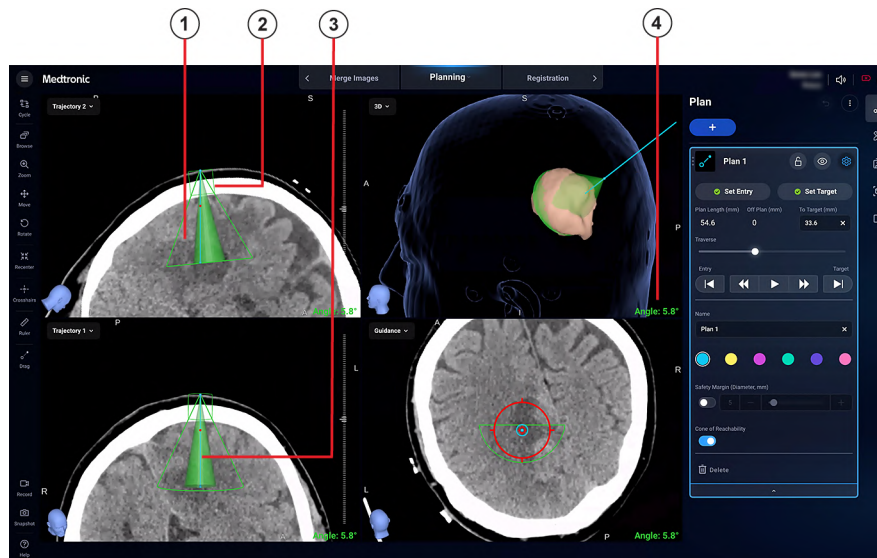
The image views display the angle between the surgical plan and the center of the cone of reachability.

- Feasible plans with an angle less than 25° display the cone of reachability in green.

**Note:** You can determine the appropriate Navigus™ base from the angle of the plan in the cone of reachability. Straight Navigus™ base configurations require an angle less than 10°. Angled Navigus™ base configurations require an angle between 5° and 25°.

- Infeasible plans with an angle greater than 25° display the cone of reachability in red. To adjust the entry for the plan, see "Editing a surgical plan".

**Note:** The accuracy of the angle measurement depends on how accurately you align the cone of reachability overlays to the patient anatomy and how accurately you attach the Navigus™ base to the location represented in the surgical plan.



- 1 Cone of reachability overlay
- 2 Burr hole line
- 3 Center of the cone of reachability
- 4 Angle between the surgical plan and the center of the cone of reachability

## Editing a surgical plan

- Use the touchscreen to edit a target or entry by moving the crosshairs and tapping either **[Set Entry]** or **[Set Target]**.
- Tap the Drag button to drag the plan entry or trajectory and rotate the plan around the target in any 2D view.
- Tap the Drag button to drag the plan target and rotate the plan around the entry in any 2D view.
- While holding the <SHIFT> key, you can use the mouse to drag the plan entry or trajectory and rotate the plan around the target in any 2D view.
- While holding the <SHIFT> key, you can use the mouse to drag the plan target and rotate the plan around the entry in any 2D view.

## Editing the target

1. Select the plan you want to edit from the **Plan** drawer.
2. Tap the Skip Forward button in the drawer to go to the target and view the current position.
3. In any view, tap the location of the desired target.  
To magnify the entire image, tap **[Zoom]**, and then drag with one finger up to zoom in or down to zoom out.
4. Tap **[Set Target]**.  
The clinical application sets the target to the point indicated by the crosshairs.

## Editing the entry

1. Select the plan you want to edit from the **Plan** drawer.
2. Tap the Skip Backward button in the drawer to go to the entry and view the current position.
3. In any view, tap the location of the desired entry.  
To magnify the entire image, tap **[Zoom]**, and then drag with one finger up to zoom in or down to zoom out.
4. Tap **[Set Entry]**.  
The clinical application sets the entry to the point indicated by the crosshairs.

## Editing the plan trajectory

Follow these steps to edit a plan's trajectory to pass through or avoid structures.

1. Select the plan you want to edit from the **Plan** drawer.
2. To review your plan trajectory, traverse the plan using one of the controls in the drawer:
  - Tap the Play button to have the clinical application move along your plan.
  - Use the **Traverse** slider to manually move along your plan.
3. While holding the <Shift> key and using the mouse, or tapping the Drag button and using your finger, you can drag the plan trajectory in any view. Select a view that allows you to visualize the relevant structures. You can only drag a plan in 2D Views, not in 3D Views.

**Note:** Editing a plan's trajectory will change either the planned entry or target.

## Locking plans

Follow these steps to lock a plan to prevent changes.

1. In the **Plan** drawer, tap the plan you want to lock.
2. Tap the Lock button to lock the selected plan.
3. To unlock a plan, tap the Lock button again.

## Changing plan names and colors

1. Select the plan you want to edit from the **Plan** drawer.
2. To change the plan name, tap the Settings button, and then tap the Expand button to enter text in the **Name** field.
3. To change the plan color, tap the Expand button, and then select one of the predefined colors, or double-tap a predefined color and select a custom color.

### Displaying a margin around the plan

1. Select the plan you want to edit from the **Plan** drawer, and then tap the Settings button.
2. Tap the Expand button.
3. Turn on the **Safety Margin**.

The default diameter for the safety margin is 5 mm.

To change the diameter, tap the **[+]** or **[-]** on the diameter slider to edit the value. You can also enter a value in the diameter text field or use the slider bar.

### Deleting a surgical plan

To delete a plan do one of the following:

- Tap the Settings button, tap the Expand button, and then tap **[Delete]**.
- Swipe right and then tap the Delete button.
- Tap the More button and then tap **[Delete Multiple]** or **[Delete All]**.

## Patient registration

### Setting up the patient reference frame (optical only)

1. Secure the patient in the skull clamp.
2. Attach the Vertek™ Articulating Arm to the skull clamp.

**Note:** If you plan to use the Vertek™ Passive Biopsy Kit, connect the dual starburst to the skull clamp before attaching the articulating arm.

3. Attach the patient reference frame to the articulating arm.

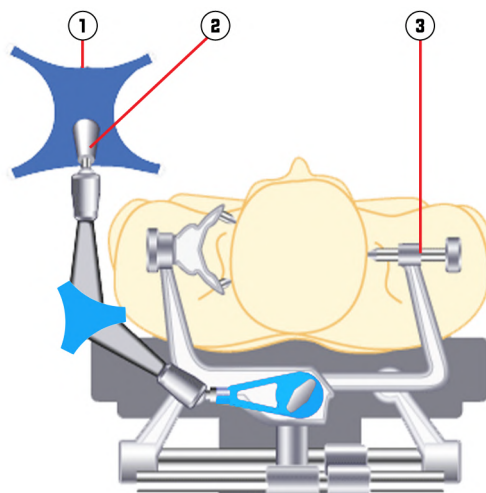
Match the mounting screw and the alignment post (on the underside of the patient reference frame) with the corresponding screw hole and alignment notch (on the swivel mount of the articulating arm). Secure the patient reference frame to the swivel mount by tightening the mounting screw.



- 1 Alignment post
- 2 Alignment notch

4. Move the patient reference frame as close as possible to the surgical field such that its location will not impede navigation.
5. Lock the patient reference frame in place.

**Warning:** Do not use the patient reference frame as a landmark for confirming navigational accuracy. The position of the anatomy is defined by the position of the patient reference frame. If the patient reference frame moves with respect to the anatomy after you register the patient, navigation becomes inaccurate. Make sure that the Vertek™ Articulating Arm is locked securely in position. If the arm slips after registration, re-secure it and register the patient again before navigating.



- 1 Patient reference frame
- 2 Vertek™ Articulating Arm
- 3 Skull clamp

## Setting up the patient reference frame (EM only)

Select a set up method, depending on your procedure and patient tracker:

|                                             |                                                                                   |
|---------------------------------------------|-----------------------------------------------------------------------------------|
| Minimally invasive procedures only          | "Option 1: Non-Invasive Patient Tracker (NIPT) (EM only)", unpinned configuration |
| Craniotomy/craniectomy or a large skin flap | "Option 2: Skull-Mount Patient Tracker (SMPT) (EM only)", unpinned configuration  |
|                                             | "Option 3: Non-Invasive Patient Tracker (NIPT), pinned configuration (EM only)"   |

### Option 1: Non-Invasive Patient Tracker (NIPT) (EM only)

(Minimally-invasive procedures only)

**Note:** The Non-Invasive Patient Tracker has a divot and arrows for use in the ENT clinical application. These features are disabled in the Cranial clinical application.

- Position the patient on the surgical table and rest the patient's head on a gel headrest or similar support.
- Review the warnings and other instructions given in the package insert that accompanies the Non-Invasive Patient Tracker (9734887).
- Select a site on the patient's skin to affix the tracker.

**Warning:** Carefully consider the location where you will place the Non-Invasive Patient Tracker. Choose a skin site that fulfills these criteria:

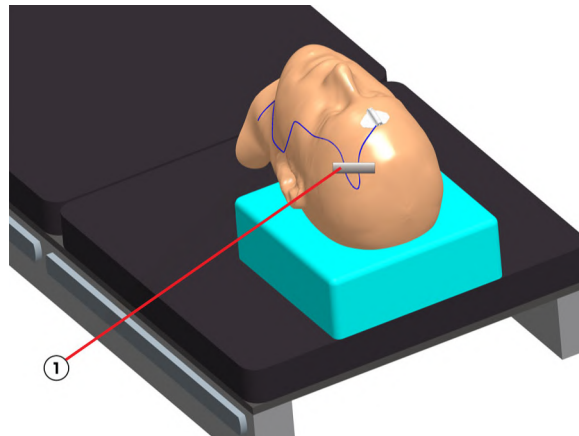
- Will lie safely within the localizer detection field during navigation.
- Will not result in stress on the tracker as a result of patient positioning.
- Will not hinder access to the surgical site.
- Will not be in the immediate vicinity of any large metal objects.
- The skin is taut and smooth, not loose, baggy, or uneven. The tracker's position relative to the operative anatomy may not be allowed to change during the procedure. The patient's forehead is the recommended application site for the Non-Invasive Patient Tracker.

**Warning:** The Non-Invasive Patient Tracker may not exhibit the same accuracy as reference frames using rigid fixation.

- If necessary, shave the site to provide maximum contact with the patient's skin.

5. Clean the site on the patient's skin with alcohol to strengthen adhesion and allow the site to dry thoroughly before applying the tracker.
- Note:** If the patient is perspiring, you may want to apply benzoin or a similar product to improve adhesion.
6. Peel away the release liner from the tracker and adhere the tracker to the patient's skin.
7. Create a U-shaped bend in the tracker's cable and tape this bend to the patient approximately 5 cm (2 in) below the tracker. This creates a strain relief mechanism that helps prevent the tracker from being dislodged accidentally.
8. Connect the patient tracker communication cable to any open port on the EM instrument interface.
9. Verify that the tracker is secure and cannot be accidentally dislodged during the procedure.

**Warning:** Do not bump or reposition the patient reference frame after registration. Movement of the patient reference frame relative to the patient anatomy will result in inaccurate navigation. If the patient reference frame moves in relation to the patient anatomy at any time after registration, you must re-register.



1. Tape the cable to the patient to create a strain relief.

### Option 2: Skull-Mount Patient Tracker (SMPT) (EM only)

This option is recommended for procedures requiring a large skin flap, such as tumor resections.

**Warning:** The safe use of the SMPT has not been demonstrated with pediatric patients.

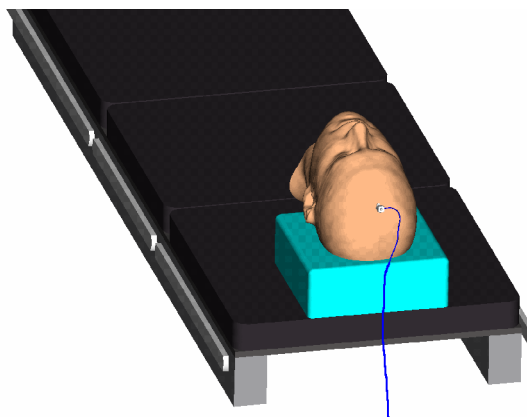
**Warning:** Before attaching the SMPT carefully examine the patient image data to verify that the skull thickness at the intended attachment site is adequate for securing the SMPT and screws, and that no arterial or venous structures lie beneath the site. When fully tightened, the SMPT screws penetrate 2.5–3.0 mm below its lower surface. Never place the SMPT at a site where the skull bone is less than 4.0 mm thick.

1. Position the patient on the surgical table and rest the patient's head on a gel headrest or similar support.
2. Review the other warnings and instructions given in the package insert that accompanies the sterile, disposable SMPT (9735300, 9731132).
3. Select and prepare the attachment site.
4. Make a small scalp incision (14–16 mm) down to the bone at the attachment site.
5. Set the sterile SMPT at the site and tighten the screws using the sterile screwdriver (reusable or disposable). Because the screws are self-tapping and self-drilling, no pilot holes are necessary.

**Caution:** Make sure that the screws are secure so the patient tracker cannot slip or shift at any time during the procedure. However, be careful not to over-tighten the screws.

6. Connect the patient tracker communication cable to any open port on the EM instrument interface.

**Warning:** Do not bump or reposition the patient reference frame after registration. Movement of the patient reference frame relative to the patient anatomy will result in inaccurate navigation. If the patient reference frame moves in relation to the patient anatomy at any time after registration, you must re-register.

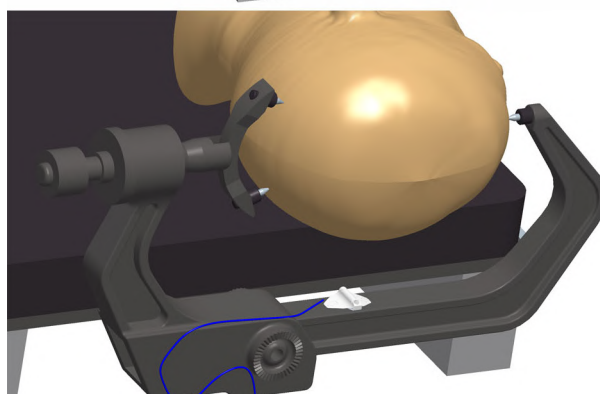
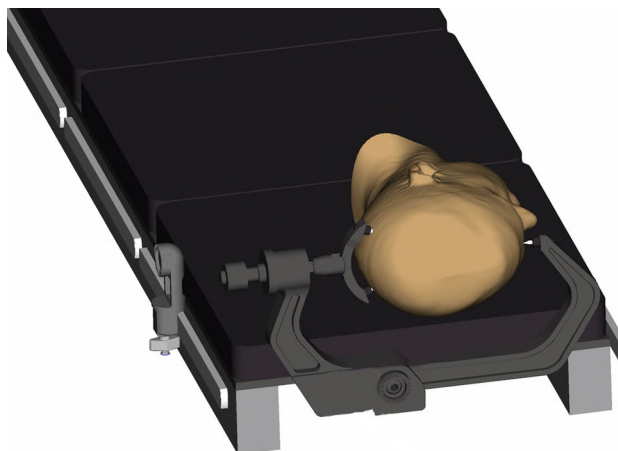


### Option 3: Non-Invasive Patient Tracker (NIPT), pinned configuration (EM only)

This option is recommended for procedures requiring a large skin flap, such as tumor resections.

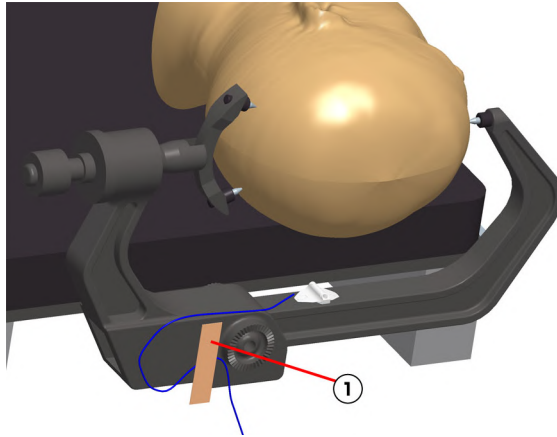
**Note:** The Non-Invasive Patient Tracker has a divot and arrows for use in the ENT clinical application. These features are disabled in the Cranial clinical application.

1. Position the patient on the surgical table and secure the patient's head in the radiolucent skull clamp.
2. Review the warnings and other instructions given in the package insert that accompanies the patient tracker (9734887).
3. With an alcohol swab, clean a space on the horizontal surface of the adjustable arm on the single-pin radiolucent skull clamp. Choose a space that is free from stickers, labels, and other obstructions.
4. Remove the backing from the adhesive pad on the patient tracker.
5. Affix the patient tracker to the space you cleaned on the skull clamp.



6. Connect the tracker to any port on the EM instrument interface.

7. Create a bend in the tracker cable and tape the cable to the skull clamp. This creates a strain relief mechanism that helps prevent the tracker from being dislodged accidentally.



1. Tape the cable to the skull clamp to create a strain relief.

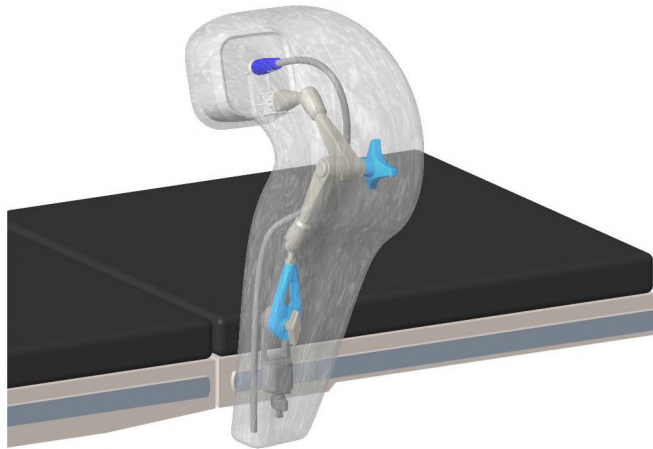
8. Take other necessary precautions to ensure the tracker will not be dislodged accidentally after registration.

**Warning:** Do not bump or reposition the patient reference frame after registration. Movement of the patient reference frame relative to the patient anatomy will result in inaccurate navigation. If the patient reference frame moves in relation to the patient anatomy at any time after registration, you must re-register.

### Draping the side emitter (EM only)

**Warning:** The side emitter and the arm assembly that supports the side emitter are not intended for sterile contact or patient interaction. If you place these items in or near the sterile field, bag or drape them. Failure to drape these accessories can compromise the sterile field.

1. Drape the side emitter, arm assembly, and bedrail adapter.
2. If you are using the Medtronic drape (9736701), cuff the drape using the tabs.



3. Leave some space in the drape around the locking knob to allow for repositioning.

**Warning:** Use care to avoid damaging the drape when re-adjusting the Vertek™ articulating arm.

## Setting up and positioning the side emitter (EM only)

Connect the side emitter cable to the emitter port on the workstation cart localization I/O panel. Set up the side emitter where it will not interfere with the surgeon's access to the patient during surgery.

**Warning:** Visually inspect all accessory devices before use. Do not use accessory devices if they are visibly damaged.

**Warning:** Do not place the side emitter in contact with the patient because the emitter temperature may reach 48°C (118°F).

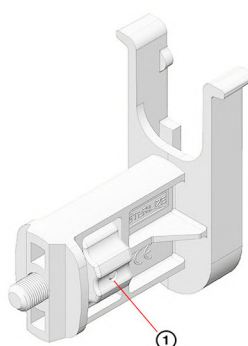
**Caution:** Procedures performed with the side emitter require a surgical table featuring side rails because the arm assembly that supports the emitter connects to the side rails.

**Note:** To reduce electromagnetic tracking distortions, use titanium or plastic retractors (not stainless steel).

1. Secure the Short Bedrail Adapter (9732566) to the rail of the surgical table.

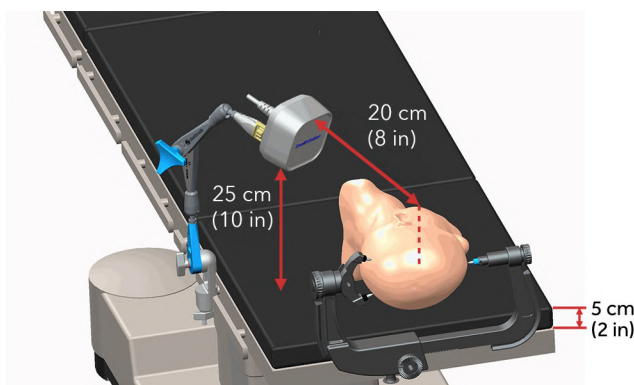
**Note:** Typically, if the surgeon is right-handed, set up the emitter on the left-hand side. For a left-handed surgeon, set it up on the right-hand side. Exact placement of the emitter depends on many factors including the type surgical table, procedure, patient location, and patient size. Use a configuration that suits your situation.

2. Connect the Emitter Support Bracket (9735547) to the Vertek™ Articulating Arm (9734252) using the thumb screw.



1 Thumb screw

3. Lock the articulating arm knobs and attach it to the bedrail adapter. Make sure that the starburst teeth mesh together.
4. Place the back side of the side emitter flush against the emitter support bracket with the logo oriented up and slide the emitter into the bracket until the emitter snaps into place.
5. While supporting the emitter, release the articulating arm locking knob and adjust the position of the emitter.
  - a. Position the emitter about 25 cm (10 in) above the metal on the table and about 20 cm (8 in) from the center of the patient's head. As a general rule, the space between the patient's head and the emitter face should be approximately equal to the width of your fist.



- b. Aim the emitter at the center of the patient's head as if aiming a flashlight.

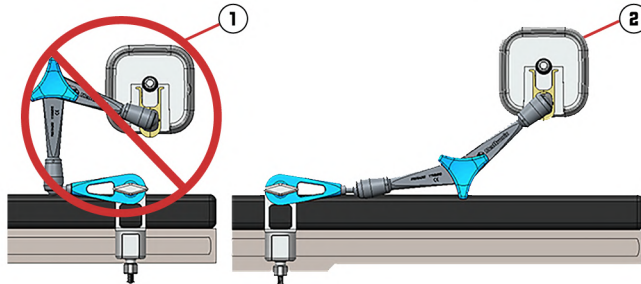
**Caution:** When adjusting the Vertek™ Articulating Arm, remove or hold on to the side emitter before loosening the adjustment knobs. Dropping the side emitter could void its calibration.

**Caution:** Always loosen the knobs before adjusting the articulating arm.

**Caution:** Moving parts may create pinch points. Use care to maintain control of the object you are moving and be aware of potential pinch points when moving articulating arm joints.

6. Tighten the locking knob of the articulating arm.

**Caution:** If the surgical table to be used with the side emitter has a metal frame or metal side rails, position the side emitter at least 25 cm (10 in) away from the metal. Positioning the emitter closer may result in reduced navigational accuracy and volume.



- 1 Example of incorrect setup (too much metal near the navigation field)
- 2 Example of correct setup

### Draping the flat emitter (EM only)

**Warning:** The flat emitter is not intended for sterile contact or interaction. If you place the flat emitter in the sterile field, bag or drape it.

Drape the flat emitter before placing the patient on the emitter.

1. Place a drape over the flat emitter.
2. Make sure that the drape completely covers the flat emitter.

**Note:** For patient comfort, you can place a sterile head cushion on the center of the flat emitter. Make sure that the cushion is centered on the etched crosshairs on the flat emitter. A sterile gel pad can also be placed below the flat emitter for shoulder positioning.

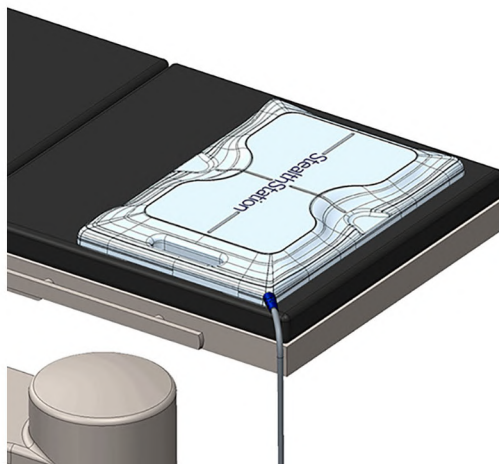
### Setting up the flat emitter (EM only)

**Caution:** Do not exceed the maximum weight load of 6.5 kg (14 lbs) on the flat emitter. Exceeding the maximum weight load may cause damage to the emitter.

Connect the flat emitter cable to the emitter port on the workstation cart localization I/O panel.

1. Place the flat emitter at the head of the surgical table and position it at least 5 cm (2 in) above the metal of the surgical table with the logo oriented up.

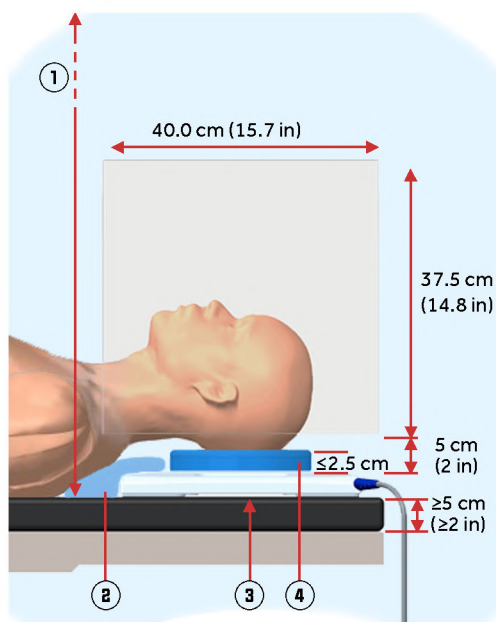
**Note:** Do not place the flat emitter directly on the metal of the surgical table, and do not place the flat emitter over the pedestal of the surgical table.



2. Make sure that the flat emitter cable rests off to the side of the surgical table and does not interfere with the surgeon's access to the patient during surgery.

**Note:** Make sure that there is no tension on the flat emitter cable.

3. For best results, place the surgical area of interest near the center of the navigation field.



- 1 Low-energy magnetic field: Metal objects, such as surgical lights, in this area will reduce the size of the navigation field and impact navigational accuracy.
- 2 Towel for patient comfort
- 3 Location to place additional padding, if necessary, to increase the separation distance between metal rails on the table and the flat emitter
- 4 Thin pad: 2.5 cm (1 in) or less

### Refining emitter position (EM only)

Before setting up for sterile navigation, check the geometry error metric in the clinical application. For best results, minimize the geometry error metric by adjusting the emitter's position or by positioning the patient closer to the emitter.

1. Connect an instrument to any port on the EM instrument interface.
2. In the **Instruments**, **Registration**, or **Navigation** task, tap the Emitter button to open the tracking view, if it is not already open.
3. Tap **[Open Tracking Details]** to display the tracking details.
4. Confirm that the emitter and EM instrument interface are connected to the system.
5. Make sure that the clinical application can track the instrument and the patient reference frame throughout the range of locations required for registration and navigation.
6. Adjust the position of the emitter as necessary until the instrument and patient reference frame display in the tracking view.  
For the flat emitter, if your volume is smaller than expected, decrease the padding thickness between the patient and the emitter, or add padding between the emitter and the metal on the surgical table. Gradually increase the separation distance between the emitter and the metal by a centimeter or more while maintaining proper patient positioning, and re-evaluate the volume.

## Registration

### Registration overview

To navigate, you must match the patient's exam to the patient's anatomy using a process called registration. You can use an automatic registration method (O-arm™ registration), or you can use manual registration methods (trace and touch registration). Automatic registration is available with optical localization only. If you plan to use an automatic registration method, you do not need to use a manual registration method.

For O-arm™ registration, see “Performing an O-arm™ Imaging System registration (optical only).”

For manual registration, you may use touch registration, trace registration, or a combination of both. Both manual registration methods require a registration instrument, see optical and EM instrument lists below.

Optical only:

- Passive Planar Blunt Probes (960-556 and 9734881\*)
- Navigus™ Probe (9733157)
- Vertek™ Probe (9733158)
- Touch-n-Go Probe (9732176)

EM only:

- EM Navigation pointer (9735519)
- EM Touch-N-Go™ Probe (9735318)
- EM Tracer Pointer (9735317)

**\*Note:** The Passive Planar Blunt Probe 9734881 and the Microscope Probe 9734889 are for sale in the European Union only. The Passive Planar Blunt Probe 960-556 and the Microscope Probe 9730269 are for sale outside of the European Union.

For the trace and touch registration methods, only use image modalities containing anatomic information. The clinical application automatically creates a three-dimensional model using the patient's images. Before you begin a manual registration, verify that the 3D skin model accurately represents the patient and that all skin fiducial markers are clearly visible if using touch registration. If the 3D model contains artifacts or scatter, edit the model to make it as accurate as possible. See “Editing a 3D registration model.”

**Note:** If you edit the 3D model after you begin a registration, you will need to restart the registration process.

- In trace registration, you move the tip of the probe over regions of the skin that can be seen within the 3D skin model. For trace registration instructions, see “Performing trace registration.”
- In touch registration, you place the tip of the probe on specifically defined landmarks on the patient's skin. For touch registration instructions, see “Performing touch registration.”

**Warning:** Do not bump or reposition the patient reference frame after registration. Movement of the patient reference frame relative to the patient anatomy will result in inaccurate navigation. If the patient reference frame moves in relation to the patient anatomy at any time after registration, you must re-register.

If you merged images and want to change the image used for registration, you can change the image before starting manual registration. If you change images after registration, you will lose your current registration and must restart the registration. To change the image, tap the Settings button, and then tap **[Change Image]**.

## Touch and Trace registration methods

The system uses all of the information that you enter on the **Touch** and **Trace** tabs to compute a single registration. You can use any combination of the touch and trace registration methods that fit your preferred work flow.

Follow these best practices when registering:

- Provide the system with as many registration points as you can reasonably include, but avoid tracing the same area repeatedly. Include points over a wide area that surround the surgical region of interest.
- Use a combination of touch and trace points if you are unable to surround the surgical region of interest using a single registration method.
- Ensure that the registration probe and patient reference frame are visible in the tracking view.
- Ensure that all the points you store on both the **Touch** and **Trace** tabs are accurate.
  - Check for inaccurate touch points and tracing.
  - Delete or undo any touch points or traces that you know are inaccurate. For traces, you can undo only the portion that you traced after the last batch of points was collected.
- After registration, verify the registration across multiple known anatomical landmarks. The **Registration Accuracy** value is an indication of the amount of error in your registration. For more information about **Registration Accuracy**, see “Registration accuracy.”

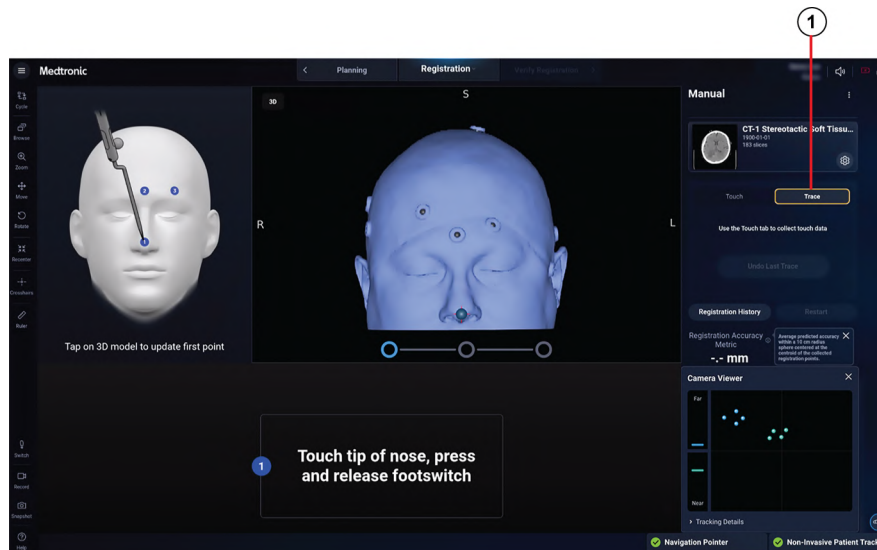
## Performing 3-point touch to orient before tracing

**Note:** You have the option to use 3-point touch to orient the registration before you begin tracing. You can enable or disable the 3-point touch registration option using the **3 Point Touch** setting on the **Navigation** tab in your surgeon profile. See “Editing a procedure.”

**Note:** Depending on the **Registration Footswitch Behavior** setting, you can collect registration points by holding the registration probe steady for 2 seconds on a touched point or by pressing the footswitch.

Follow these steps to perform 3-point touch before tracing:

1. A blue dot will appear in the 3D model view. If necessary, adjust the position of the blue dot to the most anterior position on the patient (tip of the patient's nose). You cannot adjust any other blue dots after this step.
2. Touch the tip of the registration probe to the tip of the patient's nose, press and release the footswitch or hold the instrument steady for 2 seconds until you hear a verification sound.
3. Touch the second point on the patient's skin (center of the forehead or between the eyebrows), press and release the footswitch or hold the instrument steady for 2 seconds until you hear a verification sound.
4. Touch the third point on the left side of the patient's forehead, press and release the footswitch or hold the instrument steady for 2 seconds until you hear a verification sound.



## 1 Trace

### Performing trace registration

Follow these best practices when performing trace registration:

- Create the full trace as a series of small trace segments by starting and stopping movement of the registration probe on the patient's anatomy. Each time you stop tracing, the clinical application adds the new segment to the registration and updates the **Registration Accuracy** value.
- If the last segment of the trace is inaccurate, delete it by tapping **[Undo Last Trace]**. You can remove up to the last 4 segments by tapping **[Undo Last Trace]**. If you need to remove more than 4 stored segments you must restart the registration.
- The **Registration Accuracy** value is an indication of the quality of the registration. Disregard small changes in the **Registration Accuracy** value between traces.
- When the registration is acceptable, verify the registration by touching known anatomical landmarks. See “Verifying registration.”

Follow these steps to perform a trace registration:

1. In the **Registration** task, tap the **Trace** tab.
2. To begin trace registration, place the tip of the registration probe on the patient's anatomy and press and hold the footswitch. Move the tip of the probe lightly around the firm and bony areas of the patient's nose, brow, mastoids, and scalp to collect segments for the registration.

**Note:** If the **Stationary Probe** setting is enabled for this procedure on the **Navigation** tab in your surgeon profile, you can start and stop collecting trace segments by holding the probe steady on the patient's skin for 2 seconds. See “Editing a procedure.” While tracing, the system emits an audio signal when the tracing state changes. One beep indicates the start of tracing. Two beeps indicate that tracing has stopped. One high-pitched beep indicates that initialization is complete. When initialization is complete, the progress bar prompts you to verify the registration or continue collecting trace points.

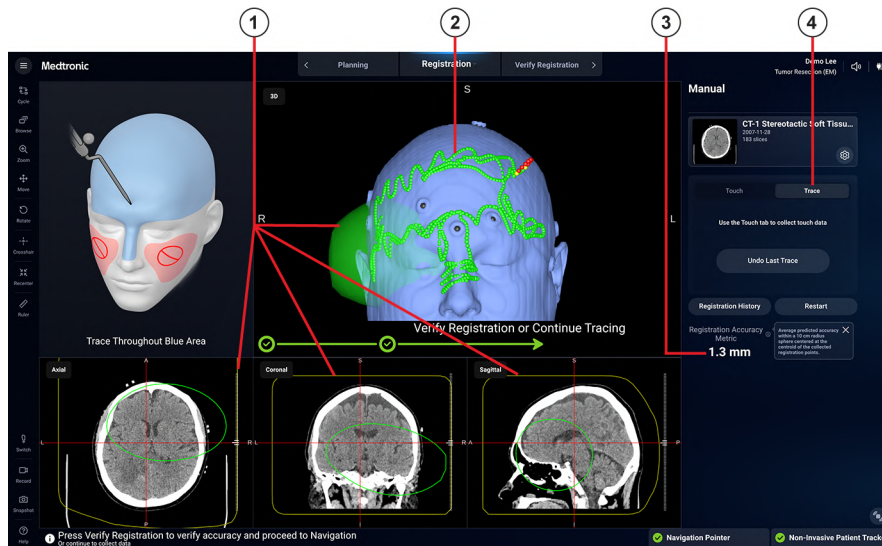
Follow these best practices for tracing:

- Gently maintain contact with skin at all times while tracing. If you need to lift the registration probe to move around an obstruction, release the footswitch (or hold the probe steady for 2 seconds if the **Stationary Probe** setting is enabled) before you lift the probe. Then reposition the probe before you press the footswitch, or hold the probe steady for 2 seconds, to start tracing again.
- Include as many uniquely-shaped areas as possible, but only trace anatomy that is included in the patient exam.
- Make sure that you collect segments both around the target anatomy and on the opposite side of the head.
- Do not collect segments in areas where the tissue is deformable or in areas such as the cheeks or neck where the skin may be loose.

When the **Minimum Trace** progress bar is full, the clinical application displays the points on the 3D model and displays the **Registration Accuracy** value. Lower numbers indicate a more accurate registration.

**Note:** If you started registration with 3-point touch, your trace segments will display blue until you reach the minimum amount of collected points needed for trace registration.

For more information about the **Registration Accuracy** value and the Zone of Predicted Registration Accuracy, see “Registration accuracy.”



- 1 Zone of Predicted Registration Accuracy
- 2 Collected trace segments
- 3 Registration Accuracy
- 4 **Trace** tab

### 3. Review your trace registration.

- If you are satisfied with the Zone of Predicted Registration Accuracy and the **Registration Accuracy** value, tap **[Verify Registration]** to advance to the registration verification task.
- If you are not satisfied with the Zone of Predicted Registration Accuracy and the **Registration Accuracy** value, try the following to refine the trace:
  - Attempt additional tracing until you have collected segments over a broad area. Each time you stop tracing, the clinical application adds the new segment to the registration and updates the Zone of Predicted Registration Accuracy and the **Registration Accuracy** value.
  - If additional tracing does not produce a usable registration, tap **Touch** and then define and touch additional landmarks. See “Performing touch registration.” After touching the registration landmarks, tap **Trace** and continue tracing.
  - If the combination of trace and touch registration does not produce a usable registration, tap **[Restart]**.

If the **Registration Accuracy** exceeds 5.0 mm, the clinical application does not allow you to advance to **Navigation**. In that case, trace over a broader area or, if your trace contains many yellow or red segments, tap **[Restart]** to start the registration process again. Previous registrations are saved to **Registration History**.

### Performing touch registration

#### Defining registration landmarks

The clinical application automatically detects IZI skin fiducial markers in the images and stores them as touch registration landmarks. You can also store anatomical landmarks for use in touch registration. For more guidance on fiducial placement, see *Stealth AXiS™ Imaging Protocols for Navigated Procedures*.

To store anatomical landmarks or undetected fiducial markers, follow these steps:

1. In the **Registration** task, tap the **Touch** tab.
2. Tap or click an identifiable anatomical landmark on the 3D model in the clinical application.

Suggested landmarks:

- Right tragus
- Left tragus
- Inner canthus
- Outer canthus
- Columella
- Nasion
- Tip of the nose

3. Tap **[+]**.

The clinical application adds a marker at that location on the 3D model and a touch registration point in the landmarks table.

4. After a touch registration point is added, if necessary, refine the position of the selected point in the 2D views using the directional arrows and tap **[Save]**.

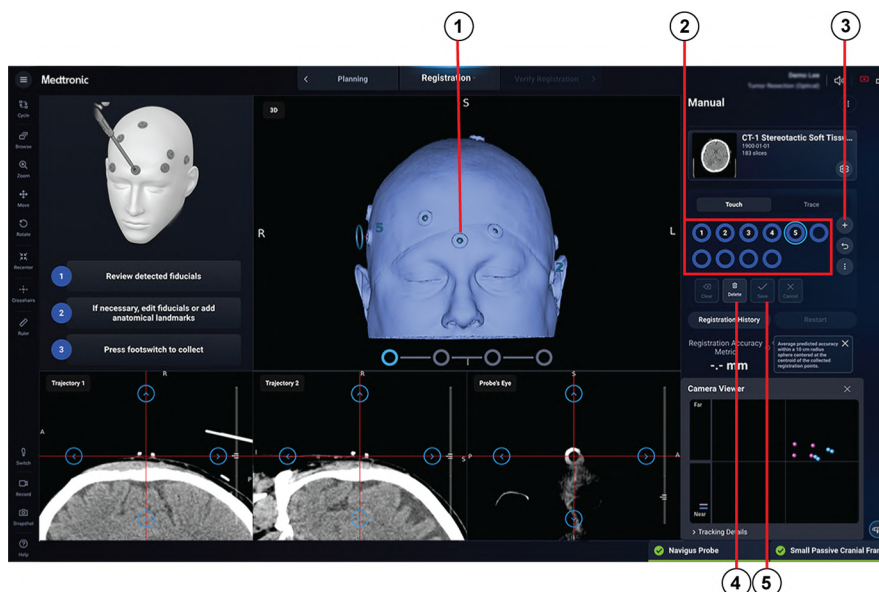
**Note:** You can cycle between two 2D image view layouts in the **Registration** task: **Trajectory 1 / Trajectory 2 / Probe's eye** and **Axial / Sagittal / Coronal**. If you tap **[Cycle]** to change the image view layout in the **Registration** task, the layout you select remains in the **Registration** task even if you exit and re-enter the procedure.

5. Repeat these steps until you have selected up to 12 well-distributed landmarks. At least 4 landmarks are required to calculate a touch registration.
  - Select an asymmetrical set of landmarks.
  - Select landmarks that vary in distance from one another.
  - Select a set of landmarks that define a 3D space, rather than a straight line or a 2D plane.
  - Select anatomical or fiducial characteristics that are identifiable in both the clinical application image and on the patient (such as moles).
  - Select a set of points near the surgical area of interest.

**Note:** If you initialized the registration using trace registration, there is no minimum number of touch points.

6. You can use optional controls to manage the set of stored landmarks.
  - To delete a landmark, select it and tap **[Delete]**.
  - To assign a number to a detected fiducial marker, select its landmark and tap **[Assign]**.

**Note:** If a landmark was added manually, then a number will be automatically assigned.



- 1 Landmark
- 2 Landmarks table
- 3 [+]
- 4 [Delete]
- 5 [Save]

### Editing registration landmarks

Check the locations of all stored fiducial landmarks to make sure that the locations are correct. Follow these steps to edit landmarks if necessary.

**Note:** Fiducial landmarks should be stored on the surface of the patient's skin in the center of the fiducial marker.

1. Tap a stored landmark in the **Touch** tab or on the 3D model.
2. In any 2D view, stretch to zoom in to the precise location of the stored landmark.
3. Use the following methods to move the stored landmark to the correct location.
  - Tap a point in the 2D or 3D views. (Tapping near a stored landmark or detected fiducial marker in the 3D views snaps the cursor to the landmark or detected fiducial marker location.)
  - To refine the landmark position, use the directional arrows in the 2D views.
4. When the landmark is in the desired location, tap **[Save]**. Alternatively, tap **[Cancel]** to restore the landmark to its original location.

### Touching landmarks

**Warning:** Do not bump or reposition the patient reference frame after registration. Movement of the patient reference frame relative to the patient anatomy will result in inaccurate navigation. If the patient reference frame moves in relation to the patient anatomy at any time after registration, you must re-register.

Follow these steps to touch the landmarks on the patient:

1. Place the tip of a registration probe that you previously verified on the patient at a landmark. Do not indent the skin. You can touch the landmarks in any order.
2. Depending on your surgeon preferences, do one of the following to store a point:
  - Hold the registration probe steady and continue to hold the probe for 2 seconds.
  - Hold the registration probe steady and press the footswitch.
3. Depending on your surgeon preferences, do one of the following to store the next point:
  - If you are using the **Stationary Probe** registration preference, hold the registration probe steady on the point. Once you hear a verification sound or see the landmark circle turn color, move the registration probe to the next point.

- If you are using the footswitch to store points, release the footswitch before removing the probe from the point.
4. Repeat this process for the remaining landmarks.

When you have touched enough landmarks to compute a registration, the clinical application displays the **Registration Accuracy** value. Lower numbers indicate a more accurate registration.

After a registration is calculated, the touched landmarks change color. The color indicates the distance between the stored landmark and the matching touched point.

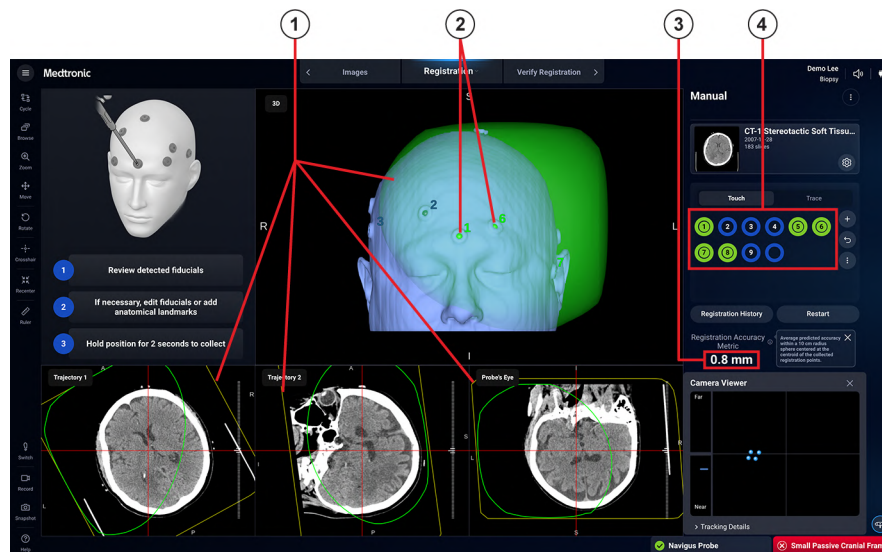
- Green landmarks: Less than or equal to 2.5 mm
- Yellow landmarks: Greater than 2.5 mm up to 5.0 mm
- Red landmarks: Greater than 5.0 mm
- Blue landmarks have not been matched to a touched point.

You can retouch a previously touched landmark by holding the registration instrument's tip on it and holding for 2 seconds or by pressing the footswitch. Retouch previously touched landmarks if you believe that the first touch was incorrect (for example, if your hand or the probe slipped). Additionally, you can clear all registration data for a landmark by selecting the landmark in the landmarks table and tapping **[Clear]**. You can clear all registration data for all landmarks by tapping the More button and then tapping **[Clear All]**.

If there are multiple red or yellow landmarks, retouch multiple landmarks, including some green landmarks. If the registration accuracy does not improve to your satisfaction, check the placement of stored landmarks in the 2D image views and adjust the placement as necessary.

The clinical application also displays a Zone of Predicted Registration Accuracy as a green shape in the 3D view and as green and yellow outlines in the 2D views.

For more information about the **Registration Accuracy** value and the Zone of Predicted Registration Accuracy, see "Registration accuracy."



- 1 Zone of Predicted Registration Accuracy
- 2 Landmarks that have been touched on the patient anatomy
- 3 Registration Accuracy
- 4 Points touched

5. Tap **[Verify Registration]** to advance to the registration verification task.

If the **Registration Accuracy** exceeds 5.0 mm, the clinical application does not allow you to advance to **Navigation**. In that case, add more landmarks, or if your touch registration contains red and yellow points, retouch multiple landmarks, including some green landmarks. If the registration accuracy does not improve to your satisfaction, consider checking the placement of the stored landmarks in the 2D image views and adjusting as necessary. Tap **[Restart]** to start the registration process again. You can also tap the **Trace** tab and collect more points by tracing on the anatomy. See "Performing trace registration."

## Registration accuracy

When you have collected enough points for the clinical application to compute a registration, the clinical application displays the **Registration Accuracy** value and the Zone of Predicted Registration Accuracy.

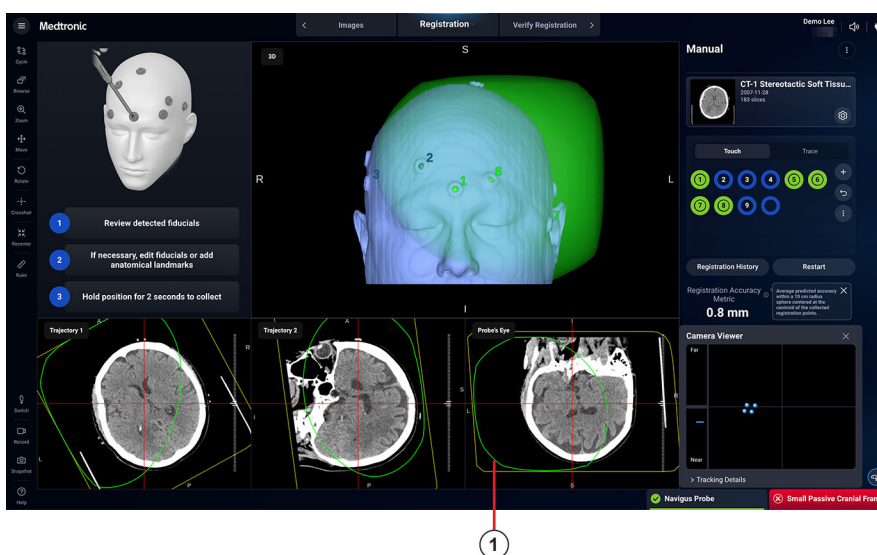
The **Registration Accuracy** value indicates the estimated accuracy due to the registration process only. It is not the overall, system-wide navigation accuracy. Lower numbers indicate a more accurate registration.

**Note:** **Registration Accuracy** is a calculation of the average probable error throughout a sphere with a 10 cm radius. The center of the sphere is located at the center of the collected registration points, and that point is typically near the center of the Zone of Predicted Registration Accuracy.

Regions of the image or 3D model within the green Zone of Predicted Registration Accuracy are predicted to have less than 1.0 mm of error due to registration. Regions of the image within the yellow outlines are predicted to have less than 2.0 mm of error due to registration. The clinical application displays this zone in the 3D view and displays cross sections of the zone as green outlines in the 2D views.

As you collect more points during registration, the clinical application will refine the zone. Do not alter your collection of registration points in order to make a very small improvement in the **Registration Accuracy** value. Make sure that the registration within the surgical area of interest is satisfactory before navigating.

**Note:** The Zone of Predicted Registration Accuracy displays only to the edge of the image volume. Large zones or zones near the edge of the exam may display flat sides and corners.



- 1 Cross-sections of the Zone of Predicted Registration Accuracy: The area within the green outlines is estimated to have a registration error of 1.0 mm at most.

## Verifying registration

**Warning:** Verify the accuracy of the registration on several known anatomical landmarks before using the registration for navigation. Failure to verify can result in inaccurate navigation.

To verify registration, follow these steps:

1. In the **Registration** task, tap **[Verify Registration]**.
2. Use the registration instrument to touch an identifiable point on the patient's anatomy.
3. Compare the location of this point to the centers of the crosshairs in the images.
4. Repeat the process for at least two more points.
5. Optionally, perform the following tasks to view additional registration information:
6. To view the registration accuracy at a specific location on the image, do one of the following:
  - Pause navigation and place the crosshair at that location to observe the **Registration Accuracy at Crosshair** value that displays on the right side of the screen.

- Tap the Plans button displayed below the **Registration Accuracy at Crosshair** to open the **Plan** drawer, and then drag the **Traverse** slider to see the **Registration Accuracy at Crosshair** along your plan.

**Note:** The **Registration Accuracy at Crosshair** will not update while navigating.

7. If you are satisfied with the registration, proceed to **Navigation**.

If the registration is not acceptable, take one of the following actions:

- Return to the **Registration** task and collect more points.
- Tap **[Restart]** to start the registration process again.

**Note:** **[Restart]** deletes collected points but retains the 3D model and stored landmarks. **[Restart]** also saves your current registration as a previous registration.

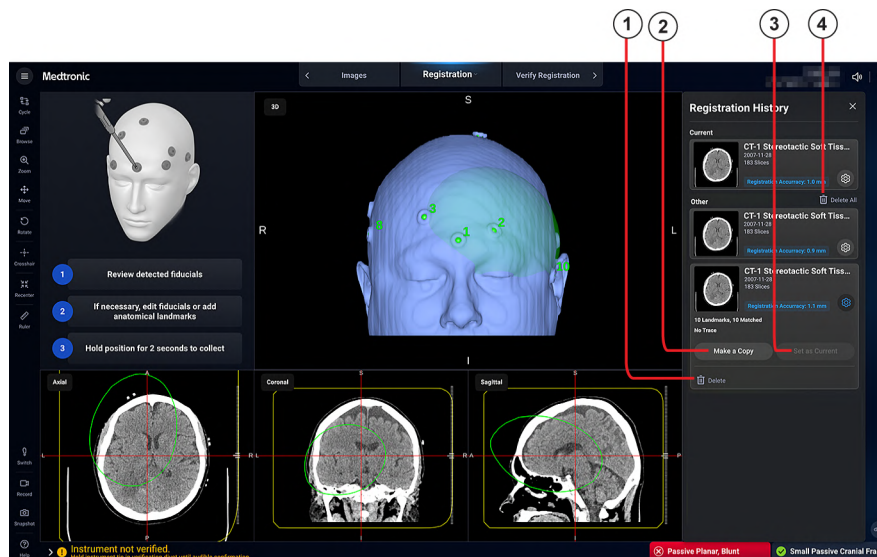
- If you have a previous registration that is accurate, tap **[Previous Registrations]** to open the **Registration History**. Then select the previous registration and tap **[Set As Current]**.

## Restoring previous registrations

If you have more than one registration, tap **[Registration History]**. The current registration appears at the top of the list, and older registrations display in the **Other** list in order from the newest to oldest.

- To make a previous registration the current registration, tap the Settings button and then tap **[Set As Current]**. This moves the registration into the Current position at the top of the list. If a registration is deactivated, the **[Set As Current]** button is disabled for that registration. Registrations are deactivated for the following reasons:
  - The patient reference frame has changed since that registration was created.
  - The registration was created in an incompatible procedure.
- To delete a registration in the **Other** list, tap the Settings button on the registration you want to delete, and then tap **[Delete]**. You cannot delete the current registration.
- To delete all registrations in the **Other** list, tap **[Delete All]**, and then tap **[Confirm]**. You cannot delete the current registration.
- To copy a registration, tap the Settings button on the registration you want to copy, and then tap **[Make a Copy]**. The original registration is preserved, and you can modify the copy.

**Note:** **[Make a Copy]** is disabled if the registration is deactivated.



- 1 **[Delete]**
- 2 **[Make a Copy]**
- 3 **[Set As Current]**
- 4 **[Delete All]**

## Performing an O-arm™ Imaging System registration (optical only)

The O-arm™ Imaging System is a mobile X-ray system designed for 2D fluoroscopic and 3D imaging for high contrast objects and anatomic structures.

Acquire a scan from the O-arm™ Imaging System in the **Registration** task.

The **Registration** task only allows O-arm™ system registration if an O-arm™ system is in the equipment list for the procedure. If necessary, click the Menu button, select the **Equipment** task, and add the O-arm™ system to the procedure.

**Note:** To transfer an unregistered scan from the O-arm™ system to the Stealth AXiS™ system, refer to the user manual that accompanies the O-arm™ Imaging System.

**Caution:** The software version on the O-arm™ Imaging System must have the navigation interface to allow communication and automatic registration with the Stealth AXiS™ clinical application. The Stealth AXiS™ Surgical System is not compatible with the O-arm™ 1000 system.

1. Make sure that the O-arm™ system is properly connected to the Stealth AXiS™ system by a network cable, and that the power is turned on.

When the systems are communicating, **O-arm System Communicating** displays in the Stealth AXiS™ clinical application.

2. Center the anatomy of interest in the center of the gantry or at isocenter.
3. Press the **3D** button on the O-arm™ pendant.
4. Make sure that both the O-arm™ tracker and the patient reference frame are visible to the Stealth AXiS™ camera.  
The Stealth AXiS™ clinical application displays **O-arm Tracker Visible** and **Patient Tracker Visible** when the trackers are in the camera's field of view.
5. Make sure that the O-arm™ system, the Stealth AXiS™ camera, and the patient reference frame are all motionless.
6. Follow the O-arm™ system instructions for acquiring the patient scan.

When the acquisition is complete, the O-arm™ system automatically sends the scan to the Stealth AXiS™ system.

The Stealth AXiS™ system automatically loads the scan and registers the patient. The scan appears in the list of **Patient Images**.

If the Stealth AXiS™ system does not automatically load the scan, see the user manual that accompanies the O-arm™ Imaging system for a different method of transferring a scan with a registration.

**Warning:** Any motion of the patient due to breathing, tremor, or any other voluntary or involuntary motion during the O-arm™ system scan may lead to registration inaccuracies.

7. Tap the patient image and review the different views.
  - a. Make sure that the anatomy of interest is captured in the images.
  - b. Make sure that the images do not contain geometric distortion or intensity distortion (intense brightness in parts of the image compared to the rest of the image). If images contain distortion, see "O-arm™ system scan troubleshooting (optical only)."

**Warning:** Verify that all aspects of the patient images are correct, including patient data and image orientation. Make sure that the image orientation matches the patient's anatomy.

If you need to adjust the level or width, tap the Settings button on the image in the **Patient Images** list.

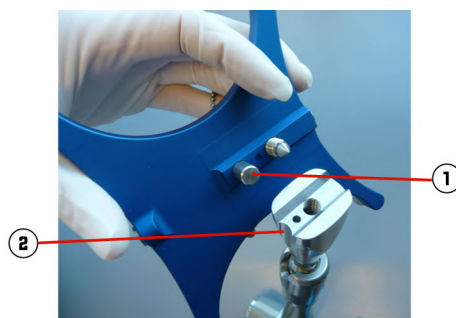
8. Before using the registration for live navigation, verify that the O-arm™ system registration is accurate by navigating to known anatomical landmarks in the **Navigation** task.
  - If there are selected pre-operative images of the patient, tap **[Merge Images]** and perform the Merge process. See "Using image merge." If there are no selected pre-operative images, tap **[Navigation]** to advance to the **Navigation** task.
  - If the O-arm™ system registration is not successful, acquire another O-arm™ system scan or select a different registration method. However, if the issue is due to a failed image transfer, the images can be manually pushed from the O-arm™ system to the Stealth AXiS™ system to avoid re-scanning. See the *O-arm™ Imaging System User Manual* for more information.



## Navigation

### Setting up for sterile navigation (optical only)

1. Remove the non-sterile patient reference frame and drape the patient.  
Note the orientation of the Medtronic logo on the surface of the patient reference frame. When you attach the sterile patient reference frame after draping the patient and going sterile, make sure the logo resides in the same orientation.
2. Drape the Vertek™ Articulating Arm swivel mount with a thin, clear plastic drape.
3. Pull the drape tight so that there are no wrinkles on the surface of the swivel mount that might create a gap at the interface between the patient reference frame and the swivel mount surface.
4. Match the mounting screw and the alignment post (on the underside of the sterile patient reference frame) with the corresponding screw hole and alignment notch (on the swivel mount of the Vertek™ arm).
5. Puncture the plastic drape with the mounting screw and secure the patient reference frame to the swivel mount by tightening the mounting screw.



- 1 Alignment post
- 2 Alignment notch

### Navigating

During navigation, the clinical application updates the 2D views in real time to reflect changes in the position of the surgical instrument's tip on the anatomy. The crosshairs in the 2D views represent the position of the instrument's tip. When navigation is active, the crosshairs display green and the instrument displays in the tracking view. When navigation is paused, or if an instrument is not tracking, the crosshairs display red. If you are using optical localization and the crosshairs display yellow, navigation is still active but the camera cannot detect all the spheres on the instrument or the patient reference frame.

You can press the footswitch or tap the Footswitch button on the screen to pause navigation.

**Note:** You can only use the footswitch to pause navigation when an instrument is tracking in the navigation field.

**Warning:** Frequently confirm navigation accuracy and system responsiveness during live navigation. Use the navigation instrument to touch several bony anatomical landmarks and confirm that the locations identified on the images match the locations touched on the patient. If accuracy degrades, re-register the patient.

**Warning:** (optical only) Before navigating, verify and confirm instrument accuracy for all instruments, including probes, SureTrak™ II, and Stealth-Midas™ MR8™ instruments. To confirm system accuracy, touch the instrument to several bony anatomical landmarks and confirm that the locations identified on the images match the locations touched on the patient. If the accuracy is not acceptable, repeat the instrument verification steps, or repeat instrument calibration steps for the SureTrak™ II instrument. Make sure that the tip of the instrument is in the bottom of the verification divot, and keep the instrument perpendicular to the divot while ensuring the spheres are rotated toward the camera.

**Warning:** Do not verify with the Stealth-Midas™ MR8™ dissecting tool while the handpiece is running to prevent device damage and to prevent metal shavings from entering the surgical site.

**Warning:** Do not navigate the tip of the Stealth-Midas™ MR8™ dissecting tool while the handpiece is running.

**Note:** Do not touch electrocautery with metal instruments.

### Instrument trajectories (optical only)

Some views, such as **Guidance**, **Trajectory**, and **Look Ahead**, include information based on an instrument trajectory, which is a calculated extension of the instrument beyond the instrument tip. For these views, the clinical application calculates the trajectory for each instrument as follows:

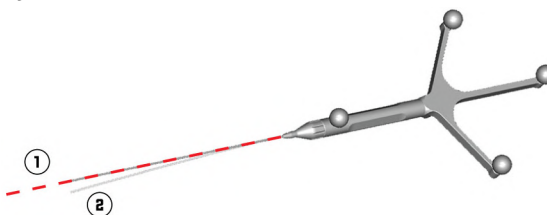
- For the Passive Planar Blunt Probes (960-556 and 9734881\*), Navigus™ Probe (9733157), and Vertek™ Probe (9733158), the trajectory follows the bayonet.



- For calibrated instruments such as the Microscope Probes (9730269 and 9734889\*), the trajectory follows the bayonet. However, because the tip of these instruments is calibrated and the trajectory is not calibrated, there may be a slight variation between the navigated trajectory and the physical trajectory of the instrument.



- For the Passive Catheter Introducer (9731116), the trajectory follows the bayonet. Because the Passive Catheter Introducer is a semi-rigid instrument and the tip can temporarily bend, there may be a variation between the navigated trajectory and the physical trajectory of the instrument.



- 1 Navigated trajectory of the Passive Catheter Introducer
- 2 Passive Catheter Introducer with a temporary bend

- For ENT instruments such as the eNV™ Straight Suction (9731234), eNV™ Curved Suction (9731250), and eNV™ Ostium Seeker (9732374), a trajectory view is not available.

**\*Note:** The Passive Planar Blunt Probe 9734881 and the Microscope Probe 9734889 are for sale in the European Union only. The Passive Planar Blunt Probe 960-556 and the Microscope Probe 9730269 are for sale outside of the European Union.

## Instrument trajectories (EM only)

Some views, such as **Guidance**, **Trajectory**, and **Look Ahead**, include information based on an instrument trajectory, which is a calculated extension of the instrument beyond the instrument tip. For these views, the clinical application calculates the trajectory for each instrument as follows:

- For the Navigation Pointer (9735319, 9733766), Tracer™ Pointer (9735317, 9733367), and Touch-n-Go Pointer (9735318, 9733368), the trajectory follows the bayonet.



- For the Stylet (9735428, 9733675, 9660204), which is a semi-rigid instrument, the trajectory follows the direction of the stylet tip. Therefore, when the stylet temporarily flexes, the navigated trajectory follows the stylet tip.



- 1 Navigated trajectory of a stylet with no flex
- 2 Navigated trajectory of a stylet with a temporary flex

## Aligning an instrument with a surgical plan in guidance view

The purpose of **Guidance** view is to help you aim and guide a navigation instrument along a surgical plan. You must create a surgical plan to use **Guidance** view.

### Choosing target guidance or trajectory guidance

The guidance view can be configured to display either target guidance or trajectory guidance.

- Target guidance assists with aiming the navigation instrument at the target of the surgical plan but not necessarily along the same trajectory as the surgical plan. The guidance view shows the plane that contains the target from the perspective of someone looking straight down the instrument trajectory. The most superior point on the patient exam remains at the top of the view.
- Trajectory guidance assists with aligning the navigation instrument along the surgical plan such that the instrument aligns with both the entry point and target point of the surgical plan. The guidance view shows the plane that contains the target from the perspective of someone looking straight down the surgical plan. The anatomy displayed at the top of the view is determined by the front face of the instrument. The image orientation changes as you rotate the instrument.

**Note:** Target guidance requires a plan target. Trajectory guidance requires a target and entry point.

Follow these steps to change the default guidance option for a procedure:

1. Tap the Menu button, and then tap **[Preferences]**.
2. Tap the desired procedure.
3. Tap the **Views** tab.
4. Under **Guidance View**, select either **Target Guidance** or **Trajectory Guidance**.
5. Proceed to the next desired task.

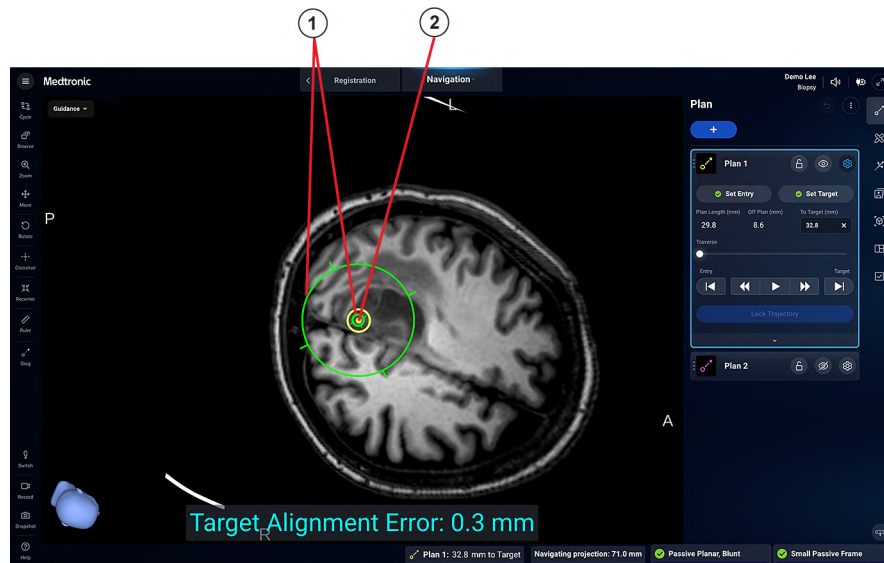
### Using the guidance view

To help align an instrument with a surgical plan, the software includes special views such as **Guidance**, **Trajectory 1**, **Trajectory 2**, and **Probe's eye**. Follow these instructions to use the guidance view:

1. To replace one of the 2D views with the guidance view, tap the image selection drop-down menu and select **Guidance** from the list.
2. Replace the other 2D views with **Trajectory 1**, **Trajectory 2**, and **Probe's eye** (see "Trajectory views" for a description of these views).
3. Place the tip of the navigation instrument at the entry point of the surgical plan.

The target guidance view shows indicators to assist with alignment:

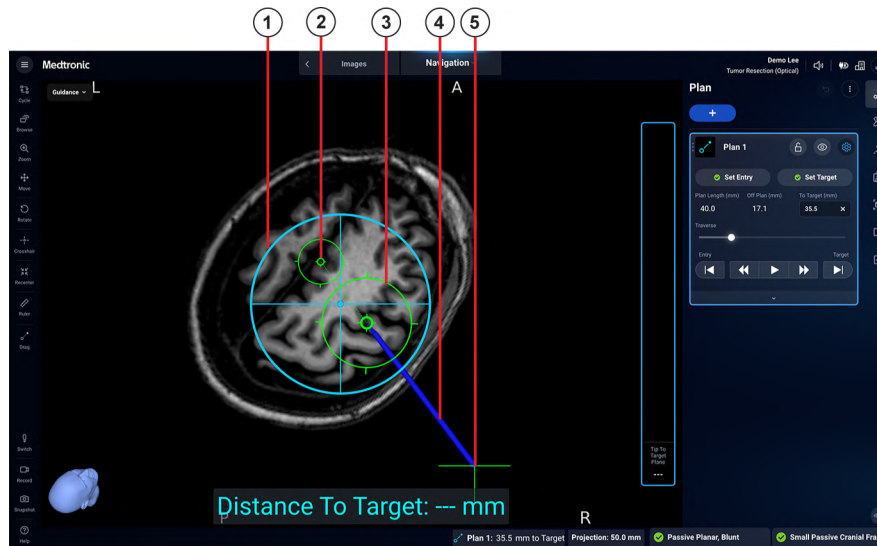
- The target is a small circle that is the same color as the selected surgical plan.
- The aiming circle around the instrument tip is a circle with four directional marks.



- 1 Target guidance: Instrument aiming circles
- 2 Target guidance: Plan target

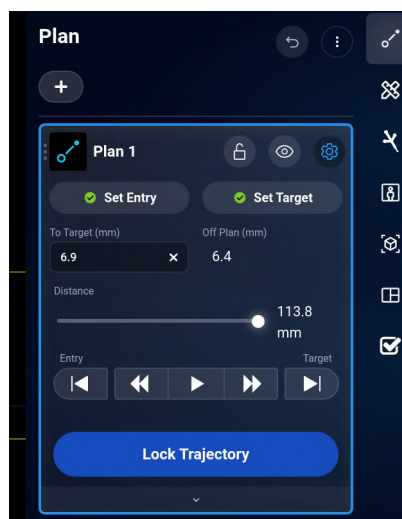
The trajectory guidance view shows additional indicators:

- The hind of the instrument is a large plus sign (+).
- The aiming circle around the instrument tip is a circle with four directional marks.
- The plan target is a dot within a circle, both in the same color as the plan.
- The trajectory from the instrument tip is a dashed line with a circle at the end. The circle represents the point at which the projected tip of the instrument intersects the target plane.

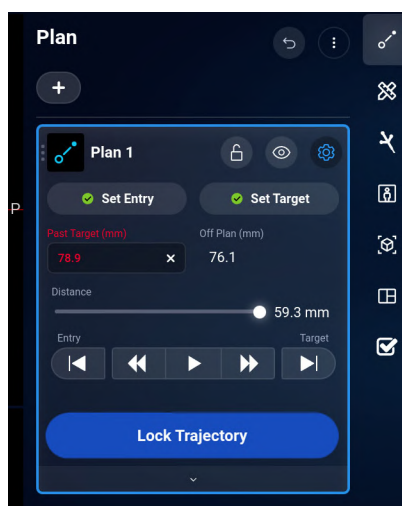


- 1 Trajectory guidance: Current plan target.
  - 2 Trajectory guidance: Projection crosshair in the intersection of the trajectory and target plane. Circle size is configured in the preferences settings.
  - 3 Trajectory guidance: Tip crosshair shows the currently navigated position. This is the point at which the projected tip of the instrument intersects the target plane.
  - 4 Trajectory guidance: Instrument representation.
  - 5 Trajectory guidance: Hind crosshair shows at 150mm away from the currently navigated position along the instrument trajectory.
4. Use the guidance view to align the guidance indicators.
- Each indicator highlights (becomes thicker) when its target alignment error is less than 2 mm. For example, the aiming circle around the instrument tip is highlighted when the instrument tip is aligned within 2 mm of the target.
- While you are aligning with the target, if you want to reset the entry point of your surgical plan at the tip of the instrument, tap **[Set Entry]**.
  - If you are using target guidance, align the center of the aiming circle with the target.
  - If you are using trajectory guidance, align the target, the center of the aiming circle, and the hind of the instrument.
5. Keep the dot centered inside the small circle as you advance the instrument along the surgical plan.
- Tip to Target Plane** appears in the depth gauge. **Tip to Target Plane** is the calculated distance from the tip of the navigated instrument to the image plane that contains the target. The system tracks the location of the instrument tip and reports the distance to target as the instrument advances.
6. In the **Plan** drawer, you can view the calculated distance.
- The **Tip to Target Plane** text changes to **Tip Past Target Plane** when the instrument tip passes the target.
- Note:** The buttons **[Lock Trajectory]** and **[Unlock Trajectory]** that display on a plan are only available for biopsy procedures.

The **To Target** text changes to **Past Target** when the instrument tip passes the target.



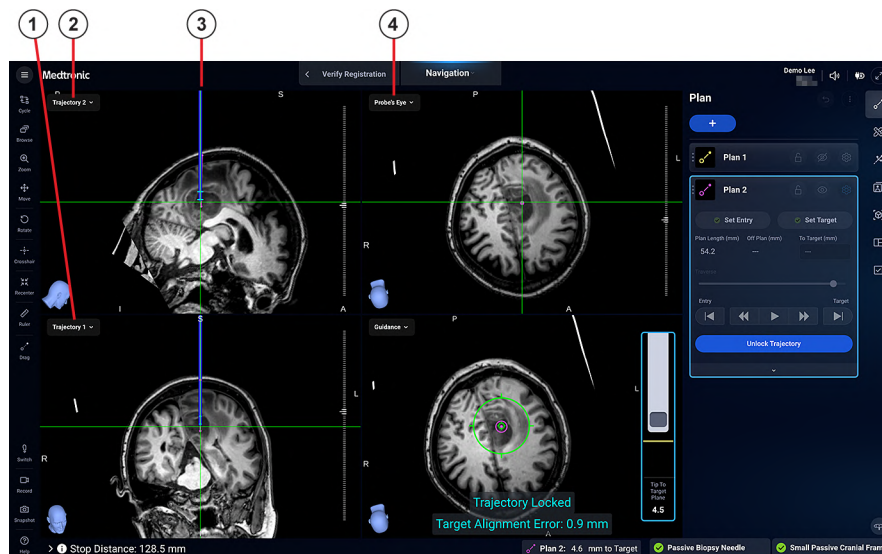
The **Past Target** text is red when the tip is past the target plane.



## Trajectory views

The **Trajectory 1** and **Trajectory 2** views show image planes that contain the instrument trajectory. The two views are perpendicular to each other. These views can help to position the tip of an instrument at the plan entry point and to align the instrument with the plan.

Both **Trajectory 1** and **Trajectory 2** contain representations of the instrument. The representation of the instrument displays at the top of the trajectory views and points to the center of the views in the **Planning** and **Navigation** tasks. When you are scrolling a plan but not navigating, the **Trajectory 1** and **Trajectory 2** views show image planes that contain the plan trajectory.



- 1 Trajectory 1 view
- 2 Trajectory 2 view
- 3 Representation of an instrument
- 4 Probe's Eye view

## Probe's eye view

The **Probe's eye** view shows the plane at the instrument tip that is perpendicular to the instrument trajectory. This perspective shows what the surgeon would see if they looked along the shaft of the instrument. When you are scrolling a plan but not navigating, the **Probe's eye** view shows the image planes that are perpendicular to the planned trajectory.

## Using Look Ahead view

The look ahead view gives you a preview of the anatomy you will encounter as you advance the instrument along the current trajectory. To display the **Look Ahead** view:

1. Tap the Image Selection button in the upper-left corner of a view to display the list of available views.
2. Tap **Look Ahead** in the list.

The look ahead view displays four probe's eye views, which shows what the surgeon would see if they looked along the shaft of the instrument. When navigating, the upper left view shows the plane at the instrument tip that is perpendicular to the instrument trajectory. The other three views are 5 mm, 10 mm, and 15 mm beyond the instrument tip.

**Note:** To change the look ahead distance, tap the menu button and then tap **[Preferences]**. Select a procedure and then tap the **Views** tab. In the **Look Ahead Distance** section, use the - or + slider buttons to adjust the look ahead distance increments. Changing the distance affects three views. For example, if the distance is set to 7 mm, the three views are 7 mm, 14 mm, and 21 mm beyond the instrument tip.

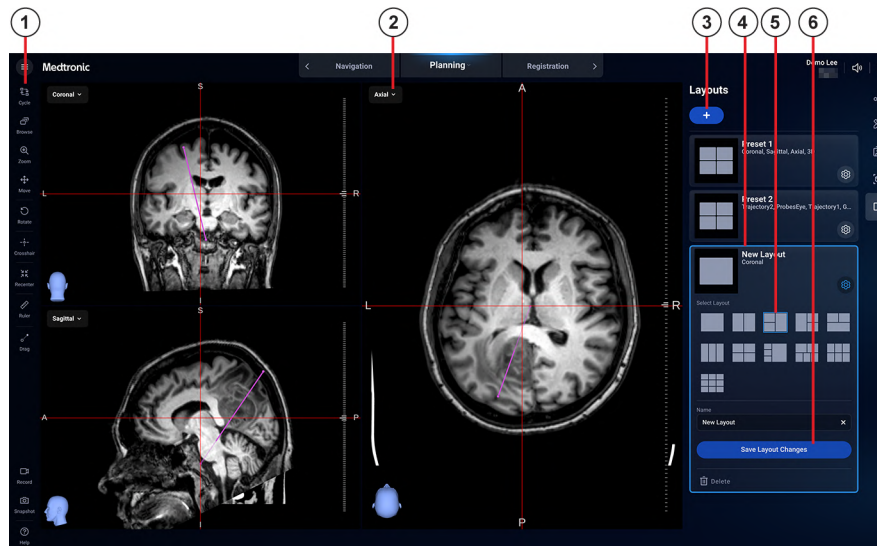


1 Look Ahead view

## Adjusting layout configurations

The default image view layouts are based on the selected procedure and task (for example, four quadrants that display the **Axial**, **Sagittal**, **Coronal**, and **3D** views). You can select one of the preconfigured layouts, modify an existing layout and save it, or create your own configuration. Layouts you modify or create are saved to the current surgeon profile and selected procedure.

- To cycle through the saved layouts, tap **[Cycle Layouts]**.
- To display a different screen layout, tap one of the layout cards in the **Layouts** drawer.
- To temporarily change a multiple view layout to a one view layout, double-tap the Image Selection drop-down menu in any view. Double-tap again to return to the previous layout.
- To change the display of a view, select the desired view from the Image Selection drop-down menu.
- To save a layout view configuration that can be reused:
  1. Tap the **Layouts** button.
  2. Tap the **Settings** button on an existing layout to edit it, or tap **[+]** to create a new layout configuration.
  3. Tap a **Select Layout** card.
  4. To select a different view type, tap the Image Selection drop-down menu in the desired view.
  5. Tap **[Save Layout Changes]**.
  6. Name your layout and tap **[Confirm]**.



- 1 [Cycle Layouts]
- 2 Image Selection drop-down menu
- 3 Create a new layout configuration
- 4 Layout view
- 5 One of the layout buttons
- 6 [Save Layout Changes]

## Catheter placement: Guiding the catheter (optical only)

**Note:** The guidance view is the recommended method for aligning the probe along the desired trajectory. However, the guidance view is functional only if you have created a surgical plan with at least a target.

For more information about the guidance views and guidance indicators within the views, see “Aligning an instrument with a surgical plan in guidance view.”

1. To replace the 2D views with **Trajectory 1**, **Trajectory 2**, and **Probe’s eye** views, tap the Image Selection drop-down menu.
2. Replace the **3D** view with the **Guidance** view.
3. Verify the Passive Catheter Introducer (PCI) by placing its tip in the bottom of the divot in the patient reference frame.

**Warning:** The PCI is a semi-rigid instrument, so the tip can temporarily bend. Make sure that the tip does not bend during verification.

4. Slide a catheter of appropriate length onto the PCI tip until the closed end of the catheter is even with the end of the PCI tip.

The recommended catheter inner diameter is greater than or equal to 1.3 mm, and the recommended length is less than or equal to 15.0 cm. Always verify catheter compatibility with the PCI before use.

5. Place the tip of the PCI at the entry point you stored earlier.
6. Use the guidance view to align the guidance indicators.
7. Keep the dot centered inside the small circle as you advance the instrument along the surgical plan.

**Warning:** The PCI is a semi-rigid instrument, and the tip can temporarily bend. Therefore, there may be a variation between the navigated trajectory and the physical trajectory of the instrument. Take the potential variation into account while navigating with the PCI.

(Refer to “Instrument trajectories (optical only)” for more information about the PCI trajectory.)

In the guidance view, the tip distance to target plane metric appears. **Tip Distance to Target Plane** is the calculated distance from the tip of the navigated instrument to the image plane that contains the target. The system tracks the location of the instrument tip and reports the distance to target as the instrument advances.

The **Tip Distance to Target Plane** message changes to **Tip Distance Past Target Plane** when the instrument tip passes the target. The message changes to red after the tip is more than 1 mm below the target plane.

## Catheter placement: Guiding the catheter (EM only)

**Warning:** The point identified by the crosshairs in the 2D views represents the tip of the instrument.

**Note:** The guidance view is the recommended method for aligning the probe along the desired trajectory. However, the guidance view is functional only if you have created a surgical plan with at least a target.

1. To replace the 2D views with **Trajectory 1**, **Trajectory 2**, and **Probe's eye** views, tap the Image Selection drop-down menu.
2. Replace the **3D** view with the **Guidance** view.
3. Insert the sterile stylet through the sterile catheter.

Any ventricular catheter lumen used with the stylet must be shorter than the length of the stylet and have an inner diameter greater than or equal to 1.2 mm. Always verify equipment compatibility with the stylet before use.

4. Place the tip of the stylet at the entry point you stored earlier.
5. Use the guidance view to align the guidance indicators.
6. Keep the dot centered inside the small circle as you advance the instrument along the surgical plan.

**Note:** The stylet is a semi-rigid instrument, and the trajectory follows the direction of the stylet tip. Therefore, when the stylet temporarily bends, the navigated trajectory follows the stylet tip.

In the guidance view, the tip distance to target plane metric appears. **Tip Distance to Target Plane** is the calculated distance from the tip of the navigated instrument to the image plane that contains the target. The system tracks the location of the instrument tip and reports the distance to target as the instrument advances.

The **Tip Distance to Target Plane** message changes to **Tip Distance Past Target Plane** when the instrument tip passes the target. The message changes to red after the tip is more than 1 mm below the target plane.

## Biopsy (optical only)

### Attaching and aiming the guidance system (optical only)

The guidance systems that can lock a trajectory for a passive biopsy procedure are the Trajectory Guide Kit, the Vertek™ Passive Biopsy Kit, and the Stealth Autoguide™ system.

- If you plan to use the Trajectory Guide Kit, see “Option 1: Attaching the Trajectory Guide Kit (optical only).”
- If you plan to use the Vertek™ Passive Biopsy Kit, see “Option 2: Attaching the Vertek™ Passive Biopsy Kit (optical only).”
- See the *Stealth Autoguide™ System Manual* for instructions on setting up and aiming the Stealth Autoguide™ system.

#### Option 1: Attaching the Trajectory Guide Kit (optical only)

1. Use the Passive Planar Probe to identify your entry point, and then mark the entry point on the scalp where you will affix the base.

**Warning:** Before attaching the base, carefully examine the patient image data to ensure that the skull thickness and density at the intended attachment site are adequate for securing the base and screws, and that no arterial or venous structures lie beneath the site. When fully tightened, the screws penetrate 4.5–5.0 mm below the base's lower surface. Patient injury or death could result if a screw perforates the patient's skull cavity.

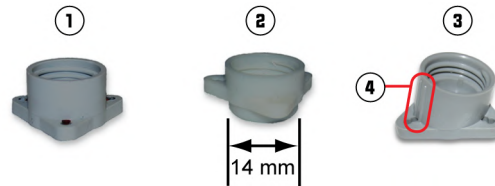
2. Select the appropriate type of Navigus™ base.

Consider both the location of the entry point and the required angle of the needle when deciding whether to use the standard base or the angled base.

With a 14 mm burr hole, the needle's range of motion in the standard base is about 10° from vertical in any direction. The needle's range of motion in the angled base is 5° to 25° (depending on direction). Select the appropriate base to accommodate your expected angle.

3. Create a scalp incision large enough to accept the base you selected.
4. Strip the periosteum away from the surface of the skull.
5. Create a burr hole of adequate diameter. The maximum recommended diameter is 14 mm.

**Note:** For an internal base, the burr hole must be 14 mm in diameter to accommodate the portion of the base that projects below the base plate.



- 1 External base
- 2 Internal base
- 3 Angled external base
- 4 Tactile markers

6. Position the selected base over the burr hole.

**Note:** Orient the angled base so that the vertical tactile markers are parallel to the intended path of the biopsy needle.

7. Anchor the selected base to the skull using three surgical implant-grade, self-drilling titanium screws (1.6 mm × 8.0 mm) included in the kit.

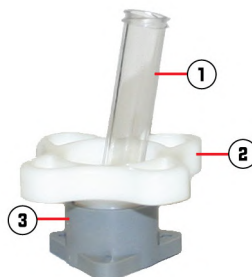
**Warning:** Using the titanium screws in high density cortical bone might result in screw failure during insertion or removal. In high density cortical bone, you might need to predrill, but do not drill through the base. The titanium screws are 1.6 mm in diameter.

**Warning:** To avoid screw stripping or breaking, align the axis of the screwdriver with the axis of a screw and turn the screw one full turn until it is engaged in the bone. Repeat this process with the remaining two screws so that all three screws are engaged. Then evenly tighten the screws just enough so that the base is snug, but do not overtighten because excessive torque could strip or break the screws.

**Warning:** The screws are for single use and are intended to be inserted and removed only one time. Any reuse increases the risk of screws stripping or breaking. Any reuse creates a risk of contamination, which could result in patient injury, illness, or death.

**Warning:** Do not overtighten the screws when the base is on a curved area. Overtightening the screws might cause the base to crack or the screws to fail, particularly if they are tightened unevenly. A damaged base could result in debris and potential debris embedded in tissue.

8. Snap the spherical end of the guide stem into the base.
9. Thread the locking ring onto the base, and tighten the locking ring so that the guide stem remains in place when released.



- 1 Guide stem
- 2 Locking ring
- 3 Base

### Aiming the trajectory guide (optical only)

**Note:** The guidance view is the preferred method for aligning the probe along the desired trajectory.

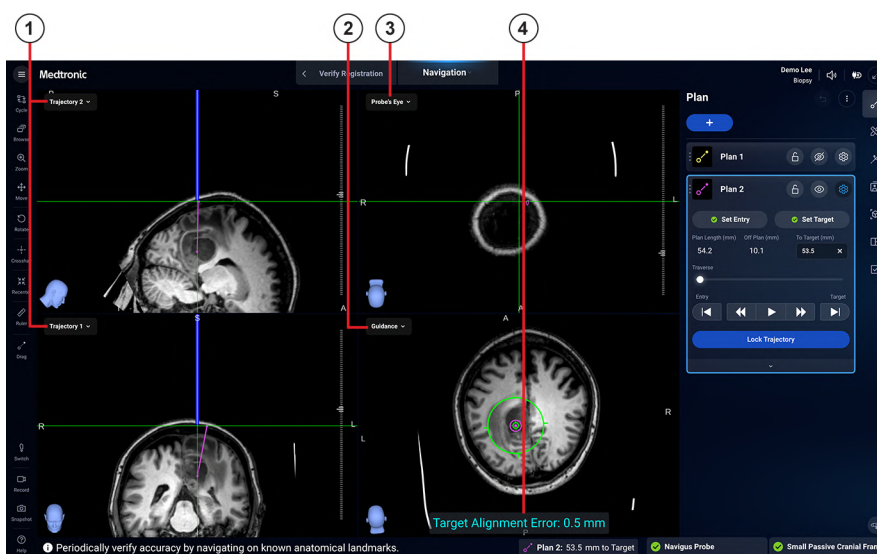
1. If you have not verified the Navigus™ Probe, hold the tip of the instrument in the bottom of the patient reference divot until you hear a positive verification sound (chime).
2. Select an existing plan or create a new plan.

3. Insert the Navigus™ Probe in the guide stem and tap **[Set Entry]** in the **Plans** drawer to update the entry point to its actual location.
4. Hold the Navigus™ Probe in the guide stem, loosen the locking ring, and use the trajectory views and guidance view to aim the guide assembly along the planned surgical trajectory.

**Warning:** Failure to loosen the locking ring before attempting to align the guide stem along the surgical plan will result in stress on the assembly and inability to navigate.



For more information about the guidance views and guidance indicators within the views, see “Aligning an instrument with a surgical plan in guidance view.”



- 1 Trajectory views
  - 2 Guidance view
  - 3 Probe's eye view
  - 4 Target Alignment Error
5. Use the guidance view to align the guidance indicators.

Each indicator highlights (becomes thicker) when its target alignment error is 2 mm or less. For example, the aiming circle around the instrument tip is highlighted when the instrument tip is aligned within 2 mm of the target.

- While you are aligning with the target, if you want to reset the entry point of your surgical plan at the tip of the instrument, tap **[Set Entry]**.
  - If you are using target guidance, align the center of the aiming circle with the target.
  - If you are using trajectory guidance, align the target, the center of the aiming circle, and the hind of the instrument.
6. Tighten the locking ring to secure the assembly so the guide stem cannot move during the procedure. Ensure that after the locking ring is tightened, that the trajectory has not changed. Verify by removing pressure from the assembly by lightly supporting the Navigus™ Probe within the guide stem.

**Warning:** Tighten the locking ring to secure the guide stem. Failure to tighten the locking ring may result in unexpected movement of the guide stem during the procedure.

7. While lightly supporting the Navigus™ Probe and ensuring it is fully seated in the guide stem, press the footswitch or tap **[Lock Trajectory]** to lock the trajectory.
8. Remove the Navigus™ Probe and insert the reducing tube into the guide stem.

**Warning:** Always use the 2.2 mm reducing tube with the passive biopsy needle to avoid incorrect positioning or unexpected motion.

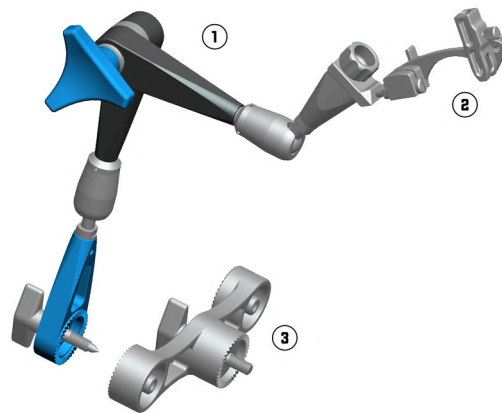
**Warning:** Make sure the Navigus™ Probe is fully seated in the guide stem before locking the trajectory. After locking the trajectory, make sure to fully seat the reducing tube in the guide stem. Failure to fully seat the probe and the reducing tube may lead to discrepancies between the stop distance value provided by the clinical application and the desired stopping position on the biopsy needle.

Proceed to “Performing the guided biopsy (optical only).”

### Option 2: Attaching the Vertek™ Passive Biopsy Kit (optical only)

1. Use the Passive Planar Probe to identify your entry point, and then mark the entry point on the scalp.
2. Connect the Vertek™ Articulating Arm to the dual starburst. Puncture the surgical drape with the base attachment screw of the articulating arm to complete the attachment.
3. Attach the precision aiming device (PAD) to the instrument swivel mount on the articulating arm.
4. While supporting all arm segments, loosen the locking knob on the articulating arm and position the arm such that the precision aiming device sits above the entry point for the biopsy. Tighten the locking knob securely.

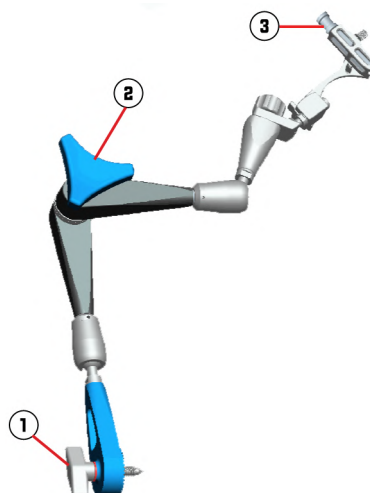
**Warning:** Tighten all joints on the articulating arm. Failure to tighten all joints may result in unexpected movement of the articulating arm during the procedure.



- 1 Vertek™ Articulating Arm
- 2 Precision aiming device (PAD)
- 3 Dual starburst attachment

5. Remove the PAD and set it aside in the sterile field.
6. Create a scalp incision and strip the periosteum away from the surface of the skull.
7. Create a burr hole of adequate diameter.

8. Re-attach the precision aiming device to the articulating arm.
9. Verify that all joints are tight and that the assembly is stable and does not slip.



- 1 Base attachment screw
- 2 Locking knob
- 3 Reducing tube

#### Aiming the precision aiming device (PAD) (optical only)

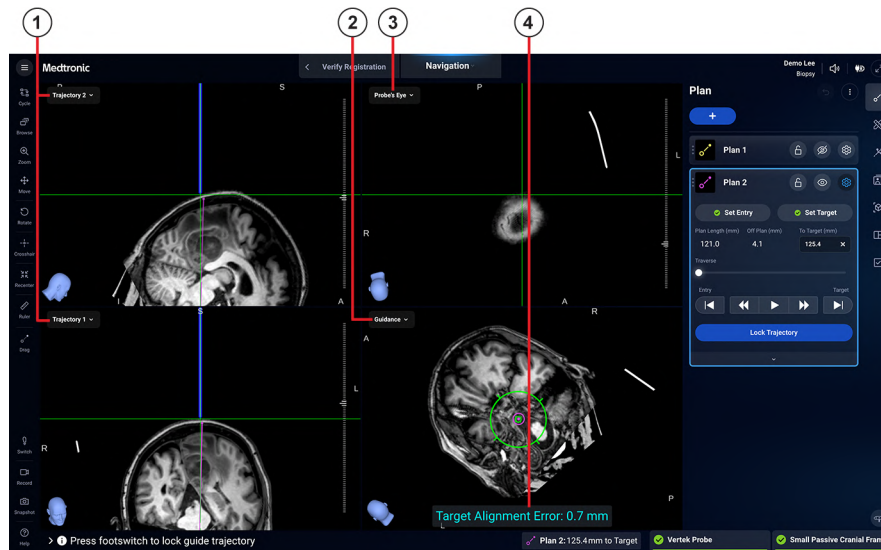
**Note:** The guidance view is the preferred method for aligning the probe along the desired trajectory.

1. If you have not verified the Vertek™ Probe, hold the tip of the probe in the bottom of the patient reference divot until you hear a positive verification sound (chime).
2. Insert the Vertek™ Probe in the PAD and tap **[Set Entry]** on the clinical application screen to update the entry point to its actual location.
3. Hold the Vertek™ Probe in the aiming device, loosen the two adjustment screws, and use the trajectory views and guidance view to aim the guide assembly along the planned surgical trajectory.

**Warning:** Loosen the adjustment screws on the precision aiming device before using the Vertek™ probe to aim the guide assembly. Failure to loosen the adjustment screws before attempting to align the guide assembly along the surgical plan will result in stress on the assembly.



For more information about the guidance views and guidance indicators within the views, see "Aligning an instrument with a surgical plan in guidance view."



- 1 Trajectory views
- 2 Guidance view
- 3 Probe's eye view
- 4 Target Alignment Error

4. Tighten the adjustment screws to secure the assembly so that the aiming device cannot move during the procedure. Ensure that after the screws are tightened, the trajectory has not changed. Verify by removing pressure from the assembly by lightly supporting the aiming device.

**Warning:** After aligning the precision aiming device with the surgical plan, tighten the adjustment screws on the precision aiming device. Failure to tighten the adjustment screws may result in unexpected movement of the aiming device during the procedure.

5. While lightly supporting the Vertek™ Probe and ensuring it is fully seated in the aiming device, press the footswitch or tap **[Lock Trajectory]** to lock the trajectory.
6. Remove the Vertek™ Probe and insert the reducing tube into the aiming device.

**Warning:** Always use the 2.2 mm reducing tube with the passive biopsy needle to avoid incorrect positioning and unexpected motion.

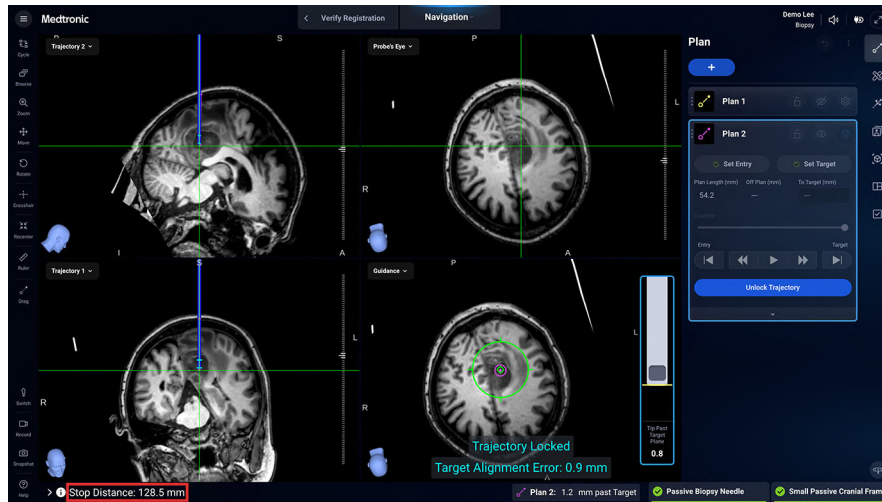
**Warning:** Make sure the Vertek™ Probe is fully seated in the aiming device before locking the trajectory. After locking the trajectory, make sure to fully seat the reducing tube in the aiming device. Failure to fully seat the probe and the reducing tube may lead to discrepancies between the stop distance value provided by the clinical application and the desired stopping position on the biopsy needle.

### Performing the guided biopsy (optical only)

**Warning:** Refer to the stop distance when setting the depth stop on the biopsy needle to avoid incorrect positioning. The clinical application automatically calculates the stop distance. The stop distance includes the distance from the top surface of the adapter (or reducing tube) to the biopsy target. This distance assumes that the aiming probe was fully seated in the aiming device when the trajectory was locked and that the reducing tube (adapter) is fully seated in the aiming device before the biopsy needle is inserted.

**Caution:** Only use the Medtronic passive biopsy needle for navigated biopsy procedures. Use of other biopsy needles will result in no navigation information in the clinical application.

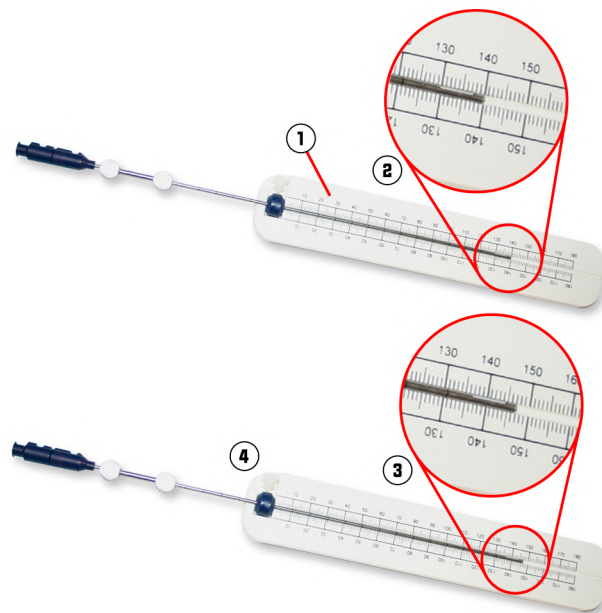
While the trajectory is locked, the system displays the stop distance necessary to deliver the measured location to the target point.



### 1 Stop Distance

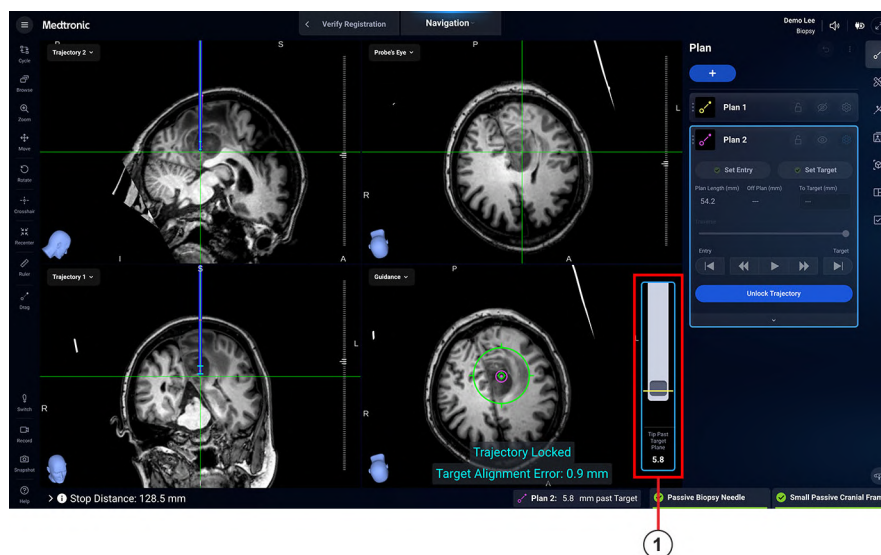
#### Setting the biopsy needle stop (optical only)

1. Decide which part of the needle you want to deliver to the target point: the tip of the needle or the center of the cutting window.
2. Use the stop adjustment tool to set the needle stop so that either the needle tip or the cutting window is at the **Stop Distance** (depth stop).  
Make sure that the needle stop is inside the stop adjustment tool **(4)**.
3. Tighten the needle stop to hold it at the desired position.



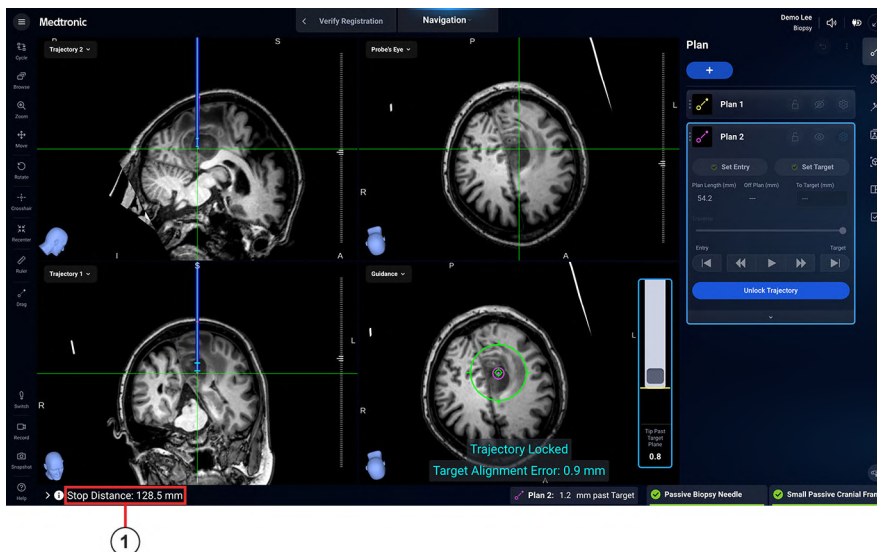
- 1 Stop adjustment tool: Use this to position either the needle tip or cutting window at the **Stop Distance** (depth stop).
- 2 Example: Needle tip at 140 mm **Stop Distance** (depth stop)
- 3 Example: Center of cutting window at 140 mm **Stop Distance** (depth stop)
- 4 Correct placement of the needle stop inside the stop adjustment tool

**Note:** If you decide to perform the biopsy using the center of the cutting window, then set the needle stop so that the cutting window center matches the stop distance. In this case, the navigation will show that the tip of the needle is approximately 6-7mm past the target point. Confirm the target indicator line in the clinical application is centered in the cutting window.



- 1 Example of the center of the cutting window at the **Stop Distance**. Needle tip is past the target.

**Note:** If you decide to perform the biopsy using the needle tip, then set the needle stop so that the tip matches the stop distance. Confirm the target indicator line in the clinical application is aligned with the needle tip.



1 Example of the tip of the needle at the **Stop Distance**

### Navigating the biopsy (optical only)

**Caution:** When the notches at the top of the biopsy needle and inner cannula are aligned, the cutting window is open. To close the cutting window, rotate the inner cannula 180° from the open position. Visually confirm that the cutting window is closed before advancing or retracting the needle through the needle adapter. Moving the needle while the cutting window is open could cause unnecessary patient trauma.



1 Notches are aligned, so cutting window is open.

1. Advance the biopsy needle through the adapter (or reducing tube) until the depth stop contacts the top of the adapter (or reducing tube). The system tracks the location of the needle tip and reports the distance to target as the needle advances.

**Note:** The system displays a graphical representation of the needle cutting window in the guidance view and trajectory views to assist in visualization of the cutting window position.

2. Acquire the biopsy sample by opening the cutting window, drawing suction, and closing the cutting window.

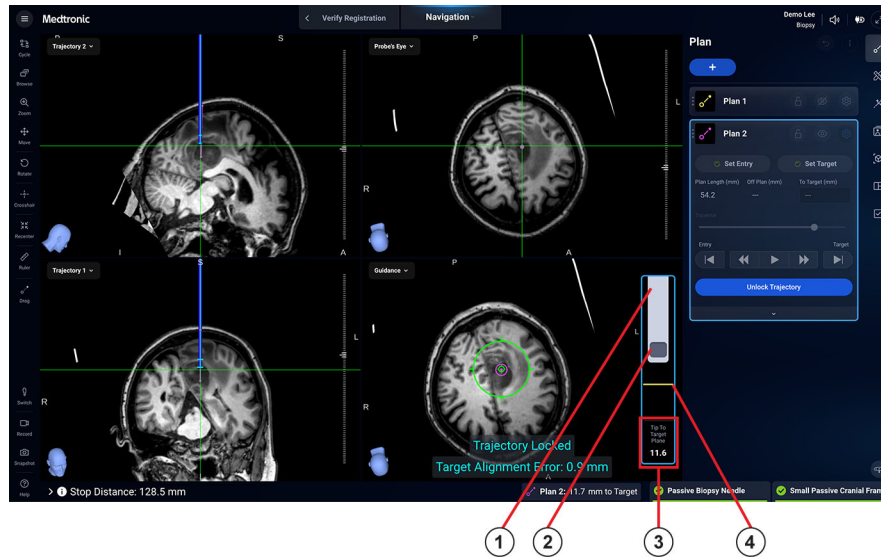
**Note:** Suction may be drawn using the aspirator tube and a syringe, or by attaching a syringe directly to the inner cannula hub.

3. Make sure the biopsy window is closed, and then retract the inner cannula of the biopsy needle to retrieve the sample.
4. If another sample is needed, slide the inner cannula of the biopsy needle into the outer needle sheath. Repeat the previous steps to acquire a biopsy sample.

**Note:** Do not remove the guide assembly from the patient until a satisfactory sample is acquired.

5. When finished, make sure the inner cannula is inserted in the biopsy needle and that the cutting window is closed.
6. Remove the biopsy needle from the patient.
7. Unscrew the Navigus™ base from the patient's skull, if you are using the Navigus™ trajectory guide kit.

**Warning:** After removing the base, remove loose tissue and loose screw or base debris from the patient.



- 1 Needle moving toward target
- 2 Cutting window
- 3 Distance to target
- 4 Target

### Working with a microscope (optical only)

To use a microscope, the microscope must be added to the procedure in the **Equipment** task. See “Setting up and managing equipment” to add the microscope to the procedure.

**Warning:** Confirm navigational accuracy using anatomical landmarks and accuracy checkpoints before navigating.

**Warning:** When navigating with a microscope, keep the crosshair in the same ocular that was used to calibrate the microscope.

- See “Displaying video input if you are setting up microscope output for the first time.”
- To display the microscope output in the image views, tap the Image Selection drop-down menu and select **Video**.
- If you use the video injection to view the Stealth AXIS™ clinical application in a microscope ocular, inject video into the same ocular that was used to calibrate the microscope.
  - First, display a model, tracts, or plan from the HUD options. Then when a 3D model, tracts or plan is shown in the 3D views on the Stealth AXIS™ clinical application, it is also shown in the microscope ocular.
  - Models always display in glass mode and tracts always display as lines.
  - Plan lines are shown as solid lines. The segment of the plan above the current focal plane is semi-transparent. The segment of the plan below the focal plane is opaque.

For additional information about working with a microscope, refer to the instructions that accompany the microscope tracker.

For additional information on using the HUD controls, see the *ZEISS KINEVO® Microscope Interface for Stealth AXIS™ Surgical Systems with Cranial Quick Reference Guide*.

## Export

### Exporting patient data

Follow these instructions to export patient data (including images, merges, and plans), snapshots, or video to media or network locations:

**Note:** The system is not meant for long-term data storage. Patient data is to be transferred to the system for the patient case. After case completion, data can be transferred to a PACS or another customer-managed storage device for long-term storage, or the data can be deleted. Data transfer and handling is managed by the customer.

1. Tap the Menu button and tap **Export**.
2. Select one or more patients whose data you want to export.
3. Select an export destination, such as a USB, CD/DVD-ROM, the filesystem, PACS, or another configured Stealth AXIS™ system from the **Destination** drop-down menu.  
Select a location for the export, if necessary.  
See “Adding a PACS server” to set up a PACS server.
4. Tap the tab for the type of data you want to export.
  - Tap the **All Patient Data** tab to export all data associated with the patient (for example, CT and MR images, snapshots, and videos) and planned data (for example, annotations, plans, and merge data).

**Note:** This selection archives CT and MR images in the Stealth format. Select the options described in the next step to modify aspects of the exported data.

**Note:** To view and edit snapshots before exporting, see “Viewing snapshots.”

- Tap the **Snapshots and Videos Only** tab to export snapshots and videos only.
  - Tap the **Patient Coordinate File Only** tab to export only the patient coordinate file.
5. Select the desired export options. The options that appear depend on the data type selection and the data that exists for the selected patients.

#### DICOM

- If desired, tap the **Description** field and change the text. This text appears in the Patient Name column of the export.
- To anonymize the export, turn on the **Anonymize** toggle and enter prefix text in the **Anonymization Prefix** field. When used, this text replaces the Description text in the Patient Name column of the export and changes patient identifying information in the DICOM tags of the export.

**Note:** Anonymizing does not delete identifying information in the images themselves, such as user-entered identifying information in any text input fields.

**Note:** Snapshots are not anonymized.

- Select **Preserve Series Description** to retain the text in the **Description** field in the export. Clear the option to change the Series Description of the export to the entered **Anonymization Prefix**.
- To add an optional accession number attribute to the export, turn on the **Define Accession Number** toggle and enter an accession number in the **Accession Number** field.
- Select **Include Original DICOM Format Images** to add the original DICOM images to the patient data export. When the **Anonymize** toggle is on, the original DICOM image files are anonymized.

#### File Options

- Select **Include Ultrasound Sequences** to include ultrasound sequences, in mp4 format, in the export.
- If the selected patients have snapshots or videos, they will be included in the export by default, in jpg and mp4 formats respectively. Turn off the **Include Snapshots and Videos** toggle to exclude these files. If you also want to include snapshots and videos in DICOM format, select **Add Snapshots and Videos in DICOM Format**.

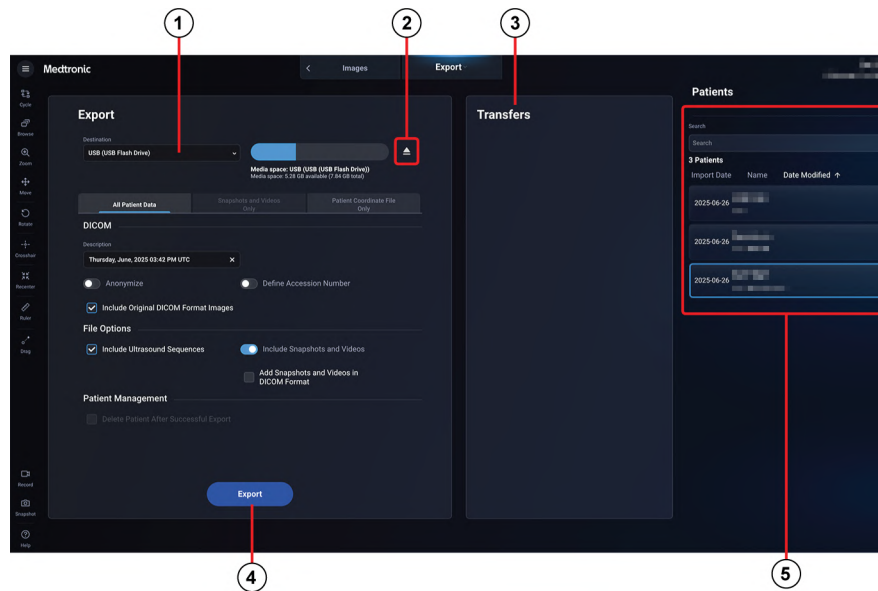
#### Patient Management

Select **Delete Patient After Successful Export** to automatically delete patient data when the export successfully completes.

6. Tap **[Export]**.  
The **Transfers** list shows export progress.
  - Tap **[Cancel]** to stop an export that is in progress.
  - Tap **[X]** to close a patient name whose transfer is complete.

Unfinished transfers will be canceled if you navigate away from the **Export** task while an export is in progress.

**Note:** When you export to a USB drive, the Eject button is displayed on the clinical application screen. After the file transfer is complete, tap the Eject button. If the USB drive has a light, wait for the light to stop flashing before you remove the USB drive from the system.



- 1 Destination drop-down menu
- 2 Eject button
- 3 Transfers
- 4 [Export]
- 5 Patients

## Viewing snapshots

Snapshot viewer allows you to view and edit snapshots before exporting them.

1. Tap the Menu button.
2. Tap **[Patient Admin]**.
3. Tap the **Snapshots** tab.
4. Double-tap the snapshot that you want to view or edit. The snapshot enlarges.
  - To rename the snapshot, enter a new name in the **Rename** field, tap **[Rename]**, and then tap **[Confirm]**.
  - To delete the snapshot, tap **[Delete]**.

## Managing patients and images

To clone, move, delete, or combine selected patient information, use the **Patient Admin** task. Follow these steps to open the **Patient Admin** task:

1. Tap the Menu button.
2. Tap **[Patient Admin]**.

The **Patient Admin** task displays.

## Cloning patients

Cloning a patient includes images, planned data, and any modifications you have made during a session.

Follow these steps to clone selected patients in the **Images** task:

1. Find the desired patient in the **Patients** list.  
**Note:** To search for a patient, enter the patient name or MRN into the text field.
2. If the patient card is not selectable, it is currently in use by someone else. Tap the Clone button.  
**Note:** This is only applicable when using the planning server.
3. If the patient card can be selected, it is not currently in use. Select the patient card to expand it and tap **[Clone Patient]**.
4. On the **Clone Patient** dialog, enter a unique patient name and select the check boxes to include edit history, snapshots, and videos.
5. Tap **[Clone]**.

The new patient card displays in the **Patients** list.

**Note:** You cannot recombine edits on a cloned patient with the original patient.

## Combining patients

Follow these steps to combine two or more selected patients in the **Patient Admin** task:

1. Select patients to combine.  
**Note:** To search for a patient, enter the patient name or MRN into the text field.
2. Tap **[Combine Selected Patients]**.
3. Select the target patient into which the selected patients will be combined.
4. Tap **[Combine]**.

## Renaming patients

Follow these steps to rename selected patients in the **Patient Admin** task.

**Note:** If a patient is used in the current procedure, that patient cannot be renamed.

1. Select a patient.  
**Note:** To search for a patient, enter the patient name or MRN into the text field.
2. Tap the Settings button next to the patient name.
3. In the **Name** field, enter the new patient name.

## Deleting a patient

Follow these steps to delete a patient from the system.

**Note:** If a patient is used in the current procedure, that patient cannot be deleted.

**Note:** All data associated with the patient, including videos and snapshots, are deleted.

In the **Patient Admin** task:

1. Select a patient.  
**Note:** To search for a patient, enter the patient name or MRN into the text field.
2. Tap the Settings button to expand the card.
3. Tap **[Delete]**.
4. Tap **[Delete]** on the dialog to confirm you want to remove the patient data from the system.

The patient data is removed from the system.

### Deleting multiple patients

In the **Patient Admin** task, follow these steps to delete more than one patient:

**Note:** If a patient is used in the current procedure, that patient cannot be deleted.

**Note:** All data associated with the patient, including videos and snapshots, are deleted.

1. Tap the **More** button at the top right of the **Patients** list.
2. To delete all patients, tap **[Delete All]**. On the confirmation dialog, tap **[Delete All]**
3. To delete multiple patients, tap the **[Delete Multiple]** button. In the Patients dialog, select the Delete button next to each desired patient. Click **[Done]** when finished.

All patients and patient data are cleared from the system.

### Moving images from one patient to another

Follow these steps to move images for selected patients in the **Patient Admin** task:

1. Select patients.  
**Note:** To search for a patient, enter the patient name or MRN into the text field.
2. In the **Images** tab, tap the patient images that you want to move.
3. Tap **[Move Selected Images]**.
  - a. To move the images to a new patient, enter a new patient name and MRN.
  - b. To move the images to an existing patient, select the patient name from the **Patients** drop-down menu.
4. Tap **[Move Selected Images]**.

### Deleting images

Follow these steps to delete images for selected patients in the **Patient Admin** task.

**Note:** Deleting the only image for a patient deletes the patient.

1. Select patients.  
**Note:** To search for a patient, enter the patient name or MRN into the text field.
2. In the **Images** tab, tap the patient images that you want to delete.
3. Tap **[Delete Selected Images]**.
4. Tap **[Delete]** to confirm.  
**Note:** All data associated with the patient, including videos and snapshots, are deleted.

## Exiting a case

Follow these instructions to exit a case:

1. Tap the Menu button.
2. Tap the Home button.

The **Select Procedure** screen displays.

## Exiting the clinical application

Follow these instructions to exit the clinical application:

1. Tap the Menu button.
2. Tap the Power button.
3. Tap **[Log Out]**.
4. Tap **[Log Out Now]**. Tapping **[Cancel]** will abort the action.

The login screen displays.

**Note:** If you do not tap **[Log Out Now]**, the system will automatically log out after 25 seconds.

## Shutting down the system

Follow these instructions to shut down the system:

**Caution:** Ensure that the workstation cart is shut down using the clinical application. Shutting down the system by disconnecting from an electrical outlet and letting the batteries drain or holding the power button for more than 6 seconds is not recommended and could damage the workstation cart or put the cart into a condition that requires maintenance.

Once logged into the clinical application, to shut down the system:

1. Tap the Menu button.
2. Tap the Power button.
3. Tap **[Power Down]**.  
The clinical application prompts you to confirm.
4. Tap **[Power Down Now]**. Tapping **[Cancel]** will abort the action.  
The system will fully shut down.

**Note:** If you do not tap **[Power Down]**, the system will automatically shut down after 25 seconds.

To power down the system from the login screen:

1. Tap the Power button in the lower-right corner to access the drop-down menu.  
**Note:** The Power button does not display during remote planning sessions.
2. Tap **[Power Down]**.
3. The system prompts you to confirm.
4. Tap **[Power Down Now]**. Tapping **[Cancel]** will abort the action.
5. The system will fully shut down.  
**Note:** If you do not tap **[Power Down]**, the system will automatically shut down after 25 seconds.

## Restarting the system

Follow these instructions to restart the system.

1. Navigate to the login screen by following the instructions in “Exiting the clinical application.”
2. Tap the Power button in the lower-right corner to access the drop-down menu.  
**Note:** The Power button does not display during remote planning sessions.
3. Tap **[Restart]**.  
The system prompts you to confirm.
4. Tap **[Restart Now]**. Tapping **[Cancel]** will abort the action.  
**Note:** If you do not tap **[Restart Now]**, the system will automatically restart after 25 seconds.

## Troubleshooting

### Image import problems

Find the symptom in the table below and follow the troubleshooting steps:

| Symptom                                                                          | Troubleshooting steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| You imported a series and slices are missing in the area of anatomical interest. | <ul style="list-style-type: none"> <li>• If you imported the series from a PACS, ask Radiology to make sure that all slices are present in the original scan.</li> <li>• Ensure that slice spacing and thickness remain the same throughout the scan. Sometimes gaps occur because the spacing and thickness changes in the scan based on how radiology acquires or processes the scan.</li> <li>• If you imported the series from media, use a DICOM viewer to make sure that all slices are present in the original scan.</li> <li>• If all slices are in the original scan, delete the imported series and reimport it.<br/>To delete the series, see “Deleting images.”</li> <li>• If the problem persists after you reimport the series, contact Medtronic technical services.</li> </ul> |

| Symptom                                                                                                                                                      | Troubleshooting steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The images do not contain the entire anatomical region of interest needed for the procedure.                                                                 | Adjust the scanner or the patient so that as much as possible of the region of interest is in the scanner's field of view and then rescan the patient.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| You need to correct the orientation of a series due to mislabeling in the scanner.                                                                           | Tap <b>[Correct]</b> , then tap the options to correct the orientation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| The level and width adjustments seem too sensitive (for example, they change from very light to very dark, and it may be difficult to show defined anatomy). | <ol style="list-style-type: none"> <li>1. In the <b>Images</b> task, tap the arrow under <b>Series</b> to display details about the patient series, including the original bit depth.</li> <li>2. In the <b>Patient Images</b> list, tap the Settings button on the patient series.</li> <li>3. Tap the <b>2D</b> tab.</li> <li>4. Tap the <b>Bit Depth</b> drop-down menu and select a bit depth that is lower than the original.</li> <li>5. Adjust the <b>Level</b> and <b>Width</b> sliders.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| You need to move a series from one patient name to another.                                                                                                  | See "Moving images from one patient to another" for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| You need to change the modality of a series.                                                                                                                 | Select the patient series in the <b>Patient Images</b> list, tap the Settings button, and select the correct modality.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| A warning message that includes: "Minimum of three slices required."                                                                                         | Ask radiology to review the scan to comply with the Imaging Protocol.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| A warning message that includes: "This exam is restricted from use in the tasks below: Registration."                                                        | If you want to include the series in image views for planning or navigation, you can manually merge the series with a series that is acceptable for registration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| A warning message that includes: "This may affect functionality or navigation accuracy of the system."                                                       | Ask radiology to reformat the scan to comply with the Imaging Protocol.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| When you try to search for images on a PACS, a warning message displays.                                                                                     | <p>After each of the following steps, try your search again to determine whether the step resolved the issue:</p> <ol style="list-style-type: none"> <li>1. Make sure the network cable is plugged in on both ends if using a wired configuration.</li> <li>2. If using wireless connectivity, ensure that an adequate signal is available.</li> <li>3. See "Running DICOM tests."</li> <li>4. Check the network settings in the Admin application: <ol style="list-style-type: none"> <li>a. Tap the Menu button.</li> <li>b. Tap <b>[Admin]</b>.</li> </ol> </li> <li>5. Make sure the ethernet port at the facility has an active network connection.</li> <li>6. Make sure there are no other devices on the facility's network that are using the same IP address and network configuration.</li> <li>7. If using wireless connectivity, ensure that adequate signal is available.</li> </ol> <p>If the problem persists, contact Medtronic technical services.</p> |

## Merge tips

When you merge images, the merged set is always used in the **Planning** and **Navigation** tasks.

| Goal                                                                                    | Steps                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To view only one of the merged images in the <b>Planning</b> or <b>Navigation</b> tasks | Tap <b>[Images]</b> , then tap the Show button on all the images that you do not want to view. When the Show button has a line through it, that image is not visible.                                                                                                  |
| To remove an image from the merged set                                                  | <p>Go to the <b>Images</b> task and turn off the <b>Select</b> toggle on the images that you want to remove.</p> <p><b>Note:</b> If you deselect the image that was previously the reference image, your planning data associated with that image will be removed.</p> |

## O-arm™ system scan troubleshooting (optical only)

Find the symptom in the table below and follow the troubleshooting steps:

| Symptom                                                                                  | Troubleshooting steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| An indicator message says, "O-arm System Not Communicating"                              | Make sure that the network cable between the O-arm™ Mobile View Station and the Stealth AXIS™ system is securely connected on both ends. If the problem persists, contact Medtronic technical services.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| An indicator message says, "Patient Tracker Not Visible" or "O-arm Tracker Not Visible." | Adjust the camera angle until the patient reference frame and the wireless tracker are both visible. If necessary, adjust the O-arm™ gantry tilt.<br><b>Note:</b> The clinical application allows instrument tracking over an extended region of the Stealth AXIS™ camera volume (up to 3.0 m [9 ft 10 in] from the camera). Ensure the instrument is within the camera volume.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| An indicator message says, "Patient movement detected."                                  | Make sure that the O-arm™ system, Stealth AXIS™ camera, and patient reference frame are motionless. See the <i>O-arm™ Imaging System User Manual</i> for information on pausing respiration for O-arm™ system scans.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| An error message says, "O-arm communication error."                                      | <ul style="list-style-type: none"> <li>• Make sure that the network cable between the O-arm™ Mobile View Station and the Stealth AXIS™ system is securely connected on both ends.</li> <li>• Check the O-arm™ system for the image and try to manually resend the image to the clinical application.</li> <li>• If the problem persists, remove the O-arm™ system from the procedure in the <b>Equipment</b> task and use a different registration method.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            |
| The scan did not completely transfer to the Stealth AXIS™ system.                        | <p>Try the following steps before attempting another scan:</p> <ul style="list-style-type: none"> <li>• Try to manually resend the image to the Stealth AXIS™ clinical application, and restart both the O-arm™ system and the Stealth AXIS™ clinical application.</li> <li>• Make sure that the network cable between the O-arm™ Mobile View Station and the Stealth AXIS™ system is securely connected on both ends.</li> <li>• Verify that the Stealth AXIS™ camera status indicator is green and that the O-arm™ system, Stealth AXIS™ camera, and patient reference frame are motionless.</li> <li>• Press the <b>3D</b> button on the O-arm™ pendant.</li> <li>• Make sure that both the O-arm™ tracker and the patient reference frame are visible in the tracking view.</li> </ul> <p>If the problem persists, contact Medtronic technical services.</p> |
| An indicator message says, "Unregistered O-arm exam received."                           | <ul style="list-style-type: none"> <li>• Merge the unregistered exam with a registered exam in the <b>Images</b> task.</li> <li>• Re-take the O-arm™ system scan.</li> <li>• If the problem persists, remove the O-arm™ system from the procedure in the <b>Equipment</b> task and use a different registration method to register the patient to a recent O-arm™ system scan or to another image in the <b>Images</b> task.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Images contain distortion.                                                               | <p>Make the following changes and then rescan the patient:</p> <ul style="list-style-type: none"> <li>• Remove metal and nonessential objects from the scanner's field of view (for example, remove instruments, extra tubing, and extra drapes).</li> <li>• Adjust the scanner or the patient so that as much as possible of the anatomy of interest is in the scanner's field of view and close to the center of the field of view.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Images do not contain the anatomy of interest.                                           | <ul style="list-style-type: none"> <li>• Make sure that the correct exam is selected.</li> <li>• Adjust the scanner or the patient so that as much as possible of the anatomy of interest is in the scanner's field of view and then rescan the patient.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| A recent scan (less than 24 hours old) is not in the <b>Patient Images</b> list.         | Tap <b>[View Recent Images]</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| The automatic O-arm patient registration seems inaccurate.                               | <ul style="list-style-type: none"> <li>• Make sure that the correct exam is selected.</li> <li>• Re-take the O-arm™ system scan.</li> <li>• If the problem persists, remove the O-arm™ system from the procedure in the <b>Equipment</b> task and use a different registration method to register the patient to a recent O-arm™ system scan or to another image in the <b>Images</b> task.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Planning and Navigation tips

Find the goal in the table below and follow the steps:

| Goal                                                                                                  | Steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To display dots instead of crosshairs in 2D views                                                     | <p>The following methods display crosshairs in 2D views:</p> <ul style="list-style-type: none"> <li>• Update preferences <ol style="list-style-type: none"> <li>1. Tap the Menu button, and then tap <b>[Preferences]</b>.</li> <li>2. Select the procedure.</li> <li>3. Tap the <b>Views</b> tab.</li> <li>4. Find <b>Crosshair Visibility</b> and tap <b>Dot</b>.</li> </ol> </li> <li>• Tap <b>[Crosshairs]</b>, and then tap <b>[Dot]</b>.</li> <li>• Press &lt;F4&gt; on the keyboard to switch between crosshairs and dots.</li> </ul> |
| To find an existing annotation in the 2D or 3D image views                                            | <ol style="list-style-type: none"> <li>1. Open the <b>Annotations and Alerts</b> drawer and make sure that the annotation is set to visible by tapping the Show button.</li> <li>2. Tap the annotation you want to see.<br/>The crosshairs update to that location.</li> </ol> <p><b>Note:</b> Annotations are not visible in <b>Guidance</b> view.</p>                                                                                                                                                                                      |
| To change an annotation's name or color, or to delete an annotation when those options are not active | <ol style="list-style-type: none"> <li>1. Tap the Lock button to unlock the annotation.</li> <li>2. To delete a single annotation, do one of the following:<br/>Tap the Settings button on the annotation and tap <b>[Delete]</b>.<br/>Swipe an annotation to the right and then tap the Delete button.</li> </ol>                                                                                                                                                                                                                           |
| To update the location of an annotation                                                               | Move the crosshairs where you want to place the new annotation and tap <b>[Set Location]</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Registration troubleshooting (optical only)

Find the symptom in the table below and follow the troubleshooting steps:

| Symptom                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Troubleshooting steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| An error message displays, "The current instrument cannot be used for registration."                                                                                                                                                                                                                                                                                                                                                                                          | <p>Use one of the valid registration instruments:</p> <ul style="list-style-type: none"> <li>• Passive Planar Probe</li> <li>• Touch-n-Go Probe</li> <li>• Active Linear Blunt Probe</li> <li>• Navigus™ Probe</li> <li>• Vertek™ Probe</li> </ul>                                                                                                                                                                                                                                                                                             |
| <ul style="list-style-type: none"> <li>• Registration accuracy is not sufficient or an error message displays, "To proceed to Navigation, the overall registration accuracy must be below 5.0 mm. Continue to collect points."</li> <li>• An error message displays, "Continue tracing over a larger area or add touch points."</li> <li>• An error message displays, "Continue to collect data. Registration error is above the recommended threshold of 5.0 mm."</li> </ul> | <ul style="list-style-type: none"> <li>• If fiducials are available, collect them. If trace registration initialization has not been performed, a minimum of 4 fiducials must be acquired.</li> <li>• Touch the locations of all of the stored landmarks.</li> <li>• Store and touch additional landmarks, if possible.</li> <li>• If you are using trace registration, collect more trace points.</li> <li>• If registration still doesn't initialize, or initializes with insufficient accuracy to proceed, restart registration.</li> </ul> |
| Touch initialization has not occurred.                                                                                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Touch at least 4 stored fiducials or stored anatomical landmarks.<br/>Touch as many stored fiducials and landmarks as possible.<br/>If registration does not initialize, touch the stored fiducials and landmarks a second time.</li> <li>• Collect a set of trace points that covers a region of the head.</li> <li>• If registration still doesn't initialize, or initializes with insufficient accuracy to proceed, restart registration.</li> </ul>                                               |

## Registration troubleshooting (EM only)

Find the symptom in the table below and follow the troubleshooting steps:

| Symptom                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Troubleshooting steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| An error message displays, "The current instrument cannot be used for registration."                                                                                                                                                                                                                                                                                                                                                                                          | <p>Use one of the valid registration instruments:</p> <ul style="list-style-type: none"> <li>• EM Tracer™ Pointer</li> <li>• EM Touch-n-Go Pointer</li> <li>• EM Navigation Pointer</li> </ul>                                                                                                                                                                                                                                                                                                                                                 |
| <ul style="list-style-type: none"> <li>• Registration accuracy is not sufficient or an error message displays, "To proceed to Navigation, the overall registration accuracy must be below 5.0 mm. Continue to collect points."</li> <li>• An error message displays, "Continue tracing over a larger area or add touch points."</li> <li>• An error message displays, "Continue to collect data. Registration error is above the recommended threshold of 5.0 mm."</li> </ul> | <ul style="list-style-type: none"> <li>• If fiducials are available, collect them. If trace registration initialization has not been performed, a minimum of 4 fiducials must be acquired.</li> <li>• Touch the locations of all of the stored landmarks.</li> <li>• Store and touch additional landmarks, if possible.</li> <li>• If you are using trace registration, collect more trace points.</li> <li>• If registration still doesn't initialize, or initializes with insufficient accuracy to proceed, restart registration.</li> </ul> |
| Touch initialization has not occurred.                                                                                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Touch at least 4 stored fiducials or stored anatomical landmarks. Touch as many stored fiducials and landmarks as possible. If registration does not initialize, touch the stored fiducials and landmarks a second time.</li> <li>• Collect a set of trace points that covers a region of the head.</li> <li>• If registration still doesn't initialize, or initializes with insufficient accuracy to proceed, restart registration.</li> </ul>                                                       |
| You cannot collect trace points.                                                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Begin registration more than 1 cm away from patient tracker divot.</li> <li>• Confirm that the patient tracker and registration instrument are actively tracking within the tracking view.</li> <li>• Check the footswitch behavior. To start tracing, either hold the probe steady for 2 seconds or activate the footswitch.</li> </ul>                                                                                                                                                              |

## Troubleshooting instruments (optical only)

Status messages about instruments may appear either on the bottom-left portion of the screen or in the tracking details.

To view tracking details in the **Instruments**, **Registration**, or **Navigation** task, tap on the instrument status bar, the patient reference frame status bar, or the localizer icon. Tap **[Tracking Details]**.

Check the status message for the instrument and follow the troubleshooting steps below:

**Note:** If the troubleshooting steps do not resolve the issue or if the issue persists, call Medtronic technical services.

| Instrument status message                                    | Troubleshooting steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A status message displays, "Instrument not verified."        | Hold the instrument tip in verification divot on the patient reference frame until you hear a chime, or press the footswitch.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| A status message displays, "Instrument verification failed." | <ul style="list-style-type: none"> <li>• Make sure that the tip of the instrument is in the bottom of the verification divot. Keep the instrument perpendicular to the divot.</li> <li>• Check the distance to divot displayed on the screen. If the distance exceeds 2.0 mm, try another instrument.</li> <li>• Check the instrument geometry in the tracking details. If the geometry error exceeds 0.5 mm, try the troubleshooting steps for when an instrument is not tracking.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Instrument is not tracking or instrument has red status.     | <ul style="list-style-type: none"> <li>• Make sure the instrument has been added to your procedure.</li> <li>• In the tracking view, make sure that the instrument's position marker is between the <b>Near</b> and <b>Far</b> marks. If necessary, move the camera so that the markers are not near the edges of the volume.</li> <li>• Rotate each reflective sphere on its mounting post, making sure the sphere is securely seated on the post.</li> <li>• Check for damage to the reflecting surface of the spheres. If a sphere is damaged, replace it.</li> <li>• Wipe away any dirt or fluid on the reflective spheres. Dry the spheres with a dry, lint-free cloth.</li> <li>• Clear the line of sight between the camera and the instrument.</li> <li>• Clean the camera lenses with a camera cloth.</li> <li>• Cover surfaces that may reflect light into the camera. Temporarily turn off or redirect the operating room lights.</li> <li>• Remove wrinkles in the drape covering the spheres on the patient reference frame.</li> </ul> <p>If the instrument status remains red, the instrument may be bent or damaged. Try another instrument.</p> |

## Troubleshooting instruments (EM only)

Status messages about instruments may appear either on the bottom-left portion of the screen or in the tracking details.

To view tracking details in the **Instruments**, **Registration**, or **Navigation** task, tap on an instrument name and the tracking view opens. Tap **[Open Tracking Details]**.

Check the instrument's color status and status message, and follow the troubleshooting steps below:

**Note:** If the troubleshooting steps do not resolve the issue or if the issue persists, call Medtronic technical services.

| Instrument status message                                                                                                     | Troubleshooting steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The instrument displays red status in the tracking view without a geometry error metric.                                      | <ul style="list-style-type: none"> <li>• Make sure that the tracked part of the instrument is no more than 39.0 cm (15.4 in) from the face of the side emitter or no more than 42.5 cm (16.7 in) from the face of the flat emitter.</li> <li>• Make sure that the tracker is at least 8 cm (3 in) from the side emitter or 5 cm (2 in) away from the flat emitter.</li> <li>• Move any metal objects farther away from the instrument.</li> <li>• Try a different port on the EM instrument interface.</li> <li>• Make sure that the <b>Emitter</b> toggle is turned on in the clinical application.</li> <li>• Turn the emitter off and on in the clinical application.</li> </ul> |
| The instrument displays orange status in the tracking view.                                                                   | <ul style="list-style-type: none"> <li>• The instrument is initializing with the clinical application. Allow at least one minute for the clinical application to establish communication with the tracking system and emitter.</li> <li>• Try plugging the instrument into a different port on the EM instrument interface.</li> </ul>                                                                                                                                                                                                                                                                                                                                              |
| The instrument displays red status in the tracking view with a high geometry error metric.                                    | <ul style="list-style-type: none"> <li>• Make sure that the tracked part of the instrument is no more than 39.0 cm (15.4 in) from the face of the side emitter or no more than 42.5 cm (16.7 in) from the face of the flat emitter.</li> <li>• Move any devices that emit electromagnetic noise (such as electric drills and electrocautery devices) away from the instrument.</li> <li>• Move any metal objects (such as the endoscope's camera) farther away from the instrument.</li> </ul>                                                                                                                                                                                      |
| A banner message displays "Localizer disconnected." The emitter or EM instrument interface also displays red status.          | <ul style="list-style-type: none"> <li>• The localizer has not established communication with the system. Make sure that the emitter and EM instrument interface cables are connected to the system.</li> <li>• Disconnect and reconnect the emitter and EM instrument interface cables to the system.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                   |
| The clinical application displays a "faulted" error message next to the instrument name, emitter, or EM instrument interface. | <ul style="list-style-type: none"> <li>• Disconnect and reconnect the emitter and EM instrument interface cables to the system.</li> <li>• The emitter, EM instrument interface, or instrument has encountered a fault.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| The emitter or the EM instrument interface display orange status.                                                             | <ul style="list-style-type: none"> <li>• The devices are initializing with the clinical application. Allow at least one minute for the clinical application to establish communication with the tracking system and emitter.</li> <li>• Disconnect and reconnect the emitter and EM instrument interface cables to the system.</li> <li>• Restart the clinical application.</li> </ul>                                                                                                                                                                                                                                                                                              |
| The emitter or the instruments display gray status.                                                                           | The EM system is not navigating. Make sure that the emitter and the EM instrument interface are connected to the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| A banner message displays, "Instrument not connected."                                                                        | Plug an EM instrument into the EM instrument interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| "Current instrument cannot be used for registration. Use a valid probe for registration."                                     | Use the EM Registration Probe.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| An instrument is plugged into a port and it does not display as red or green in the tracking details.                         | <ul style="list-style-type: none"> <li>• Unplug and replug the instrument. If the problem persists, try a new instrument.</li> <li>• Try plugging the instrument into a different port on the EM instrument interface.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| All instruments display red status.                                                                                           | Check for interference near the emitter. Make sure that no cables are draped over the emitter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Troubleshooting when using the Stealth Autoguide™ system

The following information is specific to troubleshooting when using the Stealth Autoguide™ system with the Cranial clinical application. See the *Stealth Autoguide™ System Manual* for more details.

In the following table, find the symptom and then follow the troubleshooting steps.

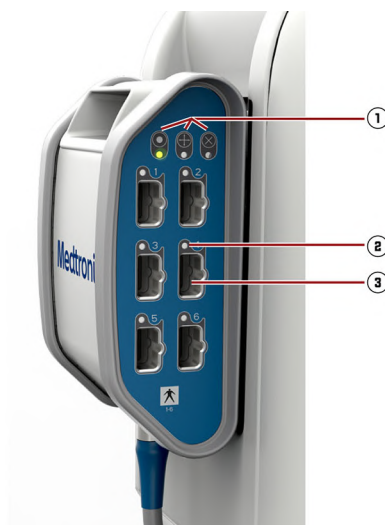
| Symptom                                                                                                                                | Troubleshooting steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The name <b>Autoguide Tracker</b> does not appear in the bottom-right corner of the Stealth AXIS™ Cranial clinical application screen. | <p>Follow these steps to add the Autoguide Tracker to the list of instruments in the procedure:</p> <ol style="list-style-type: none"> <li>1. Go to the <b>Instruments</b> task in the clinical application.</li> <li>2. Search for <b>Autoguide Tracker</b> in the Instruments list, and tap <b>[ADD]</b> to add the tracker to the Instruments in Procedure list.</li> </ol> <p>If the Autoguide Tracker is not available in the <b>Instruments</b> list, the Stealth Autoguide™ software license may not be installed. To check for the license, tap the Menu button, and then tap <b>About This Stealth</b>. If <b>Cranial Stealth Autoguide</b> is not in the list of <b>Installed Licenses</b>, contact Medtronic technical support.</p> |

## EM connection problems (EM only)

When you select an EM procedure in the Cranial clinical application the instrument interface receives power from the system. See the following table for information about each status indicator.

**Note:** When the EM instrument interface first receives power, some or all of the LEDs may light up briefly.

| Interface LED symbols                                                               | LED descriptions                                                                                                                    | Actions to take if the LED is not lit                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | Green power LED: Indicates when the system is supplying power to the EM instrument interface.                                       | <ul style="list-style-type: none"> <li>• Make sure that the cart is powered on.</li> <li>• Make sure that the EM instrument interface cable is connected to the cart.</li> <li>• Make sure that an EM procedure has been selected in the clinical application.</li> </ul>                                                                                                                                                                                               |
|  | Green navigation LED: Indicates when the EM system is in the navigation mode.                                                       | <ul style="list-style-type: none"> <li>• Make sure that the EM instrument interface cable is connected to the cart.</li> <li>• Make sure that an EM procedure has been selected in the clinical application.</li> <li>• Make sure that the emitter is turned on in the clinical application. The emitter is turned on by default, but the tracking details area in the clinical application includes a button to turn the emitter off. See "Emitter on/off."</li> </ul> |
|  | Yellow fault LED: Indicates, when lit, that the system has encountered a fault. Check the software for more diagnostic information. | If the fault LED is not lit, the system has not encountered a fault. No action is required.                                                                                                                                                                                                                                                                                                                                                                             |



- 1 EM instrument interface status indicator LEDs
- 2 Instrument indicator LED (6 total)
- 3 Instrument port (6 total)

The instrument status LED color relays real-time information about EM instruments and the EM system controller. When you plug in an EM instrument, the LED is orange until the system is ready to track the instrument. When the system is ready to track the instrument, the LED turns green.

| Instrument indicator LED color                                                                                                                                            | Actions to take                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Green - Indicates that the instrument is properly connected and can be tracked by the system.                                                                             | No action needed.                                                                                                                                                                                                                                 |
| Orange - Indicates that the system has detected the instrument, but the system is not ready to track the instrument. When the system is ready, the indicator turns green. | Instrument initialization can take several seconds. If the problem persists, plug the instrument in to a different port, or plug in a different instrument.                                                                                       |
| Not lit - Indicates that no instrument has been detected. Either no instrument is plugged in, or a failure has occurred.                                                  | Plug in an instrument, or if an instrument is already plugged in, make sure that the instrument connector is properly seated in the port. If the problem persists, plug the instrument in to a different port, or plug in a different instrument. |

## Battery backup

The workstation cart will attempt to run on backup battery power when it is not plugged into an electrical outlet. An audible indicator will sound when the system is running on battery backup, and it will increase in frequency when the battery is low. The battery status icon is displayed in the upper-right corner of the clinical application screen. After a procedure is selected, tap the battery status icon to open the **System Status Menu** and view more information about the battery status. The **System Status Menu** can be accessed during any task except while in the **Procedure Select** task.

One of the following battery icons displays in the upper-right corner of the clinical application screen:

| Battery icon                                                                     | Color           | Description                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | White           | The workstation cart is plugged into an electrical outlet and is charging.                                                                                                                                                                               |
|  | White           | The system's battery backup is in use and has a charge greater than 50%.                                                                                                                                                                                 |
|  | White           | The system's backup battery is in use and has a charge greater than 25% and less than 50%.                                                                                                                                                               |
|  | White           | The system's backup battery is in use and has a charge greater than 10% and less than 25%.                                                                                                                                                               |
|  | Flashing orange | The system's battery backup is in use and has a charge less than or equal to 10%.                                                                                                                                                                        |
|  | Flashing red    | The system's battery backup service has detected an error or the workstation cart battery has no charge. Unplugging the system's power cord from the electrical outlet may result in immediate system shutdown.<br>Contact Medtronic technical services. |

**Note:** Battery icons do not display in the clinical application on the Planning Station or in a remote planning session.

## Accessing remote support

Remote Support is a digital service that enables Medtronic to provide features such as: remote diagnostics, subcomponent level diagnostic alerts, and remote troubleshooting support using a real-time-screen-share feature. Remote Support requires an active, secured Internet connection between the Covered Product and Medtronic for connectivity. The screen-share function requires the person operating the system to initiate a screen-share session by physically opening the Remote Support application with the assistance of Medtronic Technical Support.

If requested to do so by Medtronic technical services, access remote support. For detailed information about accessing a remote support session, see the *StealthAXiS™ IT Guide*.

**Note:** Remote support is unavailable when using the remote planning feature.

## Appendixes

### A: Using image merge

The image merge feature is only available when the Cranial StealthMerge™ license is installed. Image merge enables you to merge multiple images from the same patient.

**Warning:** Merging patient images manually may impact the overall system accuracy. Always visually verify the accuracy of each merge before using it for planning and navigation. Failure to confirm accuracy can result in inaccurate navigation.

**Warning:** In some cases, dissimilar image modalities may not accurately merge with one another. Images with similar anatomical details are likely to merge accurately, whereas images with significantly different details may not merge accurately in every case. For best results:

- When merging more than two images, use the highest contrast image as the reference image. For example, use a pre-operative, high resolution, diagnostic-quality MR-T1 or CT as the reference image.
- Make sure that a significant portion of the skull base is visible in all images so that there are enough anatomical details to complete an accurate merge.

You can merge images in the **Merge Images** task. If you load or acquire more images later in the procedure, you can return to the **Merge Images** task to merge additional images. In the **Images** task, turn on the **Select** toggle in an image series to add the images to the merge set. To remove an image series from the merge set, turn off the **Select** toggle in the image series.

To merge multiple images from the same patient:

1. Tap the **[Merge Images]** task.

The **Merge Images** screen appears.

The clinical application automatically selects a reference image and starts merging the first image in the **Images** list to the reference image. To change the reference image, tap the Settings button on the image series and tap **[Set as Reference]**.

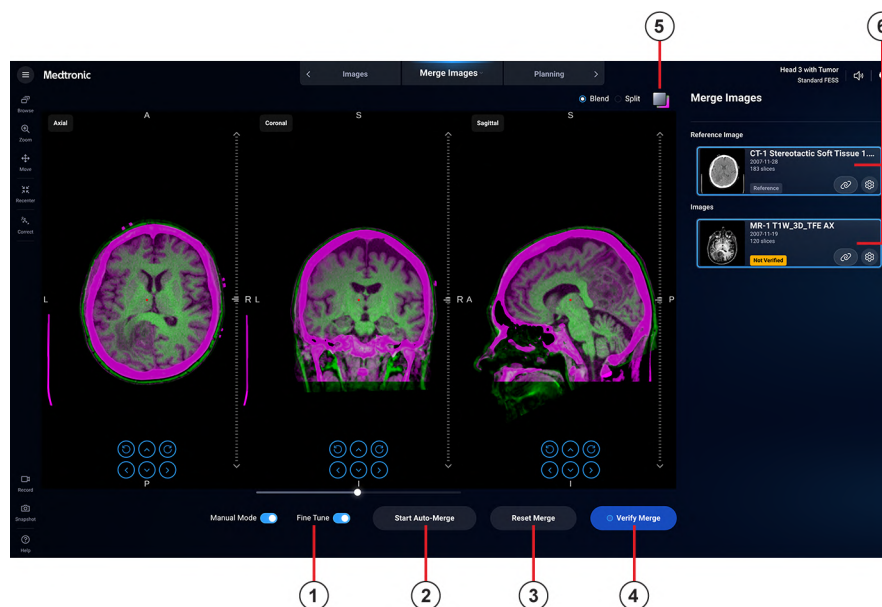
**Note:** If you are merging an O-arm™ system image with a pre-operative image, choose the pre-operative image as the reference image.

**Warning:** If you change the reference image after you merge images, the clinical application deletes any existing merges of images that had been automatically or manually merged (such as CT and MR images). You must then re-merge images to the new reference. Changing the reference image does not delete pre-merges or the diffusion merge. Failure to re-merge images to the new reference can result in the inability to plan and navigate.

2. Check the accuracy of each merged exam. To check merge accuracy, tap **Blend** or **Split** and scroll through the slices. Tap the Color button when in the split display to visually enhance features. The reference image will display magenta.
  - If the accuracy of the merge is sufficient for the procedure, tap **[Verify Merge]**.
  - If you want to fine tune the merge, tap **Manual Mode** and follow these steps:
    - a. Tap **Fine Tune**. Directional arrows and rotate buttons appear in the Axial, Coronal, and Sagittal views.
    - b. Tap the Color button when in the split display to visually enhance features.
    - c. Use the directional arrows to move the image 0.1 mm at a time.
    - d. Use the rotate buttons to rotate the image 0.1 degree at a time.
    - e. When the image to merge is aligned with the reference image, tap **[Verify Merge]**.

**Note:** You can also auto-merge again after manually adjusting. The auto-merge will include the manual adjustments as the new starting position.

- If the merge is not aligned correctly, tap **[Start Auto-Merge]**. Alternatively, you can tap **[Reset Merge]** and turn on **Manual Mode**.
  - a. Drag the image to merge into approximate alignment with the reference image.
  - b. Tap **[Start Auto-Merge]**.
- 3. Visually confirm the datasets match, then tap **[Verify Merge]**. The next merged image displays for your verification. When you merge images, the merged set is always used in the **Planning** and **Navigation** tasks. See “Adjusting views for images and models” if you want to change which images are overlaid together in the display.



- 1 Fine Tune
- 2 [Start Auto-Merge] switches to [Stop Auto-Merge] while merging
- 3 [Reset Merge]
- 4 [Verify Merge]
- 5 Color button
- 6 Selected images for merge with Link button

### Pre-merging non-anatomical images

If there are diffusion images, see the *Stealth Tractography User Manual* for merging diffusion images.

If an image series does not contain anatomical images (such as a PET or fMRI), the clinical application may automatically associate the series with an anatomical image with the same frame of reference. If the clinical application does not automatically associate the series, follow these steps to link the images:

1. Tap the image card for the image that will be the reference image.
2. Tap the Link button and select **[Pre-merge]**.
3. Tap the image card for the image you want to pre-merge to the reference image.
4. Tap **[Pre-merge]**.
5. Verify that the match between them looks correct.

To unlink pre-merged images, tap the Settings button on the desired image and follow one of these steps:

- Tap **[Unlink Pre-Merge]** on a pre-merge image to break the link to the reference image. The pre-merge image moves back into the **Images** list.
- Tap **[Unlink All Pre-merged Images]** on the pre-merge reference image to break the link to all pre-merged images. The pre-merge reference image and all other images move back into the **Images** list.

### B: Using fMRI with Stealth AXiS™ Cranial clinical application

This section provides guidance on ways to display fMRI datasets, which are post-processed by third-party fMRI analysis software. There are a number of considerations for import, merge, and display of fMRI activation maps, as well as some limitations.

**Note:** Stealth AXiS™ clinical application cannot process BOLD scans into fMRI activation maps.

The following terminology is used in the guidance table below:

- Structural (anatomical) MRI – an MR scan (for example, a T1 or T2 scan) that shows anatomy (the structure of the tissues).

- fMRI Reference MR – the structural MRI scan acquired in the same MR scan session as the BOLD scans and used as part of the post-processing in the third-party fMRI analysis software. The activation maps must be inherently aligned with this fMRI reference dataset.
- Activation maps (also called fMRI activations) – output of the third-party fMRI analysis software depicting regions of the brain that are activated during the fMRI paradigms that were executed in the MR with the patient.

**Note:** Because activation maps that are exported separately from their fMRI reference do not show structural anatomy, they must be inherently aligned (spatially co-registered) with the fMRI Reference MR in the third-party fMRI analysis software before being exported and subsequently imported into the Stealth AXIS™ clinical application. If this spatial co-registration is not done, the activation map may be shown in the wrong location relative to the patient's anatomy.

- Navigation Reference – the main structural MRI (or CT) to which other scans in the merge set are merged. It should comply with the *Imaging Protocol* instructions to be usable for patient registration and navigation.

| Methods of fMRI DICOM output from third-party fMRI analysis software                                                                                                                                                                                                                                                                                                                                                                                                     | Summary of use in Stealth AXIS™ clinical application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>BIP: An activation map is embedded as bright white burned-in pixels (BIP) in the fMRI Reference MR. This is usually done as one activation per dataset to make it easy to turn individual activations on or off.</p>                                                                                                                                                                                                                                                  | <ol style="list-style-type: none"> <li>1. Import and merge the fMRI series to the Navigation Reference and verify the merge (see "Using image merge").</li> <li>2. Display the fMRI dataset in any of the 2D or 3D views.</li> </ol> <p>You can optionally build 3D models of the activation regions using the thresholding method to colorize each activation and make them easy to turn on or off in the 2D and 3D views.</p> <ul style="list-style-type: none"> <li>• If the activation has noise regions, you can initially threshold the activation areas for the whole volume, and then tap <b>[Select]</b> to pick the activation areas that you want to keep, and then tap <b>[Retain]</b>.</li> <li>• These can be displayed in any of the 2D or 3D views.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p>Color DICOM: One color DICOM RGB file contains the gray scale fMRI Reference MR along with overlays of one or more activation maps. Each activation map is represented by a different color range.</p> <p>Turn off text annotations in the images if possible, or use a different export technique, such as activation overlays (below).</p> <p>The separate fMRI Reference MR needs to be sent to the Stealth AXIS™ system along with the DICOM RGB format file.</p> | <p>The clinical application can accept fMRI scans in color DICOM RGB format. The separate fMRI Reference MR (grayscale) associated with it is also needed.</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• The clinical application cannot accept DICOM Palette Color.</li> <li>• The scan needs to contain the DICOM tags Image Position Patient (IPP) and Image Orientation Patient (IOP).</li> <li>• If the DICOM RGB color scans include additional text annotations embedded in the volume, they will not be usable. Instead, use separate gray scale activation maps. See the "Activation overlays" method in the next section of this table.</li> <li>• DICOM RGB color scans cannot be used as the Navigation Reference (for patient registration) or in 3D model building.</li> </ul> <ol style="list-style-type: none"> <li>1. Import both the DICOM RGB and fMRI Reference MR series.</li> <li>2. In the <b>Merge Images</b> task, the fMRI Reference can be designated as the Navigation Reference if it is adequate, or it might be just another image in the merge set. <ol style="list-style-type: none"> <li>a. If the Navigation Reference and fMRI Reference are different, the clinical application will run Auto-Merge to align the fMRI Reference MR to the Navigation Reference.</li> <li>b. Verify that the match between them looks correct.</li> </ol> </li> <li>3. Follow these steps to indicate that the DICOM RGB data is already inherently aligned (that is, pre-merged) with the fMRI Reference MR: <ol style="list-style-type: none"> <li>a. Tap the image card for the image that will be the reference image.</li> <li>b. Tap the Link button and select <b>[Pre-merge]</b>.</li> <li>c. Tap the image card for the image you want to pre-merge to the reference image.</li> <li>d. Tap <b>[Pre-merge]</b>.</li> <li>e. Verify that the match between them looks correct.</li> </ol> </li> <li>4. Display the color fMRI scan in either the 2D or 3D views (there is no need to keep using the fMRI Reference MR at this point).</li> </ol> |

| Methods of fMRI DICOM output from third-party fMRI analysis software                                                                                                                                                                                                                                                                                                                                                                                                                        | Summary of use in Stealth AXiS™ clinical application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Activation overlays: The fMRI Reference MR is in one DICOM file, and the activation maps are in separate gray scale DICOM files. If activation maps are thresholded during the fMRI analysis, they will look like they contain free-floating blobs. Activation overlays are often exported as one file per activation to make them easy to turn on or off as overlays.</p> <p>The fMRI Reference and all of the needed activation overlays must be sent to the clinical application.</p> | <p>Because the fMRI activation overlays do not depict anatomy, prior to import they must already be inherently aligned with the fMRI Reference MR that shows the anatomy.</p> <ol style="list-style-type: none"> <li>1. While in the <b>Images</b> task, for each activation overlay image series card, tap the gear button, and select <b>fMRI</b> from the <b>Modality</b> drop-down menu. This indicates to the clinical application that the activation overlay image series must be pre-merged; that is, the Auto-Merge feature is disabled because there is no anatomy depicted in the dataset.</li> <li>2. Follow these steps to indicate that the activations are already inherently aligned (that is, pre-merged): <ol style="list-style-type: none"> <li>a. Tap the image card for the image that will be the reference image.</li> <li>b. Tap the Link button and select <b>[Pre-merge]</b>.</li> <li>c. Tap the image card for the image you want to pre-merge to the reference image.</li> <li>d. Tap <b>[Pre-merge]</b>.</li> <li>e. Verify that the match between each activation map and the fMRI Reference MR looks correct.</li> </ol> </li> <li>3. Use Auto-Merge to merge the fMRI Reference MR to the Navigation Reference. Verify their match to add all datasets to the merge set.</li> <li>4. Display any of the fMRI activations as overlays on top of any other image in the merge set in any of the 2D or 3D views. Changing the modality to fMRI will change the default colormap to Rainbow but that can be changed later, if desired. Also, background black areas of the scan will be transparent when the activations are displayed as overlays so the underlying image showing anatomy will also be seen.</li> </ol> |
| Activation overlays: options                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>The following are optional actions in the clinical application:</p> <ul style="list-style-type: none"> <li>• Change the display stacking order in 2D or 3D by dragging the image series cards to reorder the list.</li> <li>• Change the color of any of the activation overlays in 2D or 3D views. Tap the gear button, tap the <b>Colormap</b> drop-down menu, and select the desired coloring.</li> <li>• You can build 3D models of the activation regions using the thresholding method to colorize each activation and make them easy to turn on or off in the 2D and 3D views.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## C: Editing a 3D registration model

**Note:** The clinical application builds the registration model automatically, and then you can edit that model.

To edit a 3D registration model, follow these steps:

**Note:** If you edit the 3D model after you begin a registration, the clinical application saves the 3D model to the previous registration.

1. In the **Registration** task, tap **[Edit Model]**.  
The **Model Building** task opens.
2. Drag the **Threshold** slider to adjust the threshold.
3. Tap **[Refine]** to automatically refine the 3D model and create a closed surface for optimal registration.
4. Edit the 3D registration model as necessary. See “Model editing tools.”
5. Tap **[Done]**.

## Removing the head holder from a 3D registration model

The clinical application automatically builds a 3D registration model, which may include the head holder. To remove a head holder from the 3D registration model, follow these steps:

1. In the **Registration** task, tap **[Edit Model]**.  
The **Model Building** task opens.
2. Use any of the following methods to remove the head holder:
  - To retain only a portion of the 3D model, tap the area you want to retain and then tap **[Retain]**.
  - To delete a portion of the 3D model, tap the area you want to delete and then tap **[Delete]**.

- To remove the outside edges of the model, swipe a 2D image view into the main window and adjust the cropping square to surround only the portion that you want to keep. The model updates automatically as you adjust the cropping square.
- To erase part of the 3D model, such as the head holder, tap the **Paint** drop-down button and tap **[Erase]**, swipe a 2D image view into the main window, and then drag on the image to erase the head holder.
- To delete items that are not connected to the largest part of the model, tap the **Advanced** drop-down button and tap **[Remove Islands]**.
- To cancel the changes to the model and return to registration, tap **[Cancel]**.

You can further refine the 3D registration model. See “Editing a 3D registration model.”

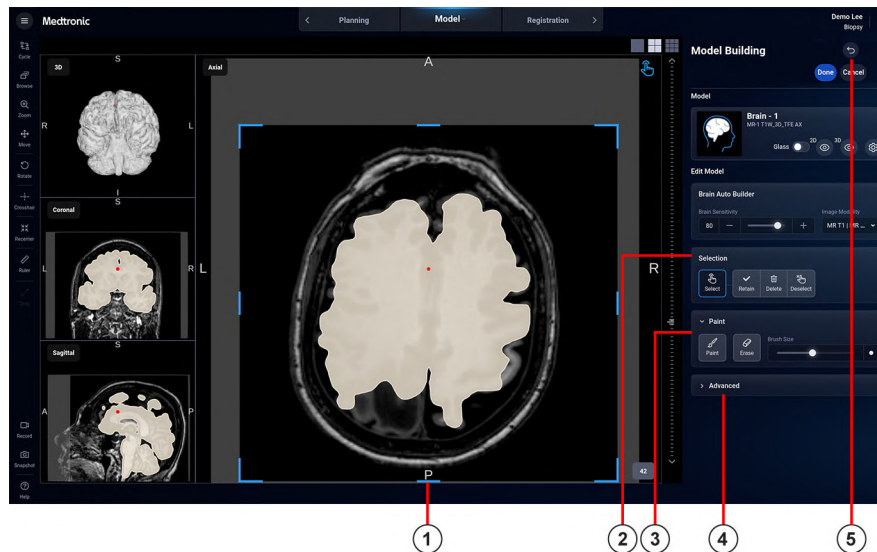
3. Tap **[Refine]** to automatically refine the 3D model and create a closed surface for optimal registration.
4. Tap **[Done]**.

## D: Building and editing models

### Model editing tools

When building and editing 3D models, you can use a variety of editing tools for each model.

| Editing option  | Behavior                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selection card  | <p>Displays the buttons for retain, deselect, and delete options.</p> <ul style="list-style-type: none"> <li>• To retain only a portion of the 3D model, tap the area you want to retain and then tap <b>[Retain]</b>.</li> <li>• To delete a portion of the 3D model, tap the area you want to delete and then tap <b>[Delete]</b>.</li> <li>• To cancel the selection of the 3D model, tap <b>[Deselect]</b>.</li> </ul>                                                                                                                     |
| Paint card      | <p>Displays the buttons for painting and erasing within a 2D image.</p> <ul style="list-style-type: none"> <li>• To use the <b>[Paint]</b> and <b>[Erase]</b> tools, tap <b>[Paint]</b>.</li> <li>• Swipe a 2D image view into the main window, adjust the <b>Brush Size</b> slider, and then drag on the image to paint or erase.</li> </ul>                                                                                                                                                                                                  |
| Advanced card   | <p>Refines the model with additional options including:</p> <ul style="list-style-type: none"> <li>• To fill small gaps or to smooth the model, tap the <b>[+]</b> and <b>[-]</b> to adjust the <b>Gap Fill</b> and <b>Smooth</b> sliders.</li> <li>• To reduce the model's size, tap <b>[Contract]</b>.</li> <li>• To increase the model's size, tap <b>[Expand]</b>.</li> <li>• To delete items that are not connected to the model, tap <b>[Remove Islands]</b>.</li> <li>• To fill holes in the model, tap <b>[Fill Holes]</b>.</li> </ul> |
| Cropping square | <p>Removes the outside edge of a model.</p> <ul style="list-style-type: none"> <li>• Swipe a 2D image view into the main window and then adjust the cropping square to surround only the portion you want to keep. The model updates automatically when you adjust the cropping square.</li> <li>• To adjust the cropping square, drag the cropping line to the preferred position. You can adjust the cropping square in any 2D view.</li> </ul>                                                                                              |
| Undo button     | Undoes all generic tool actions, all threshold sliders, and Tumor/Other Seed and 3D Paint tools.                                                                                                                                                                                                                                                                                                                                                                                                                                               |



- 1 Cropping square
- 2 Selection card
- 3 Paint card
- 4 Advanced card
- 5 Undo button

### Building and editing a tumor or other model

Follow these steps to build a tumor model, ventricle model, or any other model type.

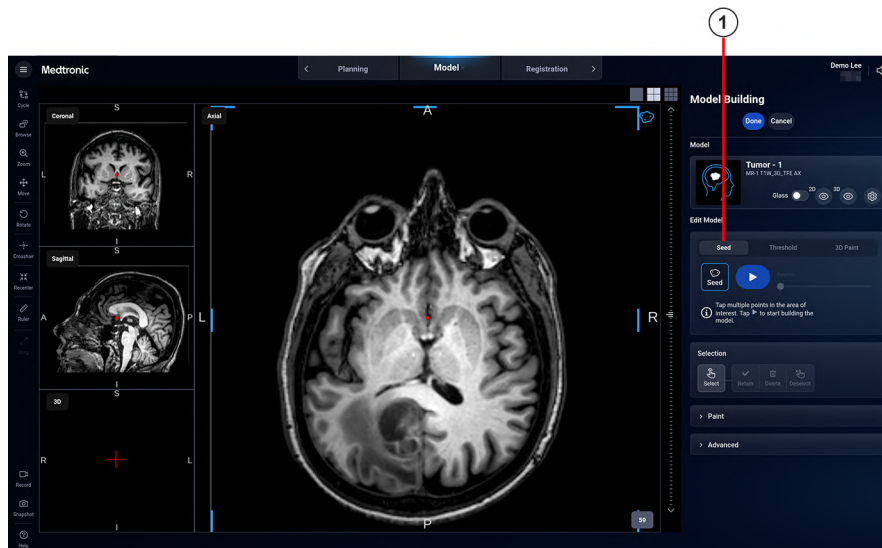
1. In the **Planning** task or **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Tumor/Other]**.
4. Select an image.
5. The **Model Building** task opens. Select one of the three methods to build a model: **Seed** (default), **Threshold**, or **3D Paint**.

**Note:** The clinical application builds the model using only one method. If you select a different method, the current model will be deleted.

The **Seed** tab is the preferred method to build a model. Select seed points and grow the model from those points.

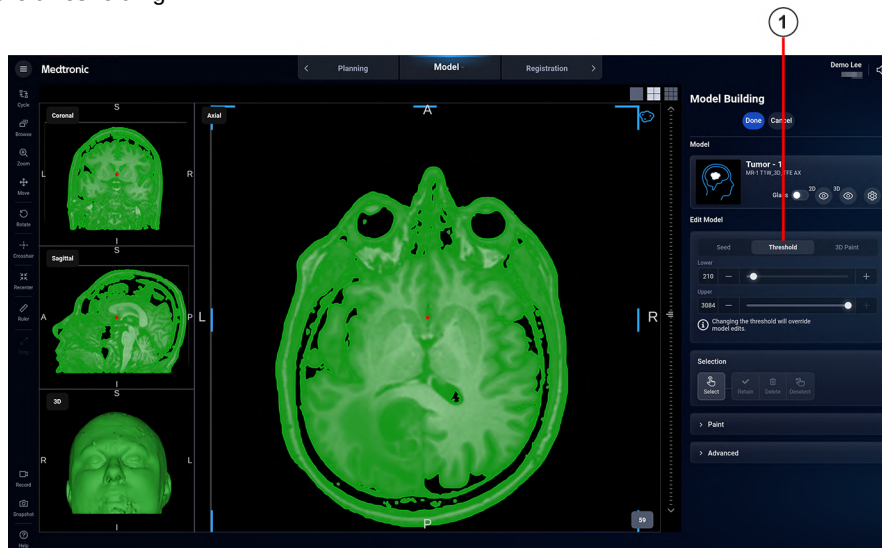
- a. Tap the **[Seed]** button.
- b. Locate a 2D image slice that contains the area of interest.
- c. Tap at least one point in the area of interest.
- d. Locate another slice that contains the area of interest and tap at least one point.
- e. Repeat the steps until you have selected at least three points on different slices.
- f. Tap the Play button to create the model.

**Note:** You can freeze seed growth by tapping the Pause button. To rewind seed growth, use the **Rewind** slider.



### 1 Seed tab

Tap the **Threshold** tab to build a model based on light and dark intensity values in the images. Use the **Lower** and **Upper** slider to adjust the thresholding.



### 1 Threshold tab

Tap the **3D Paint** tab to build a model by painting on multiple image slices that contain the area of interest:

- Tap the **[Paint]** tool.
- Locate the image slice with the smallest portion of the area of interest visible.
- Drag the cursor on the image to paint the area of interest.
- Repeat this process to paint the area of interest on multiple image slices. You must paint in image slices of the same perspective, using only all sagittal, coronal, or axial slices.

**Note:** Drag on the scroll bar to scroll through the image slices. You can skip some image slices in which the anatomy is similar.

- Tap **[Interpolate]** to create the model.



1 3D Paint tab

- To display the model in 2D or 3D views in the **Planning** and **Navigation** tasks, tap the 2D or 3D Show button.

### Building and editing a brain model

Follow these steps to build and edit a 3D brain model.

- In the **Planning** task or the **Navigation** task, tap the Models button.
- Tap the Add button.
- Under **Select Model Type**, tap **[Brain]**.
- Select an image.  
The **Model Building** task opens, and the clinical application automatically builds a brain model.
- Edit using brain specific edit options, as appropriate:
  - To change the image modality, select an option from the **Image Modality** drop-down menu.
  - To show more or less brain anatomy in the brain model, drag the **Brain Sensitivity** slider.
- Continue editing the model with the generic editing tools as necessary.
- Tap **[Done]**.

### Building and editing a bone model

Follow these steps to build and edit a 3D bone model.

- In the **Planning** task or the **Navigation** task, tap the Models button.
- Tap the Add button.
- Under **Select Model Type**, tap **[Bone]**.
- Select an image.  
The **Model Building** task opens, and the clinical application automatically builds the bone model using the **Threshold** method.
- Use the **Lower** and **Upper** sliders to adjust the threshold.
- Edit the bone model as necessary.
- Tap **[Done]**.

## Building and editing a skin model

Follow these steps to build and edit a skin model.

1. In the **Planning** task or the **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Skin]**.
4. Select an image.  
The **Model Building** task opens, and the clinical application automatically builds a skin model using the **Skin** method.
5. Tap **[Refine]**, and drag the **Threshold** slider to adjust the threshold.
6. Continue editing the model as necessary.
7. Tap **[Done]**.

## Building and editing a vessel model

Follow these steps to build and edit a 3D vessel model.

1. In the **Planning** task or the **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Vessel]**.
4. Select an image.  
The **Model Building** task opens.
5. Edit using vessel specific edit options, as appropriate:
  - To change the image modality in the **Vessel Auto Builder** method, select an option from the **Image Modality** drop-down menu.
  - Drag the **Threshold** slider to adjust the threshold.
  - To show more or less brain anatomy in the brain model, drag the **Brain Sensitivity** slider.
6. Continue editing the model with the generic editing tools as necessary.
7. Tap **[Done]**.

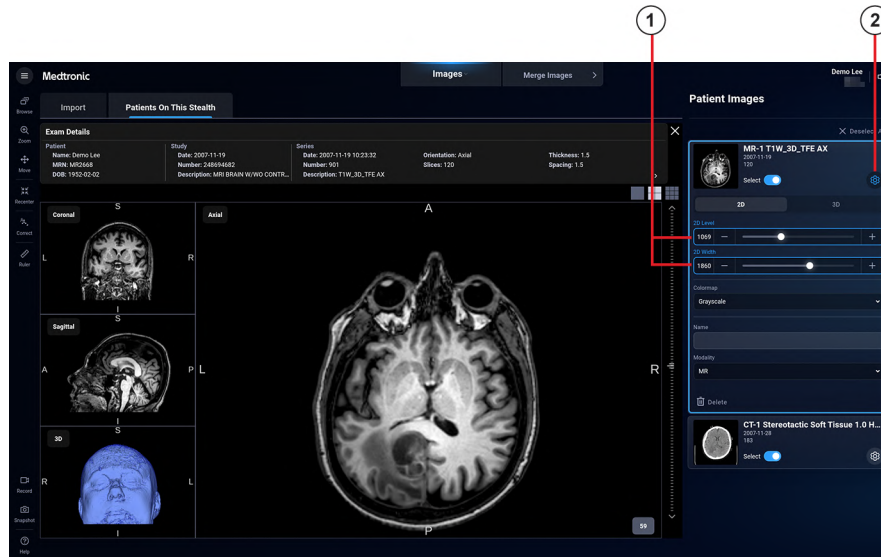
## E: Adjusting views for images and models

### Adjusting level and width

You can use the **Level** and **Width** controls to change how patient images and models display in 2D and 3D views.

Follow these steps to adjust level and width:

1. Tap the Images button to open the **Images** drawer or tap the Models button to open the **Models** drawer.
2. Tap the Settings button on the image or model you want to adjust.
3. Tap the **2D** or **3D** tab.
4. Drag the **Level** and **Width** sliders.  
While the **Level** and **Width** controls are selected, you can also adjust level and width by dragging on the views.
  - Drag up and down to adjust **Level**.
  - Drag left and right to adjust **Width**.
5. Tap the Settings button or select a different image or model to deselect the level and width controls.



- 1 Level and Width sliders
- 2 Settings button

### Displaying multiple images in 2D views

If you merged images, you can display multiple images in 2D views in the **Planning** and **Navigation** tasks. Tap the **Images** button to open the **Images** drawer, and then follow these instructions to adjust how images display:

- To manually display images on top of one another, follow these instructions:
  - Tap the **2D Show** button to display or hide an image.
  - Use the **Opacity** slider to adjust the opacity of an image so that the bottom images can be seen.
- To reorder images, tap on the left edge of the image card and drag up or down in the **Images** drawer.

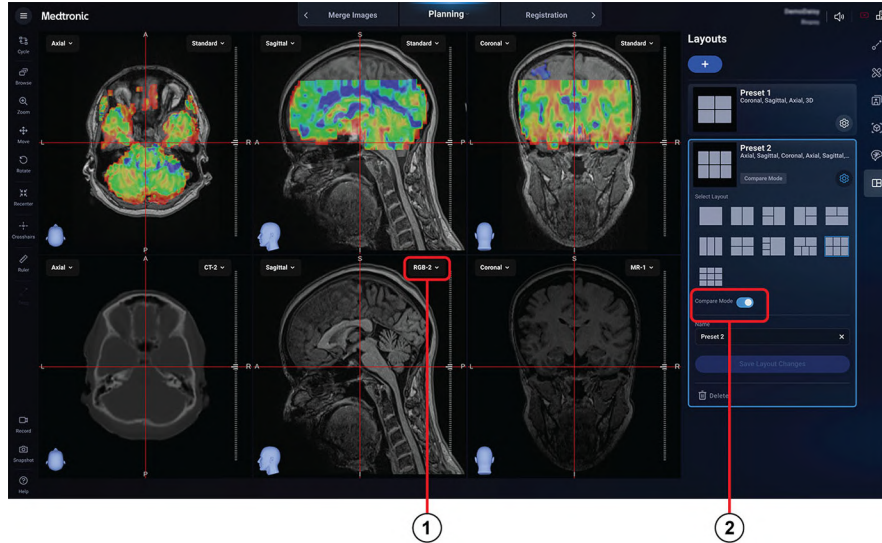
**Note:** Transparent images placed on top of opaque images will show both images. Images underneath opaque images will not be visible.

### Using the compare mode

To compare images, follow these instructions:

1. Tap the **Layouts** button to open the **Layouts** drawer.
2. Tap the **Compare Mode** toggle button on. The view layout changes to a 6-up view with Axial, Sagittal and Coronal views on both the top and bottom rows. If desired, the view layout and view types can be changed.
3. In each view, tap the **Select Image** drop-down menu to change the desired image to display.
4. To save the layout and images selected in each view:
  - Tap in the **Name** field and type a unique preset name.
  - Tap **[Save Layout Changes]** on the compare mode preset card.

The layout is saved for the surgeon profile and procedure in use.



- 1 Select image drop-down menu
- 2 Compare mode toggle

### Adjusting colormap or modality of images in 2D views

If you are displaying fMRI datasets, see “Using fMRI with Stealth AXiS™ Cranial clinical application.”

1. Tap the Images button to open the **Images** drawer.
  2. Tap the Settings button on the image that you want to modify.
  3. To adjust the colormap, open the **Colormap** drop-down menu and then select an option.
  4. To adjust the image modality, tap the Settings button and select a modality from the drop-down list.
- The image modality option is only available in the **Images** task.

### Adjusting how models display in 2D views

For a 2D view of a model, you can choose to view contours (outlines) or to fill the outlines.

1. Tap the Models button to open the **Models** drawer.
2. By default, models are displayed as outlines. Tap the **[2D Fill]** toggle to fill the outlines.  
The **2D Opacity** slider now affects the opacity of the fill. The outline remains as is.
3. To return to the default view, turn off the **[2D Fill]** toggle.

The **2D Opacity** slider now affects the opacity of the outline.

To reorder models, drag the models up or down in the **Models** drawer. Transparent models placed on top of opaque models will show both models. Models underneath opaque models will not be visible by tapping and dragging left edge of the card.

### Adjusting how models and images display in the 3D view

If you want to change how models and images display in the 3D view, follow these instructions.

1. In the **Planning** or **Navigation** task, tap the Models button to open the **Models** drawer.
2. Tap the Settings button on a model or image to open its display options.
3. Adjust how models and images display in the 3D view as necessary, using the following options:

Options for models:

- Turn on **Glass** mode.
- Adjust the **3D Opacity**.
- Choose a **Colormap** from the drop-down menu.
- Update the **Name** field.

Options for images:

- Turn on **Glass** mode.
  - Adjust the **3D Opacity**.
  - Adjust the **3D Level** and **3D Width**.
  - Choose a **Colormap** from the drop-down menu.
4. Tap the Settings button or select a different model or image to close the 3D display options.

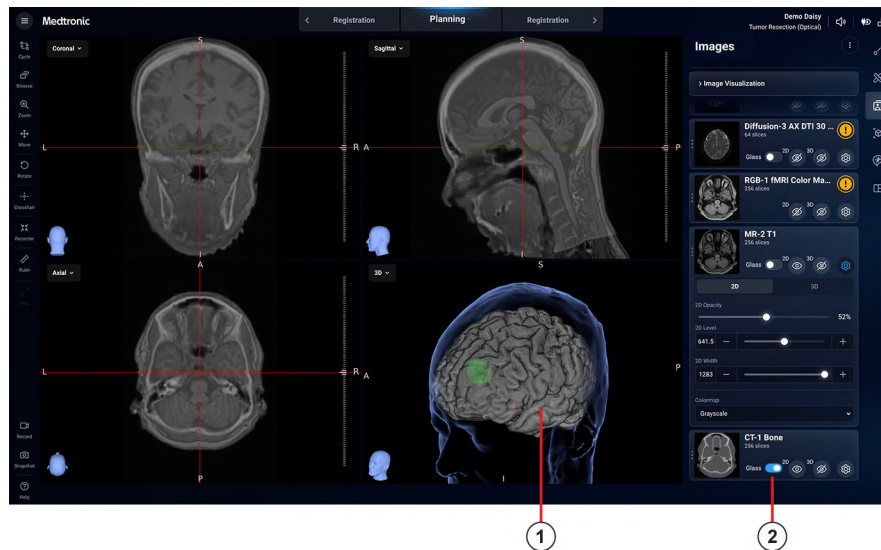
### Using Glass mode

Glass mode allows you to see through outer layers of 3D models, such as skin and bone, to increase the visibility of other models, such as tumor models.

To use Glass mode, follow these instructions:

1. In the **Planning** or **Navigation** task, tap the Models button to open the **Models** drawer.
2. Tap the Settings button on a model or image to open its display options.
3. Turn on the **Glass** toggle to enable Glass mode.

In Glass mode, the 3D Opacity is automatically set to 50%. Use the **3D Opacity** slider to adjust the opacity from 0% to 100%.



- 1 Glass mode view
- 2 Glass toggle

### Using the 3D Cut mode tool

The 3D Cut mode tool can cut away sections of images and models in the 3D view. To use the 3D Cut mode tool, follow these instructions.

**Note:** The 3D Cut mode tool does not edit the 3D model. The 3D Cut mode tool only changes the 3D view of the model.

1. In the **Planning** or **Navigation** task, tap the Models button to open the **Images** drawer.
2. Tap **Image Visualization** at the top of the drawer.
3. Turn on the **3D Cut** toggle to enable cut mode.
4. Move the crosshairs to the location you want to see in the 3D view.  
The 3D Cut mode tool cuts away a section of the model or image (for which the visibility is currently on) to show the location of the crosshairs in the 3D view.
5. Turn on the **Display 2D Image in Cut Mode** toggle to display the 2D images at the current cut plane.

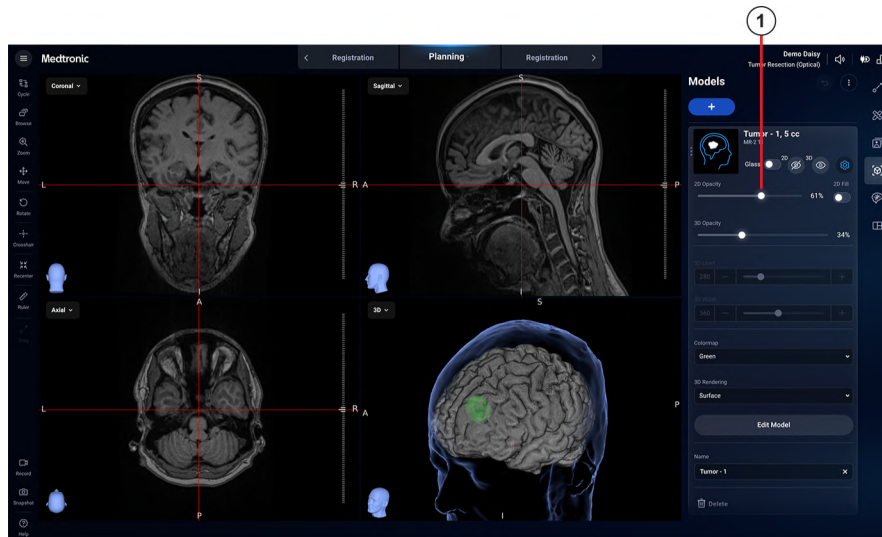
## Adjusting the opacity of images and models

In the **Planning** and **Navigation** tasks, you can adjust the opacity of images and models.

1. Tap the **Images** button to open the **Images** drawer or tap the **Models** button to open the **Models** drawer.
2. Tap the image or model that you want to adjust.
3. Use the **2D Opacity** or **3D Opacity** slider to adjust the image or model's opacity from 0 to 100 percent.

**Note:** In 2D views, you can re-order image panels to place transparent images on top of opaque images.

**Note:** For a 2D view, if the **2D Fill** toggle is off, the opacity slider affects the outline of the model. If the **2D Fill** toggle is on, the **2D Opacity** slider affects the model's fill, but not the outline.

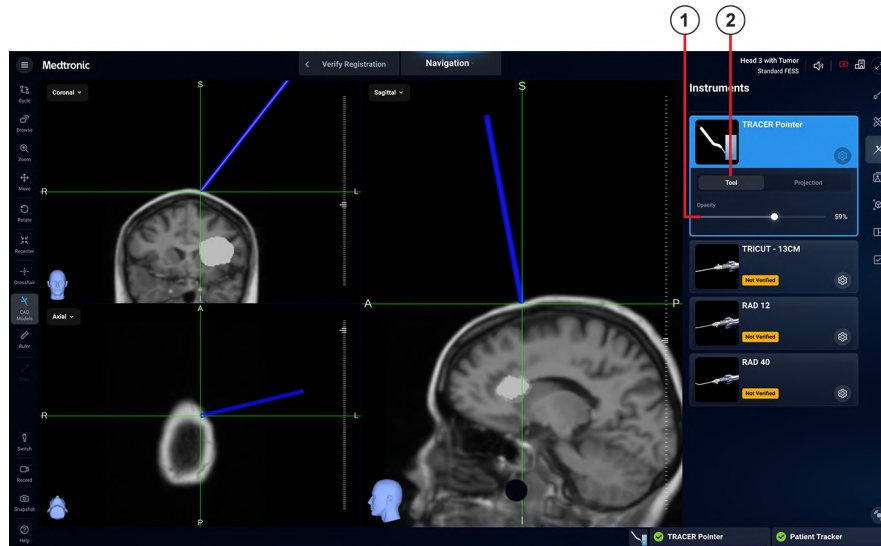


1 2D Opacity slider

## Adjusting instrument CAD model opacity

Use the **Opacity** slider to adjust the transparency of the instrument CAD model.

1. In the **Navigation** task, tap **[Instruments]**.
2. Under the desired instrument, tap **[Tool]**.
3. Define the transparency of the instrument CAD model by using the **Opacity** slider.



- 1 Opacity slider
- 2 [Tool]

## F: Creating and editing annotations and alerts

The **Annotations and Alerts** drawer allows you to store points in images and return to them later in the procedure. For example, you can save specific locations along a surgical plan that are defined in 2D and 3D views.

### Creating a new annotation

Follow these steps to create a new annotation:

1. In the **Planning** task or **Navigation** task, tap the **Annotations and Alerts** button to open the **Annotations and Alerts** drawer.
2. Tap on a 2D or 3D patient image to place the crosshairs where you want an annotation.
3. Tap the **Annotate** button to add the new annotation. An annotation is created at the location of the crosshairs.  
If you want to update the location of the annotation, move the crosshairs where you want to place the new annotation and tap **[Set Location]**.
4. Enter a name for the annotation in the **Name** field.  
**Note:** A name will automatically be assigned if it is not entered in the **Name** field.
5. To create another annotation, tap where you want the annotation on the patient image and tap the **Annotate** button again.

### Viewing and hiding annotations

- When you create an annotation, it appears in the **Annotations and Alerts** drawer and the annotation dot is visible in the 2D and 3D image views.
- If you moved away from a specific annotation in the image views and want to return to it, take one of the following actions:
  - Tap the **Annotations and Alerts** button to open the **Annotations and Alerts** drawer and then tap the specific annotation.
  - Scroll through the image slices in a 2D image view until you locate the slice that contains the specific annotation.
- To view a single annotation's name in the 2D or 3D image views, hold the mouse cursor over the annotation's dot.
- To show or hide a single annotation, tap the **Show** button.
- To show or hide all annotations, tap the **More** button and then tap the **Show All** button.  
See "Keyboard shortcuts" for information on displaying all annotation descriptions using a keyboard shortcut.

## Creating alerts

An alert can be enabled for an annotation that provides audible and visual feedback when navigating a tool within a user-defined radius. The **Alert** toggle can be turned on for any defined annotations. Follow these steps to create an alert for an existing annotation:

1. In the **Planning** task or **Navigation** task, tap the Annotations and Alerts button to open the **Annotations and Alerts** drawer.
2. Tap the Settings button on the annotation card that you are adding the alert to.
3. Tap the **Alert** toggle to turn on the alert.
4. Set the radius of the alert region using the slider. The alert will add a semi-transparent highlight of this size around the annotation in the same color used for the annotation.
5. To hide and turn off the visual and audible features of an alert, tap the **Alert** toggle. Tap the toggle again to show and turn on the alert.

## Editing, locking, and deleting annotations

1. In the **Planning** task or the **Navigation** task, tap the Add Annotations and Alerts button to open the **Annotations and Alerts** drawer.

2. Tap the Settings button of the annotation that you want to modify.

You can modify an annotation in the following ways:

- Edit the location of an annotation.
- Edit the name

**Note:** You can also edit an annotation name by double-tapping the name in the **Planning** task or the **Navigation** task in the **Annotations and Alerts** drawer.

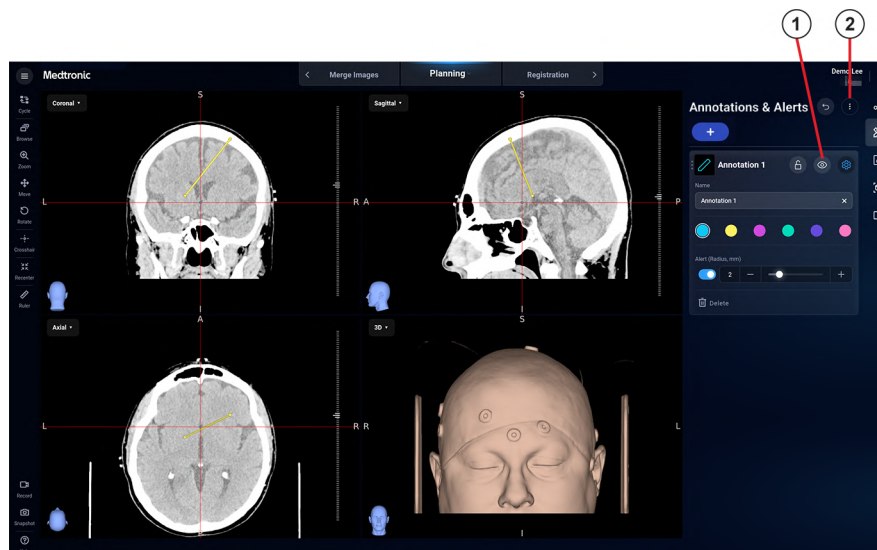
- Change the color
- Lock the annotation to prevent changes
- Delete the annotation

If you made a change to an annotation and you want to undo the change, tap the Undo button.

3. To delete one or more annotations, do one of the following:

**Note:** You cannot delete a locked annotation. To restore a deleted annotation, tap the Undo button.

- To delete a single annotation, do one of the following:
  - Tap the Settings button on the annotation and tap **[Delete]**.
  - Swipe an annotation to the right and then tap the Delete button.
- To delete multiple annotations, tap the More button then tap **[Delete Multiple]**. Tap the Delete button next to each annotation you want to remove.
- To delete all annotations, tap the More button then tap **[Delete All]**.



- 1 Show button to show or hide annotations
- 2 The More button to **[Hide All]**, **[Delete Multiple]**, or **[Delete All]**.

## G: Creating accuracy checkpoints (optional)

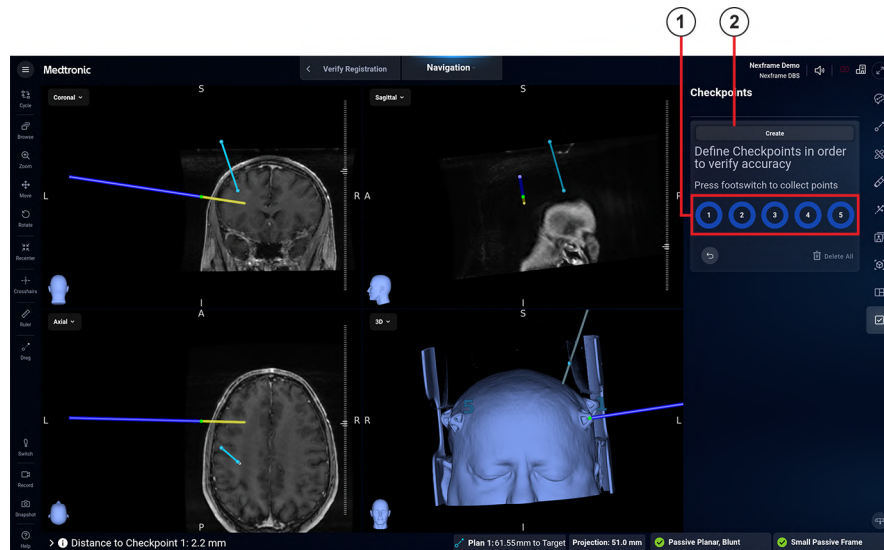
Creating five fixed, identifiable points on the anatomy in the sterile field enables you to realign the current registration if you bump or otherwise move the patient reference frame in relation to the patient anatomy. Create the checkpoints immediately after a successful registration.

**Warning:** If you do not create accuracy checkpoints and the patient reference frame moves relative to the patient anatomy at any time after surgery begins, you must re-register the patient or abandon use of the system.

Follow these steps to create accuracy checkpoints in the **Navigation** task.

1. Tap **[Checkpoints]** to open the **Checkpoints** control panel.
2. Place the sterile registration instrument on the first point and hold it steady for 2 seconds or press the footswitch. The system stores the point in the **Checkpoints** table, makes a confirmation sound (chime), and advances to the next point.
3. Repeat the steps to store 4 more accuracy checkpoints.

If you need to delete an accuracy checkpoint, tap the Undo button or **[Delete All]**.



1 Checkpoints table

2 Create tab

## Realigning to accuracy checkpoints (optional)

If you created accuracy checkpoints, you can use them in the **Navigation** task to confirm registration accuracy or to realign the registration.

- At any time during the procedure, touch a navigation instrument to any of the accuracy checkpoints and observe the **Distance to Checkpoint** metric in the clinical application banner message.
- A distance in excess of 2 mm may indicate that the patient reference frame moved in relation to the anatomy. If the distance is greater than 2 mm, check navigational accuracy by touching several anatomical points.

To realign the registration to the accuracy checkpoints, follow these steps:

- In the **Checkpoints** control panel, tap the **Realign** tab.
- Tap a stored accuracy checkpoint in the **Checkpoints** table.
- Place the tip of a registration probe on the patient at the first checkpoint. Do not indent the skin.
- Press the footswitch or hold the probe steady for 2 seconds.

The system stores the location of the checkpoint.

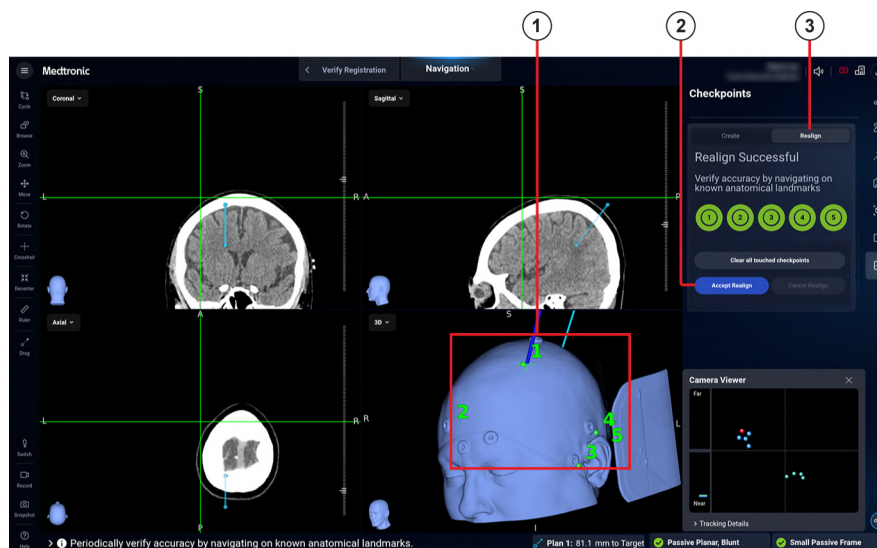
- Repeat this process for the remaining checkpoints.

When you have touched all 5 accuracy checkpoints, the system realigns the registration to the checkpoints.

- Use the registration instrument to touch identifiable points on the patient's anatomy and compare the locations to the crosshairs in the images.

- If you are satisfied with the realigned registration accuracy, tap **[Accept Realign]**.

If the accuracy of the realigned registration is not acceptable, tap **[Cancel Realign]** to discard the realigned registration.



- 1 Touched checkpoints
- 2 [Accept Realign]
- 3 Realign tab

## H: Performing SureTrak™ II calibration (optical only)

The SureTrak™ II system is an optional kit that allows you to navigate non-navigated surgical instruments. To accomplish this, you must attach a tracker to the instrument shaft and then calibrate the instrument.

**Warning:** If you attach a SureTrak™ II tracker to an instrument with interchangeable tips, you must recalibrate the instrument if you change the tip.

**Warning:** Do not use the same geometry SureTrak™ II tracker on two different instruments.

**Warning:** Do not use a SureTrak™ II tracker on flexible instruments.

**Warning:** If you move a tracker onto a different instrument or reposition a tracker on the same instrument, you must recalibrate.

**Warning:** Do not attempt to use the SureTrak™ II system if any components appear to be bent or otherwise damaged.

**Warning:** When selecting a passive tracker, choose the largest size feasible for the instrument.

To calibrate an instrument with a SureTrak™ II tracker, follow these steps:

1. Confirm that the SureTrak™ II assembly is rigidly secured to the instrument and lies within the maximum recommended distance from the tip. Refer to the instructions that accompanied the SureTrak™ II system for more information about attaching a tracker.

|                                                      |                             |
|------------------------------------------------------|-----------------------------|
| Small passive tracker (SureTrak™ II Passive Orange)  | Distances of 130 mm or less |
| Active tracker (SureTrak™ II Active Blue)            | Distances of 175 mm or less |
| Medium passive tracker (SureTrak™ II Passive Silver) | Distances of 200 mm or less |
| Large passive tracker (SureTrak™ II Passive Black)   | Distances of 260 mm or less |

**Warning:** Any movement of the assembly relative to the instrument after the instrument is calibrated results in an inaccurate display of the instrument location.

2. Add the SureTrak™ II tracker to the procedure in the **Instruments** task and hold the assembly in the camera's field of view. The clinical application displays instructions for SureTrak™ II calibration.
3. Review the on-screen warning and press the footswitch to proceed with the calibration process.
4. Choose a calibrating probe (for example, the Passive Planar Registration Probe). If necessary, verify the calibrating probe.
5. To specify the divot, place the tip of the calibrating probe in the bottom of the patient reference frame divot and press the footswitch.

6. Hold the SureTrak™ II instrument perpendicular to the divot with the instrument tip in the bottom of the divot and press the footswitch.
7. Place the tip of the SureTrak™ II instrument on the surface of the patient reference frame next to the divot. Hold the instrument at the same angle used in the previous step and press the footswitch.
8. Hold the tip of the calibrating probe to the hind of the SureTrak™ II instrument and press the footswitch.

**Note:** You can restart the calibration at any time in the calibration process by tapping **[Restart calibration]**.

When a SureTrak™ II instrument is fully calibrated, you can recalibrate it by tapping **[Recalibrate]** on the instrument's tool card in the **Instruments** drawer. Recalibration is required when you change the tip on an instrument with interchangeable tips.

## I: Using an ultrasound system (optical only)

**Note:** A Stealth AXIS™ Ultrasound License is required to use an ultrasound system with the clinical application.

You can navigate an ultrasound transducer and display live ultrasound overlaid with other patient images.

The following ultrasound systems and transducers are supported:

- bkActiv ultrasound system
- GEHC Craniotomy Transducer (9062)

The following ultrasound system modes and settings are supported:

- Modes: 2D, Power Doppler (P), and Color Doppler (C)
- Settings: Image Size 80%, Zoom 1x

**Note:** If an unsupported mode or setting is selected on the ultrasound system, navigation stops.

**Note:** If the ultrasound system screen is moved to a vertical position, navigation stops.

**Note:** The ultrasound transducer cannot be in freeze mode when initiating navigation.

**Warning:** Before each use, physically examine the Ethernet cable for damage and proper connection between the ultrasound system and the workstation cart. Connection issues could result in no navigation information.

**Warning:** Ensure the transducer tracker is securely attached in the correct orientation to the ultrasound transducer. Incorrect attachment may affect navigational accuracy.

**Warning:** Ensure that the transducer tracker maintains a rigid orientation relative to the ultrasound transducer via the transducer tracker mounting mechanism. Movement of the transducer tracker after verification could impact navigational accuracy.

**Note:** If both a digital microscope and ultrasound are used in a procedure, configure them both to the same IP setting.

## Verifying the navigated ultrasound transducer (optical only)

**Warning:** If the ultrasound transducer with the transducer tracker attached is dropped or otherwise impacted in a manner that could cause the tracker to be damaged or displaced, discontinue use and re-sterilize the ultrasound transducer before reverifying the ultrasound transducer with the transducer tracker attached. Failure to reverify the navigated ultrasound transducer may result in navigational inaccuracy.

**Caution:** You cannot proceed with navigation unless the ultrasound transducer has been calibrated by qualified Medtronic personnel. Only use a navigated ultrasound transducer that is properly calibrated and verified prior to use.

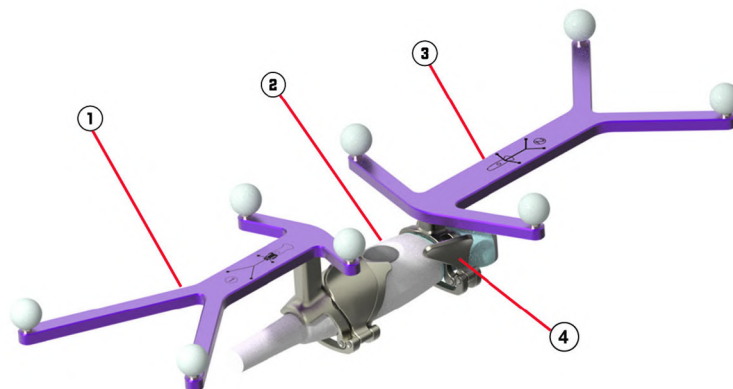
You must verify the navigated ultrasound transducer before using it for a navigated procedure. See the *GEHC Craniotomy Transducer (9062) Integration Hardware Package Insert* for instructions on attaching the craniotomy transducer tracker, attaching the craniotomy transducer verification device, and preparing the ultrasound transducer for verification.

1. Connect the ultrasound system and Stealth AXIS™ workstation with an Ethernet cable. Ensure proper ultrasound equipment is added to the **Equipment** task in the clinical application. If the ultrasound equipment card is not displayed, search for the ultrasound system in the **Add Equipment** list and tap the Add button.  
After communication between the Stealth AXIS™ workstation and the ultrasound is established, the equipment status bar and connection line displays green.
2. Ensure the proper instruments are added in the **Instruments** task in the clinical application.

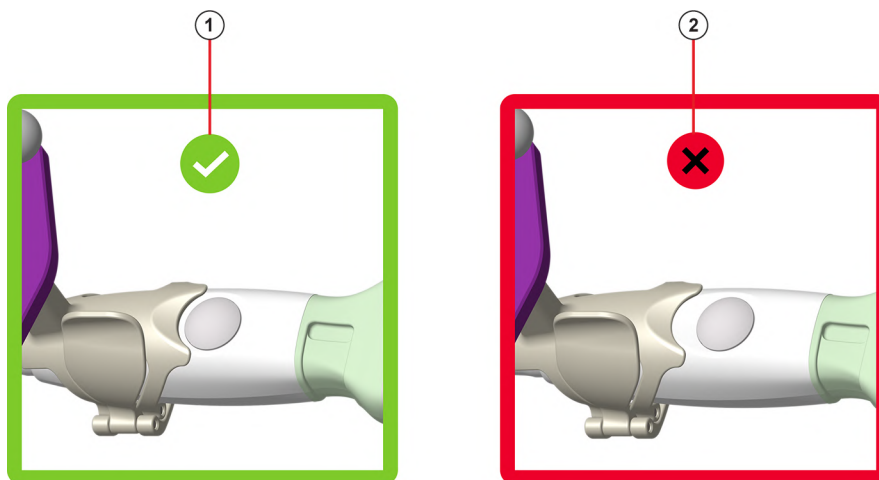
**Note:** The tracker and verification device instrument cards are automatically added when the ultrasound system equipment card is added. Ensure that the transducer tracker instrument card has a serial number displayed. The serial number is a unique identifier that matches the device calibration with the device itself. If the transducer serial number does not have a valid calibration or no transducer is connected, there will be an indication on the instrument card.

3. Ensure both the transducer tracker and transducer verification device are attached, and then face the transducer tracker and transducer verification device spheres toward the camera to verify the navigated ultrasound transducer. Hold the assembly steady.
4. Tap the Tracking View button to open the tracking view to confirm that both the transducer tracker and the transducer verification device are in the camera's field of view. If verification is not successful, see "Troubleshooting instruments (optical only)."
5. After the navigated ultrasound transducer is verified, remove the transducer verification device from the navigated ultrasound transducer by opening the latch on the transducer verification device mount.

**Note:** Do not remove the craniotomy transducer tracker until the end of the case. If the craniotomy transducer tracker is removed, the navigated ultrasound transducer should be re-verified.



- 1 Craniotomy transducer tracker
- 2 Ultrasound transducer
- 3 Transducer verification device
- 4 Transducer verification device mount



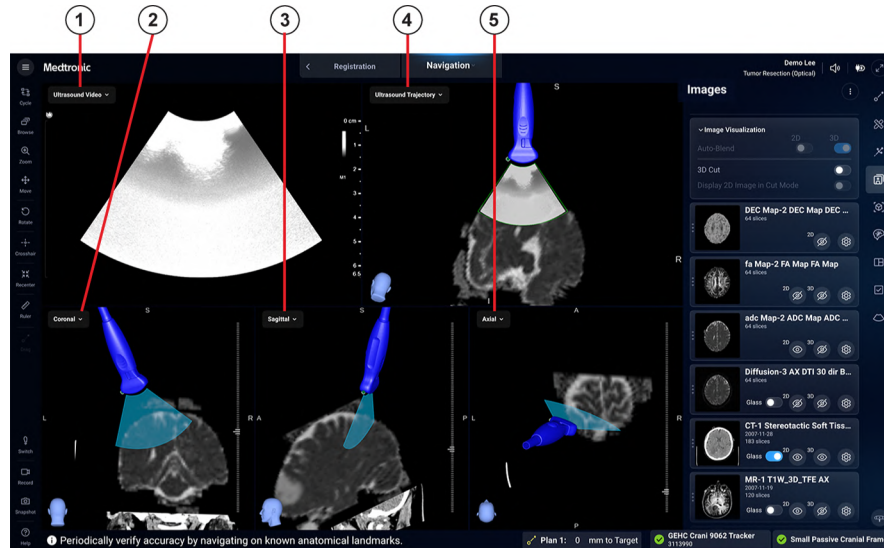
- 1 Example of correct setup
- 2 Example of incorrect setup

## Ultrasound views (optical only)

The ultrasound views are displayed in the **Navigation** task when the ultrasound navigation feature is selected.

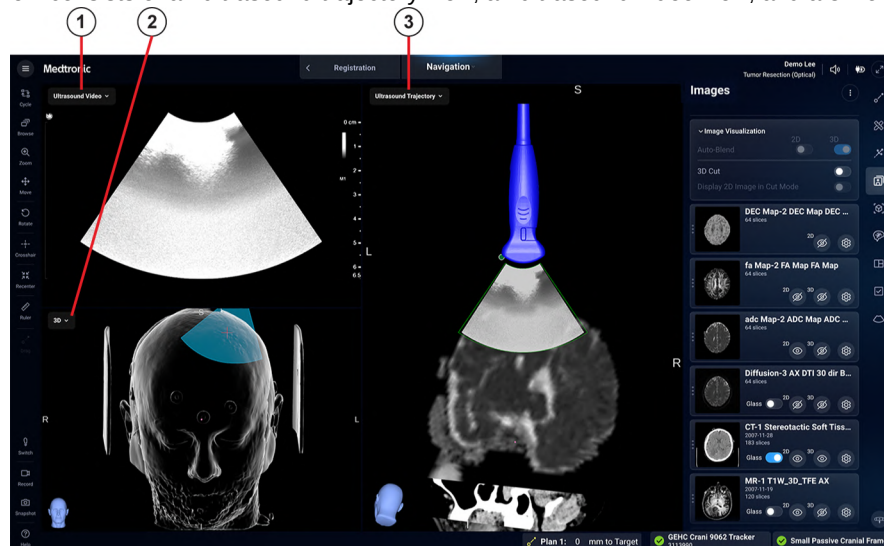
The ultrasound navigation screen has two view layout options:

The five-up layout view consists of an axial, coronal, and sagittal orientation view, an ultrasound trajectory view, and an ultrasound video view.



- 1 Ultrasound video view
- 2 Coronal orientation view
- 3 Sagittal orientation view
- 4 Ultrasound trajectory view
- 5 Axial orientation view

The three-up layout view consists of an ultrasound trajectory view, an ultrasound video view, and a 3D orientation view.



- 1 Ultrasound video view
- 2 3D orientation view
- 3 Ultrasound trajectory view

## Using an ultrasound system (optical only)

The clinical application displays the **Ultrasound** drawer in the **Navigation** and **Planning** tasks when ultrasound equipment is added to the procedure. The ultrasound drawer also displays when ultrasound video series are saved in the clinical application.

**Note:** When navigating the ultrasound transducer, the clinical application automatically updates the ultrasound views in real time. Tap a position on the screen in the ultrasound views to update the other views.

### Using the ultrasound trajectory and video views (optical only)

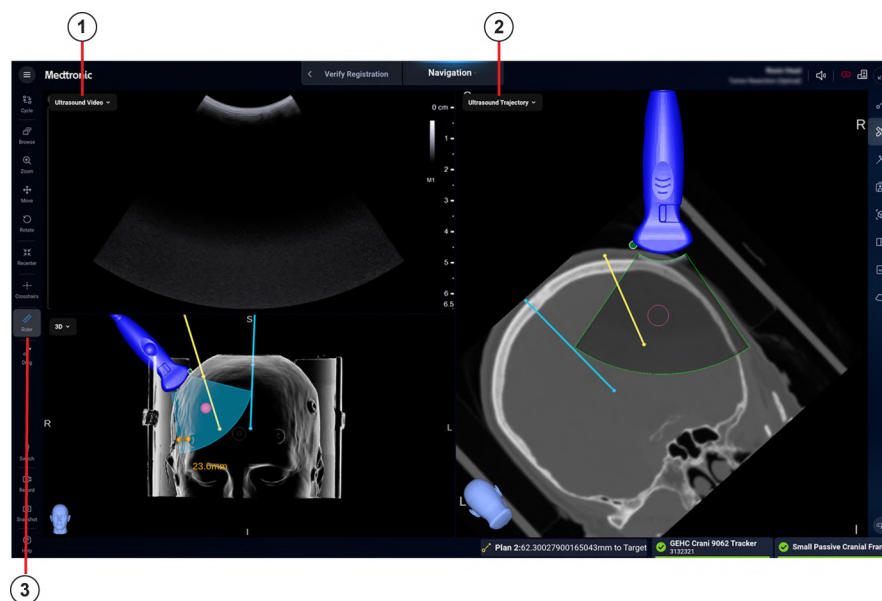
- The ultrasound trajectory view displays the ultrasound image fan overlaid onto the 2D image slice where the ultrasound transducer is positioned.
- Press and release the footswitch to freeze or unfreeze the ultrasound views.
- The navigation status outline around the fan indicates whether the ultrasound transducer is being tracked.

The ultrasound trajectory view displays annotations that are visible. You can create or modify annotations within the entire trajectory view (not just the ultrasound image area).

The ultrasound trajectory view also allows in-view measurements using the Ruler button. Tap anywhere in the view to establish a location for an annotation or a measurement. You can create or modify in-view measurements within the entire trajectory view (not just the ultrasound image area).

The ultrasound video view displays the full ultrasound image by itself. It has the same features as the ultrasound trajectory view except annotations and measurements can only be made in the ultrasound image area.

**Note:** You cannot create annotations or measurements while recording ultrasound images.



- 1 Ultrasound video view
- 2 Ultrasound trajectory view
- 3 Ruler button

### Adjusting ultrasound live video settings (optical only)

You can adjust the live ultrasound image during navigation by updating the **Live Video Settings**.

1. In the **Navigation** task, tap the **Ultrasound** drawer. The **Live Video Settings** card displays open in the drawer.
2. Use the **Opacity** slider to adjust the transparency of the ultrasound image in the views.
3. Use the **Level** and **Width** sliders to adjust the contrast of the ultrasound image in the views.
4. Tap the **Colormap** drop-down menu to choose a colormap for the ultrasound in the views.

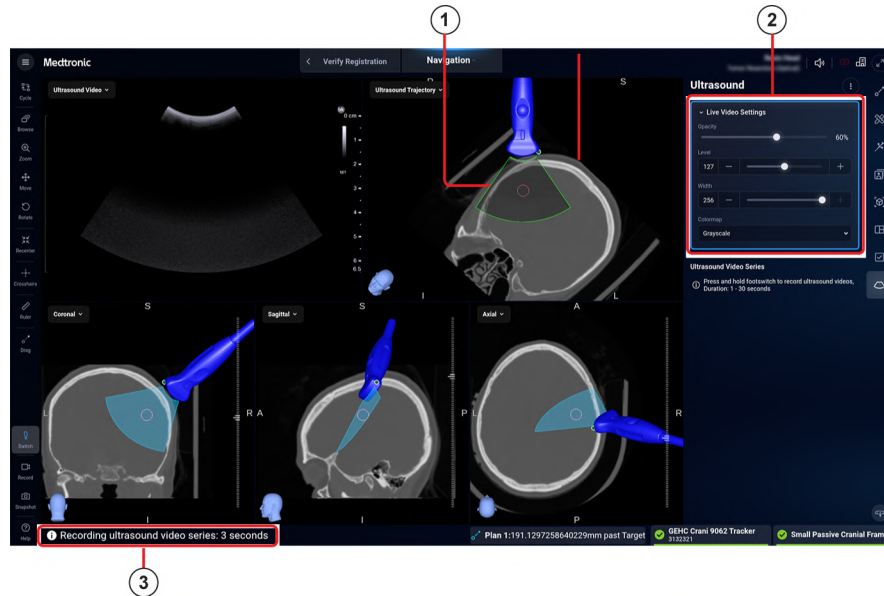
### Recording an ultrasound video series (optical only)

The recorded video series only contains ultrasound images collected when the ultrasound transducer is actively tracking and the fan outline is green or yellow. The recording will stop if the ultrasound system is changed to an unsupported mode or setting. To record an ultrasound video series, use the following steps:

1. Press and hold down the footswitch.
2. Move the ultrasound transducer slowly and evenly over the anatomy of interest to record the ultrasound images. The clinical application displays **Recording ultrasound video series** at the bottom of the screen.
3. Release the footswitch.

The video series displays as a thumbnail image in the **Ultrasound** drawer.

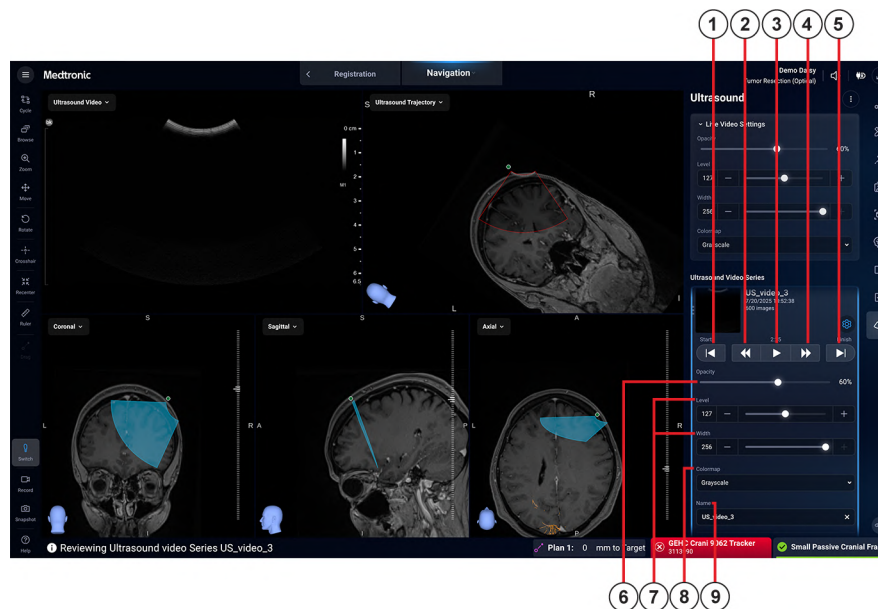
**Note:** The maximum duration of a recorded video series is 30 seconds.



- 1 Ultrasound image fan outline
- 2 Live video settings
- 3 Recording ultrasound video series

### Viewing an ultrasound video series (optical only)

- In the **Navigation** task, tap a video series card in the **Ultrasound** drawer to view an ultrasound video series.
- Tap the Settings button to perform any of the following actions:
  - To play an ultrasound video series, tap the Play button within the video controls. Tap the Forward, Backward, Skip Forward, and Skip Backward buttons to navigate within the video series.
  - Tap the **Colormap** drop-down menu to choose a colormap for the ultrasound video series.
  - Use the **Opacity** slider on the video series card to adjust the transparency of the ultrasound image in the views.
  - Use the **Level** and **Width** sliders on the video series card to adjust the contrast of the ultrasound image in the views.
  - To delete a video series, tap **[Delete]**.



- 1 Skip backward button
- 2 Backward button
- 3 Play button
- 4 Forward button
- 5 Skip forward button
- 6 **Opacity** slider
- 7 **Level** and **Width** sliders
- 8 **Colormap** drop-down
- 9 **Name** field

## J: Using tip projection

The tip projection feature allows you to localize at a point along the current instrument trajectory beyond the instrument tip, as if the selected instrument is longer than it is.

**Warning:** The end of the projection is not the same as the tip of the instrument. The projection shows where the instrument will go if you advance it along its current trajectory. Hide the projection when you want to localize at the tip of the probe or focal point of the microscope.

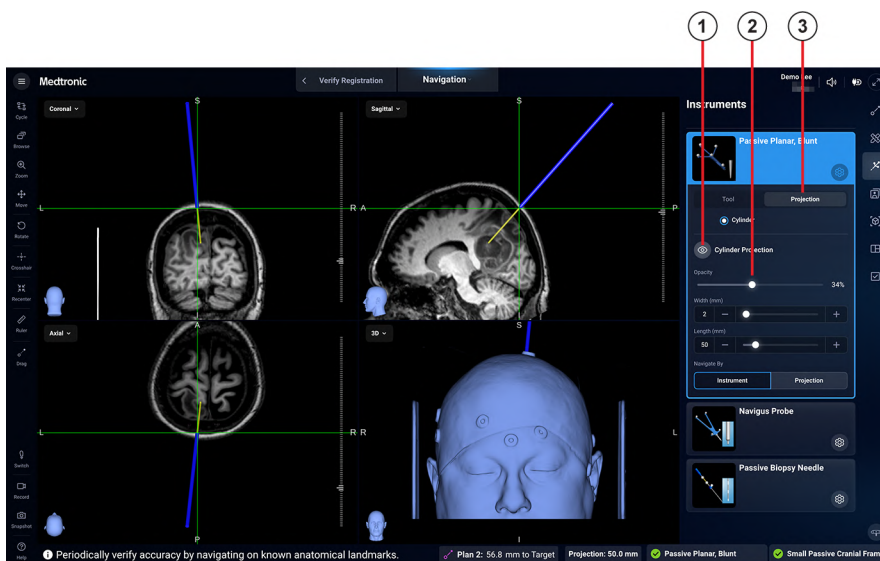
Follow these steps to use the projection feature:

1. Make sure that the instrument is verified.
2. In the **Navigation** task, tap **[Instruments]**.
3. Tap the **Projection** tab on the instrument you want to use.
4. Define the length and width of the tip projection by using the **Length** and **Width** sliders.
5. To adjust the opacity of the projection, use the **Opacity** slider.
6. Use the instrument to align the projection in the desired position.

**Note:** You can navigate the projection or the tip of the instrument by selecting either **Projection** or **Instrument**.

An extension from the end of the instrument model visually shows the cylinder projection. When you are navigating the projection, the crosshairs are located at the end of the projection and displayed distances are measured from the end of the projection. This is true even when a microscope interface is the selected instrument.

To hide a projection, tap the Show button. Tap the Show button again to display the projection.



- 1 Show button to display or hide projections
- 2 Opacity slider
- 3 **Projection** tab

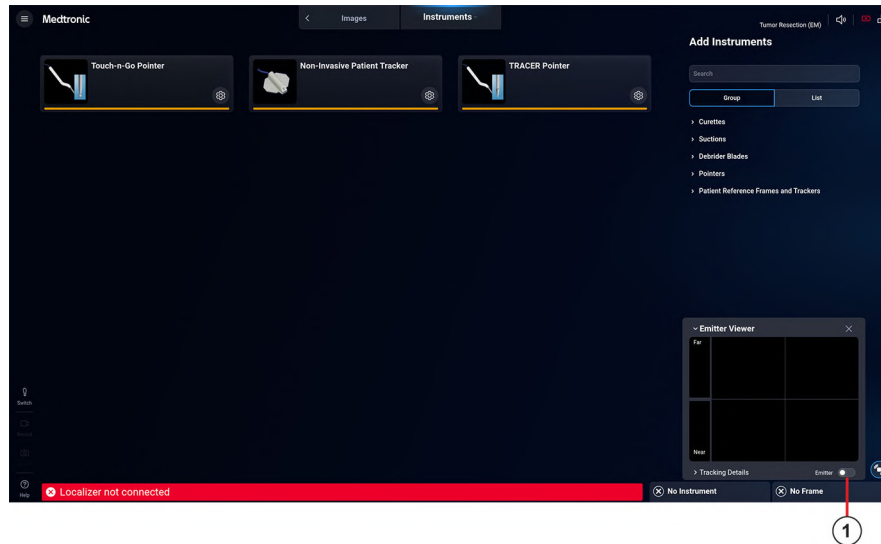
## K: Emitter on/off

Turn off the emitter temporarily to allow use of devices that are sensitive to electromagnetic fields. Live navigation is disabled while the emitter is turned off.

**Note:** When the emitter is unplugged and plugged back in, the emitter will display a connected status even if the **Emitter** toggle is turned off.

1. In the tracking view, tap **[Tracking Details]**.  
The status of the emitter displays in the tracking details.
2. To turn off the emitter, tap the **Emitter** toggle.  
**Note:** When the emitter is turned off, live navigation is disabled.
3. To turn on the emitter, tap the **Emitter** toggle again.  
The emitter displays **Connected** with a green bar under the emitter.

**Note:** The emitter takes a few minutes to resume transmitting.

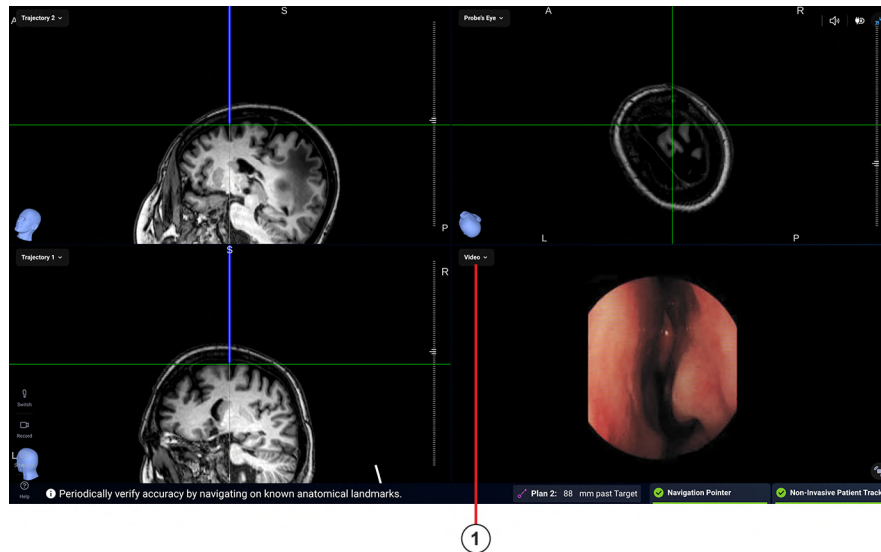


1 Emitter toggle

## L: Displaying video input

Video input from various devices can be displayed in the clinical application. Make sure that the video input device is properly connected to the system. Confirm that all critical information can be viewed on the source video monitor before importing the video stream. The system displays video input for secondary display purposes only.

- Output the video maintaining a frame rate of at least 4 Hz and a max latency of less than one second.
- Tap the **Image Selection** drop-down menu in an image view and select **Video** to display the video stream.



1 Video

## M: iMRI guidance (EM only)

Before acquiring an iMRI image, check for any EM patient trackers attached to the patient.

- If the EM tracker is MR Conditional, follow the MR conditions in the instructions that accompany the tracker.

**Caution:** The Non-Invasive Patient Tracker (NIPT) and the Skull-Mounted Patient Tracker (SMPT) are MR Conditional. See the *MRI safety information* instructions for use that accompanies these EM trackers for guidance on MR Conditional use. If the MR system exceeds the MR conditions for these EM trackers, remove the tracker from the patient before acquiring an iMRI image. Failure to follow the instructions can lead to patient harm.

- If the tracker is not MR Conditional, remove the tracker from the patient before acquiring an iMRI image.

**Caution:** Remove EM patient trackers that are not MR Conditional before acquiring an iMRI image. Failure to follow these instructions can lead to patient harm.

## N: Customizing procedures

You can add customized procedures to a surgeon profile or edit an existing procedure. Customizing procedures allows you to save preferences such as instrument and equipment selection, footswitch behavior, and viewing preferences.

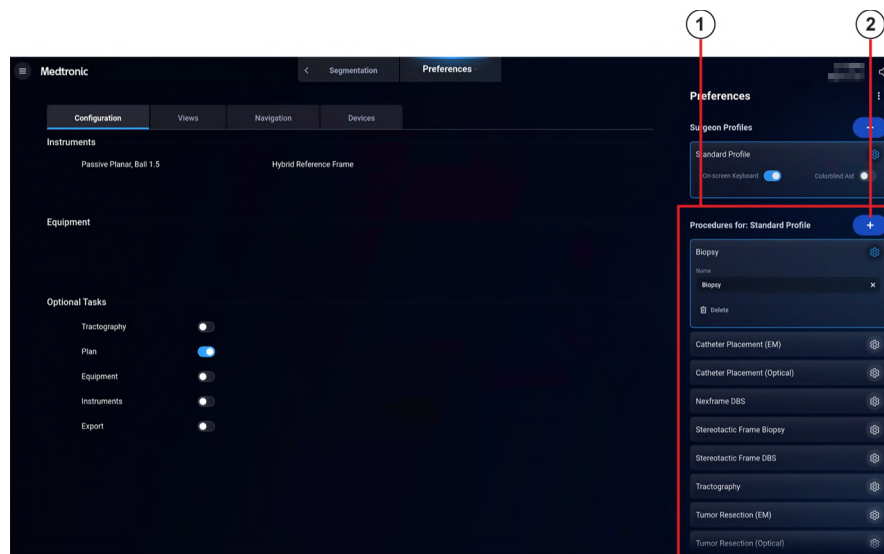
### Creating a new procedure

To create a new procedure in your surgeon profile and to customize the procedure selections and surgeon preferences, follow these steps:

**Note:** To create a new procedure, you must copy an existing procedure or surgeon profile.

- Tap the Menu button, and then tap **[Preferences]**.
- Under **Surgeon Profiles**, select the surgeon profile to display a list of available procedures that you can copy.
- Tap **[+]** next to the procedures list.
- Tap anywhere in the **Name** field and then enter a name for the procedure.
- Select a surgeon in the **Copy From** drop-down menu.
- Select a procedure from the procedure list.
- Tap **[Create]**.

The procedure is added to the procedure list for the surgeon. All instruments and equipment used in the procedure are automatically saved. You can further edit the procedure to change procedure selections and surgeon preferences. See “Editing a procedure.”



- Procedures list
- [+]** (Add New Procedure button)

### Editing a procedure

**Note:** Edits made to surgeon profiles only apply to the system the edits are made on.

You can change the procedure selections and surgeon preferences. To edit surgeon preferences for a procedure, follow these steps:

1. Tap the Menu button, and then tap **[Preferences]**.
2. Under **Surgeon Profiles**, select the surgeon profile to display a list of available procedures that you can edit.
3. Select the procedure that you want to edit.

**Note:** If you want to make an edit during a procedure, the clinical application will automatically edit the surgeon profile and procedure you are currently in.

4. Edit the selections for the behavior and preferences that you want for each tab.

The settings save automatically as you update them.

### Exporting surgeon profiles

You can export surgeon profiles when you set up a new system or if you want to update multiple systems with the same surgeon preferences.

**Note:** If surgeon profiles are imported between a planning station and a workstation cart, only shared settings will update.

1. Tap the Menu button, and then tap **[Preferences]**.
2. Tap the More button, and then tap **[Export Surgeon Profiles]**.
3. Insert media that you want to export surgeon profiles to, and then select it from the **Media** drop-down menu.

**Note:** You can export surgeon profiles to a USB, CD, or DVD.

4. Tap **[Export]**.
5. Once the export is complete, tap **[Close]**.

### Importing surgeon profiles

You can import surgeon profiles when you set up a new system or if you want to update multiple systems with the same surgeon preferences.

**Note:** If surgeon profiles are imported between a planning station and a workstation cart, only shared settings will update.

1. Tap the Menu button, and then tap **[Preferences]**.
2. Tap the More button, and then tap **[Import Surgeon Profiles]**.
3. Insert media that contains surgeon profiles you want to import.

**Note:** You can import surgeon profiles from a USB, CD, or DVD.

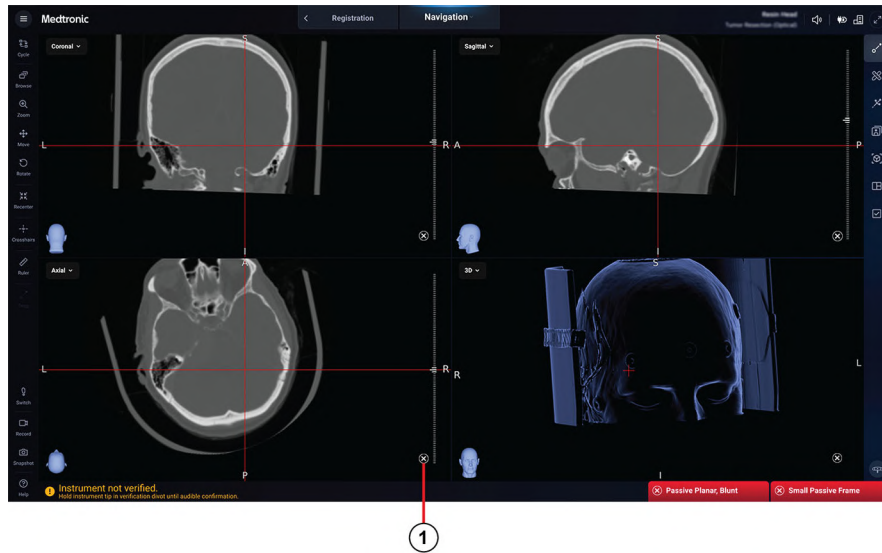
4. Select the desired surgeon profile file.
5. Tap **[Import]**.
6. Once the import is complete, tap **[Close]**.

### O: Enabling colorblind aid

The colorblind aid feature allows you to use a symbol in the views to show if the system is tracking a navigated instrument.

1. Tap the Menu button, and then tap **[Preferences]**.
2. Under **Surgeon Profiles**, select the surgeon profile that you want to edit.
3. Tap **Colorblind Aid** to switch it to the on position.

When the colorblind aid is enabled for the surgeon profile, the not navigating indicator displays in the bottom right corner of every image view.



- 1 The colorblind indicator that the instrument is not navigating.

## P: Controls

### Touch and mouse interactions

This table describes touch gestures for using the touchscreen to interact with the clinical application and the equivalent mouse interactions. For touch gestures that require only one finger, you can use a capacitive stylus instead of a finger.

| Touch gesture     | Gesture description                                                                                            | Equivalent mouse interaction                  | Clinical application action                                                                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tap               | Touch and release the screen with one finger.                                                                  | Click and release the left mouse button       | <ul style="list-style-type: none"> <li>Initiate an action with a clinical application button</li> <li>Select a menu option</li> <li>Identify a location in an image</li> </ul>           |
| One finger drag   | Place a finger on the screen and move it in a direction while maintaining contact with the screen.             | Left click and drag                           | <ul style="list-style-type: none"> <li>Rotate the 3D model</li> <li>Scroll through menu options</li> <li>Update a sliding value</li> <li>Use Paint or Erase in Model Building</li> </ul> |
| Two finger drag   | Place two fingers on the screen and move them in the same direction while maintaining contact with the screen. | Click <b>[Move]</b> and then click and drag   | Move images                                                                                                                                                                              |
| Pinch             | Place two fingers on the screen and bring them closer together to zoom out.                                    | Right click and drag down                     | Zoom out on the images                                                                                                                                                                   |
| Stretch           | Place two fingers on the screen and move them farther apart to zoom in.                                        | Right click and drag up                       | Zoom in on the images                                                                                                                                                                    |
| Two finger rotate | Place two fingers on the screen and rotate one around the other.                                               | Click <b>[Rotate]</b> and then click and drag | Rotate some 2D views                                                                                                                                                                     |
| Flick             | Quickly slide and release one finger on the screen.                                                            | Left click and drag                           | Swap images in the <b>Images</b> task                                                                                                                                                    |
| Touch and hold    | Touch the screen with one finger and hold it in place.                                                         | Left click and hold                           | To use the virtual <b>[Footswitch]</b> instead of a physical footswitch, touch and hold <b>[Footswitch]</b> until you want to release the footswitch.                                    |

| Touch gesture | Gesture description                                         | Equivalent mouse interaction | Clinical application action                                                                                                                                                                                                                                                                  |
|---------------|-------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Double-tap    | Touch the screen twice with one finger in rapid succession. | Double-click                 | <ul style="list-style-type: none"> <li>To temporarily change a multiple view layout to a one view layout, double-tap the Image Selection drop-down menu in any view. Double-tap again to return to the previous layout.</li> <li>Edit a plan name, plan color, or annotation name</li> </ul> |

## Keyboard shortcuts

The following table contains general clinical application keyboard shortcuts.

| Keyboard shortcut                                                                                                                                                                                         | Result                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <F1>                                                                                                                                                                                                      | Selects view layout preset 1, if stored.<br>At the login screen, switches between small and large text sizes.                                                                            |
| <F2>                                                                                                                                                                                                      | Selects view layout preset 2, if stored.<br>At the login screen, switches between a brighter or darker display.                                                                          |
| <F3>                                                                                                                                                                                                      | Selects layout view preset 3, if stored.<br>At the login screen, displays or hides the virtual keyboard.                                                                                 |
| <F4>                                                                                                                                                                                                      | Switches between a dot and a crosshair in the 2D views in the <b>Planning, Navigation, or Tractography</b> task.                                                                         |
| <F8>                                                                                                                                                                                                      | Displays the crosshair image coordinates panel. The coordinates shown are the current crosshair location. Allows user to specify the crosshair position and crosshair trajectory (Hind). |
| <F9>                                                                                                                                                                                                      | Creates a new annotation when the <b>Annotations</b> interface is open.                                                                                                                  |
| <Alt> + <t>: applicable for the workstation and planning carts<br><Alt> + <t>: (Windows) applicable during a remote planning session<br><Option> + <t>: (Mac) applicable during a remote planning session | Displays all annotation descriptions and saved measurements.                                                                                                                             |
| Left arrow or right arrow                                                                                                                                                                                 | Moves crosshairs by 0.1 mm along the current plan when the <b>Plan</b> drawer is open.                                                                                                   |
| Up arrow or down arrow                                                                                                                                                                                    | Moves crosshairs by 1.0 mm along the current plan when the <b>Plan</b> drawer is open.                                                                                                   |

## System Status Menu

The **System Status Menu** is in the top-right corner of the clinical application screen during a task on the workstation cart or planning station. Select the status icon area to open the **System Status Menu**.

| System Status icons                                                                                            | Description                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Volume</b><br>           | You can control the volume by selecting the Volume button.<br><b>Note:</b> The volume control is not available in remote planning sessions.                             |
| <b>Battery Status</b><br>   | On the workstation cart only, the battery status icon displays the current detected battery status. See "Battery backup" for more detailed battery status descriptions. |
| <b>Hospital Network</b><br> | The hospital network icon indicates whether you are connected to a network by Ethernet or WiFi, or if you are not connected to a network.                               |

**Note:** The **System Status Menu** does not display in the **Select Procedure** task.

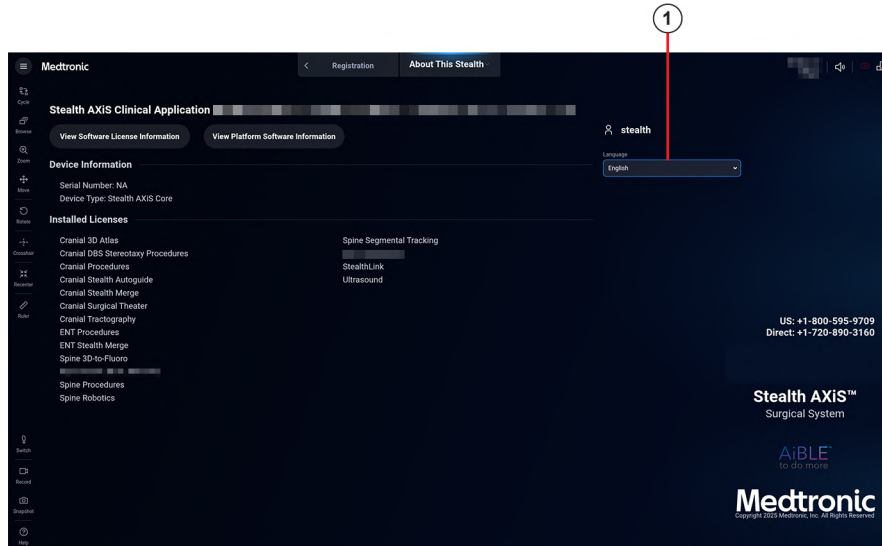
## Q: Changing the default system setup

### Changing the default language

From the **About This Stealth** window you can change the default language.

1. From the main menu, tap **About This Stealth**.
2. To change the default language settings for the application, tap the **Language** drop-down menu and select the preferred language.

Once a language is selected, the application will immediately update to the selected language.



1 Language drop-down menu

## R: DICOM Setup

The **DICOM Setup** task contains networking and DICOM settings for the Stealth AXiS™ System. You can add, edit, and delete PACS servers and other Medtronic Navigation™ systems. You can also run configuration tests and configure a system to act as a PACS server.

Use **DICOM Setup** to complete these tasks:

- Add a PACS server
- Edit a PACS server
- Delete a PACS server
- Change the DICOM settings

To open **DICOM Setup**, follow these steps:

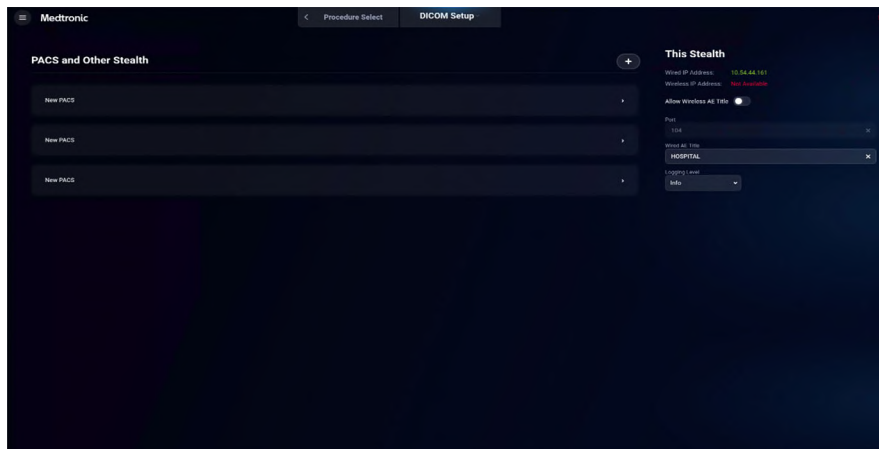
1. Tap the Menu button.
2. Tap **[DICOM Setup]**.

### Adding a PACS server

To add a new PACS server, follow these steps:

1. Open the **DICOM Setup** task.
2. Tap **[+]** to add a new PACS server.
3. Enter information for the PACS server:
  - **Name**
  - **AE Title**
  - **IP Address/Hostname**
  - **Port**
4. Select how the system will use the PACS server:

- **Query**
  - **Storage**
5. To save the PACS server as the default, tap **Default** (optional).
- Note:** Multiple servers can be configured, but the clinical application can only save one default server at a time.



### Editing a PACS server

To edit a PACS server or other Medtronic Navigation™ systems, follow these steps:

1. Open the **DICOM Setup** task.
  2. Select an existing DICOM source from the **PACS & Other Stealth** list.
  3. You can edit the following:
    - **Name**
    - **AE Title**
    - **IP Address/Hostname**
    - **Port**
  4. Select how the system will use the DICOM server:
    - **Query**
    - **Storage**
  5. Turn on the **Default** toggle (optional) to save the DICOM server as the default.
- Note:** Multiple servers can be configured, but the clinical application can only save one default server at a time.

### Deleting a PACS server

To delete a PACS server, follow these steps:

1. Open the **DICOM Setup** task.
2. Select an existing DICOM source from the **PACS & Other Stealth** list.
3. Tap **[Delete]**.
4. Tap **[Delete]** on the dialog to confirm. The server will be removed from the list of available servers.

### Running DICOM tests

The following DICOM tests can be run:

- **Local Config:** Displays information about the local configuration.
- **Ping:** Sends an ICMP echo to the IP address.
- **PACS Port:** Attempts to open the specified port.
- **Association:** Tests that the Stealth AXIS™ system can open a DICOM association to the DICOM server.
- **DICOM Echo:** Performs the C-ECHO DICOM command.

## Running all tests

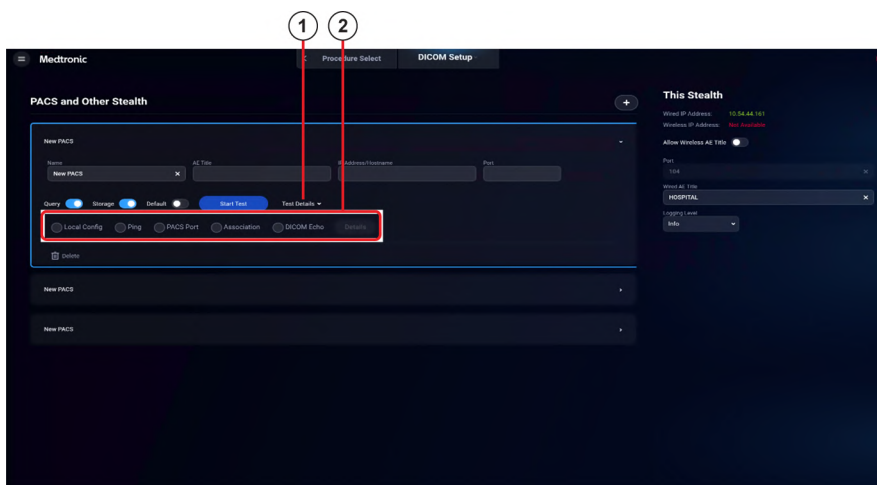
To run all tests and view results:

1. Tap **[Start Test]**.  
The **Test Details** drawer opens. As each test is executed, its corresponding button changes to a status color.
  - Green indicates the test passed.
  - Yellow indicates the test encountered a non-fatal error.
  - Red indicates the test failed with a fatal error.
2. Click **[Details]** to view the detailed output of each test run.

## Running selected tests

To select which tests you want to run and display the results:

1. Tap the **All Tests** tab.
2. In the **Test Details** drawer, tap the gray button next to each desired test.
3. Tap **[Start Test]**.  
As each test is executed, the corresponding gray button changes to a status color.
  - Green indicates the test passed.
  - Yellow indicates the test encountered a non-fatal error.
  - Red indicates the test failed with a fatal error.
4. Click **[Details]** to view the detailed output of each test run.



- 1 All Tests tab
- 2 Test Details drawer

## Changing This Stealth DICOM (PACS) settings

Update **This Stealth** DICOM (PACS) settings to configure the system to act like a PACS server for query, storage, or both.

Only one IP address (wired or wireless) can communicate on the network at any given time. Wireless settings are only shown on systems that have wireless network hardware.

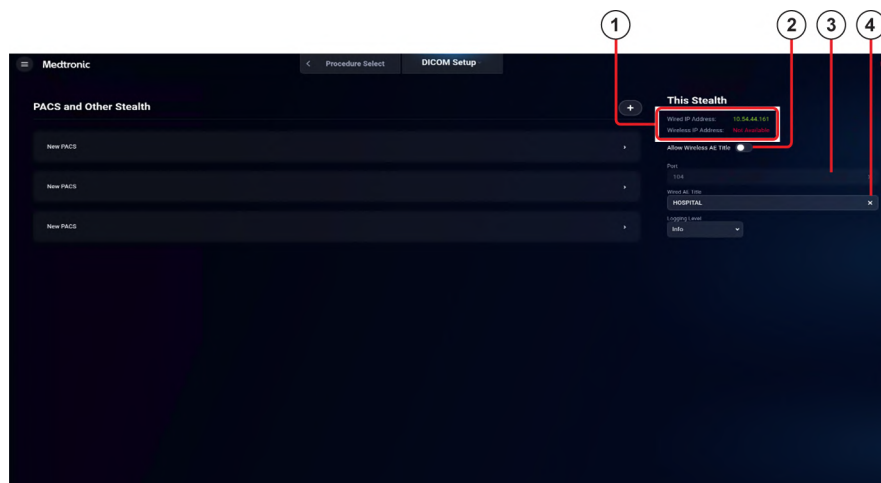
**Note:** You can use hostnames or IP addresses in the DICOM configuration. These might need to be the fully qualified domain name (FQDN) and not just the hostname by itself, depending on your network setup. Refer to the *Stealth AXiS™ IT Guide* for instructions on how to change IP/hostname.

To enter settings for the system, follow these steps:

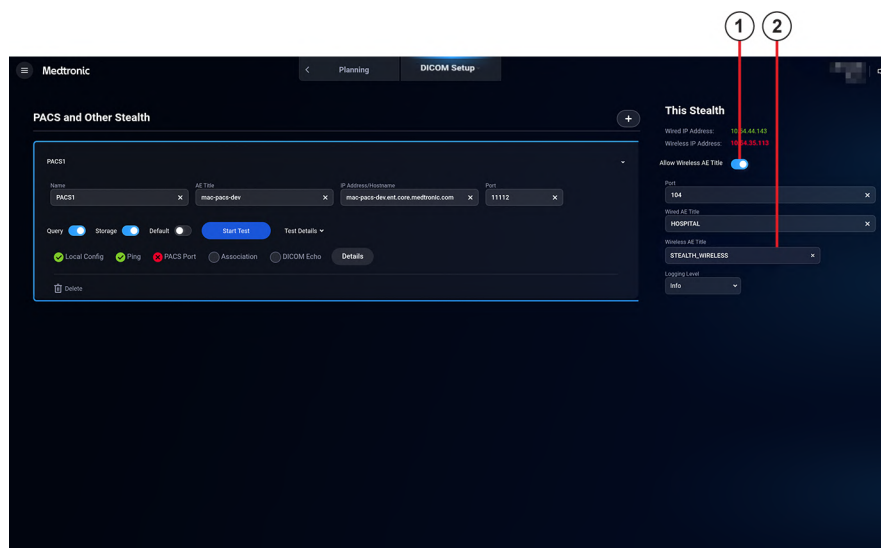
1. Open the **DICOM Setup** task.
2. Change these fields:
  - **Port**
  - **Wired AE Title**

**Note:** If you are using a workstation cart and the **Allow Wireless AE Title** option is switched on, fields appear for both **Wired AE Title** and **Wireless AE Title**. Both AE title fields can be changed.

**Note:** Only change the **Logging Level** from its default setting, "Info", at the request of Medtronic technical services.



- 1 If the system is able to communicate on the network, the IP address label is green. If the system is not able to communicate, the label is red. Only one IP address (wired or wireless) can be green at any given time.
- 2 Wireless settings are only shown on systems that have wireless network hardware.
- 3 **Port** field
- 4 **Wired AE Title** field



- 1 **Allow Wireless AE Title** toggle turned on
- 2 **Wireless AE Title** field

## S: Buttons

### Buttons in Images, Planning, or Navigation

The following buttons display in the **Images**, **Planning**, or **Navigation** tasks.



**[Cycle]** cycles through the preset image layouts.



**[Zoom]** magnifies an image. Tap **[Zoom]** and drag up or down on an image.



**[Correct]** displays options to reorient the images.



**[Switch]** controls different navigation functions. Tap **[Switch]** as an alternative to pressing the footswitch.



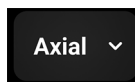
**[Browse]** scrolls through image slices. Tap this button and then drag up or down on an image.



**[Record]** captures and records a video of the screen and stores it in the patient data.



**[Crosshairs]** switches between crosshairs and dots in the 2D views.



The Image Selection drop-down menu allows you to select the content in each view. Tap this button and choose from the drop-down list (for example, **Axial**, **Coronal**, **Sagittal**, **3D**, or **Video**).



The layout buttons change the screen layout to the preset image views.



**[Move]** allows you to move an image within a view. Tap **[Move]** and then drag an image.



**[Recenter]** returns the crosshairs to the centers of the images.



**[Snapshot]** captures a snapshot of the screen and stores it in the patient data.



**[Ruler]** displays the distance in millimeters between any two selected points on an image.



**[Help]** displays relevant help information.



**[Rotate]** allows you to rotate the 3D model and the trajectory views. Drag the 3D model in the direction that you want it to rotate. In a trajectory view, tap one of the rotation buttons to turn the image 90°.



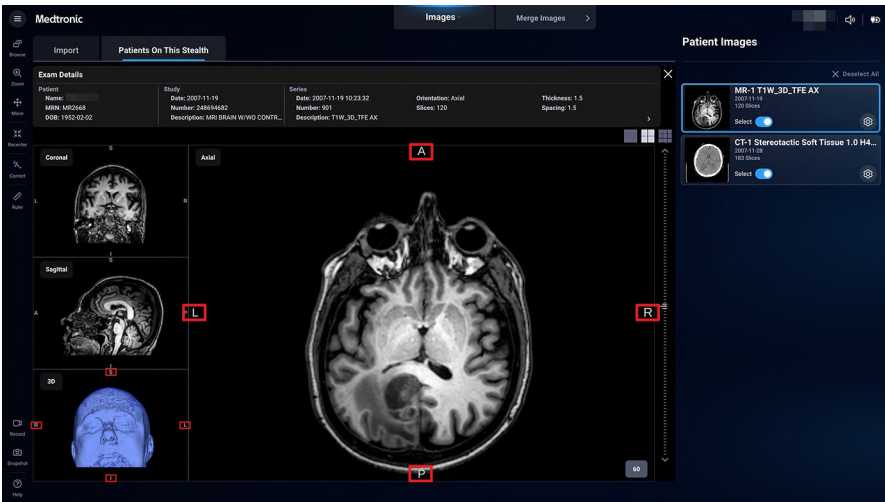
**[Drag]** allows you to drag the plan entry or target point. This enables you to update the entry and plan length.

## T: Image label translations

The clinical application displays image orientation labels on the edges of the image views to help identify the patient's orientation. The orientation labels are the initials for the six main anatomical location terms: inferior, superior, anterior, posterior, left, and right.

The image orientation labels may not be translated in the clinical application. They are labeled by the English initials I, S, A, P, L, R, and L.

| Label | Meaning   |
|-------|-----------|
| I     | Inferior  |
| S     | Superior  |
| A     | Anterior  |
| P     | Posterior |
| L     | Left      |
| R     | Right     |



U: Symbols

Explanation of symbols on package labeling

The following symbols may appear on system equipment, on system packaging, on accessories used with the system, or in this document.

|                                                                                    |                                                                                                                                                                          |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Medical Device                                                                                                                                                           |
|   | cTUVus Mark certifies the product meets the requirements of a Nationally Recognized Testing Laboratory for compliance with IEC 60601-1 and CAN/CSA-C22.2 No. 60601-1:14. |
|  | Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.                                                                                    |
|  | Caution                                                                                                                                                                  |
|  | Follow instructions for use.                                                                                                                                             |
|  | Consult instructions for use.                                                                                                                                            |
|  | Consult electronic instructions for use.                                                                                                                                 |
|  | Type BF applied part, in compliance with IEC/UL60601-1.                                                                                                                  |
|  | Type B applied part, in compliance with IEC/UL60601-1.                                                                                                                   |
|  | Single use only. Do not reuse.                                                                                                                                           |
|  | Use by date                                                                                                                                                              |
|  | Batch code                                                                                                                                                               |
|  | Serial number                                                                                                                                                            |
|  | Catalog number                                                                                                                                                           |
|  | Quantity                                                                                                                                                                 |

|                                                                                     |                                                                    |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|    | Sterile                                                            |
|    | Sterilized using ethylene oxide                                    |
|    | Sterilized using irradiation                                       |
|    | Single sterile barrier system                                      |
|    | Single sterile barrier system with protective packaging inside     |
|    | Single sterile barrier system with protective packaging outside    |
|    | Double sterile barrier system                                      |
|    | Non-sterile                                                        |
|    | Do not sterilize.                                                  |
|    | Do not resterilize.                                                |
|    | Manufacturer                                                       |
|    | Authorized representative in the European Community                |
|    | Responsible person in Great Britain (England, Wales, and Scotland) |
|    | Importer                                                           |
|  | Date of manufacture                                                |
|  | Keep dry.                                                          |
|  | Do not use if package is damaged.                                  |
|  | Fragile, handle with care.                                         |
|  | Keep upright.                                                      |
|  | Do not stack.                                                      |
|  | Power on. Connect to main power.                                   |
|  | Power off. Disconnect from main power.                             |
|  | Power on for part of the system.                                   |
|  | Power off for part of the system.                                  |
|  | Power button                                                       |
|  | Protective Earth (ground)                                          |
|  | AC Power                                                           |



Equipotentiality: identifies the terminal that when connected together, bring the various parts of the equipment or system to the same potential, not necessarily being earth potential (for local bonding).



Dead CMOS battery



Do not allow contact with patient. Temperature may exceed limits.



Atmospheric pressure limits



Humidity limits



Temperature limits



Localizer must not be used in ambient temperatures greater than 30°C (86°F).



Storage temperature between -29°C and 60°C (-20°F and 140°F)



Shipping temperature between -30°C and 60°C (-22°F and 140°F)



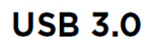
MR Conditional. Certain accessories that may be used with the system are MR Conditional. See the labeling for those devices for additional instructions on MR conditional use.



Do not disassemble.



USB 2.0 port



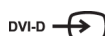
USB port: Connects the system to an external high-speed USB device.



Ethernet port: Connects the system to the site network and medical imaging devices.



DisplayPort for video out



DVI-D port for video in



Footswitch



Camera cart port: Connects the camera cart to the workstation cart.



Robot cart port: Connects the robot cart to the workstation cart.



Workstation cart port: Connects the workstation cart to the robot cart.



Cart docking location



EM instrument interface



EM emitter



Microscope: Connects the system to a surgical microscope. The orange port is for analog microscopes, and the yellow port is for digital microscopes.



SCU ports: Connects an active, wired instrument (patient reference frame or microscope tracker) to the system.



Radio frequency device. Interference may occur in the vicinity of the device.



Protect from heat and radioactive sources

IPX0

Not protected

IPX2

Protected against dripping water when tilted up to 15 degrees

IPX6

Protected against powerful water jets



Locked



Unlocked



Motion lock



Directional lock



Robot arm protective stop



Do not transport the camera cart with a camera or monitor in an undocked position. Always dock the camera and monitor before moving the camera cart. Use the handle to push the cart.



Do not transport the workstation cart with the monitor in an undocked position. Always dock the monitor and close drawers before moving the workstation cart. Use the handle to push the cart.



Do not push the cart from the back of the cart or from the labeled or adjacent surfaces.



LASER radiation emitted from aperture. Do not stare into beam. Class 2 LASER product. Maximum output: <1mW, CW: 635–670 nm, IEC/EN 60825-1 (2014), ANSI Z136.1 (2014). Complies with 21CFR1040.10 and 1040.11 except for conformance with IEC 60825-1 Ed. 3., as described in Laser Notice No. 56, dated May 8, 2019.



Do not dispose of this product in the unsorted municipal waste stream. Dispose of this product according to local regulations. See <http://recycling.medtronic.com> for instructions on proper disposal of this product.



China RoHS compliant. Environmental protection use period of 5 years. Environmental protection use period of 10 years. Environmental protection use period of 50 years.



China RoHS compliant: Electronic Information Products Pollution Control symbol. (The “e” in the logo signifies none of the restricted substances are present above permitted levels.)



Open here

**R<sub>x</sub> Only**

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Medtronic Navigation, Inc.  
200 Medtronic Drive  
Lafayette, CO 80026  
USA  
Main 1 720 890 3200  
Fax 1 720 890 3500  
[www.medtronic.com](http://www.medtronic.com)  
Technical support:  
USA 1 800 595 9709  
International +1 720 890 3160  
[rs.navtechsupport@medtronic.com](mailto:rs.navtechsupport@medtronic.com)  
[www.medtronic.com/manuals](http://www.medtronic.com/manuals)

